BIOPOLICY SOLUTIONS, LLC

September 12, 2024

Susan Carlson, Ph.D., Director
Office of Food Additive Safety (HFS-200)
Center for Food Safety and Applied Nutrition (CFSAN)
Food and Drug Administration
5001 Campus Drive
College Park, MD 20740

Re: GRAS Notice for the Use of Phoenix Oyster Mushroom (*Pleurotus pulmonarius*) Mycelium as an Ingredient in Food

Dear Dr. Carlson,

In accordance with 21 CFR 170 Subpart E (170.203 through 170.285), BioPolicy Solutions LLC (Houston, Texas, USA) is submitting this GRAS Notice as an agent on behalf of Mushlabs GmbH (Humboldtst. 59, 22083 Hamburg, Germany). The enclosed document provides notice to the agency that the food ingredient, Phoenix Oyster mushroom (*Pleurotus pulmonarius*) mycelium (POMM) described in the enclosed notification is exempt from the premarket approval requirement of the Federal Food, Drug, and Cosmetic Act because it has been determined to be generally recognized as safe (GRAS) based on scientific procedures.

Attached to this letter for review and evaluation by FDA is all the information and data upon which Mushlabs' GRAS conclusion has been reached.

Please direct any questions or concerns regarding this GRAS Notice that may arise during the review process to BioPolicy Solutions LLC to ensure a timely response. BioPolicy Solutions can be reached by e-mail at lmp@bpsllc.us (Dr. Laura Plunkett) or lxr@bpsllc.us Dr. Larisa Rudenko).

Sincerely,

[redacted]

Laura M. Plunkett, Ph.D., DABT
Co-Founder BioPolicy Solutions LLC

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The Generally-Recognized As Safe (GRAS) Determination of Phoenix Oyster Mushroom Mycelium for Use as an Ingredient in Human Food

Submitted By:

Mushlabs GmbH
Humboldtstr. 59, 22083 Hamburg, Germany

Submitted To:

U.S. Food and Drug Administration
Office of Food Additive Safety HFS-200
5001 Campus Drive College Park, MD 20740

Contact for Technical or Other Information:

Laura M. Plunkett, Ph.D, DABT
Larisa Rudenko, Ph.D
BioPolicy Solutions LLC
Houston, TX 77077

September 12, 2024

Table of Contents

PART 1. Signed Statements and Certification

In accordance with 21 CFR 170 Subpart E (170.225(c)(1)), BioPolicy Solutions LLC (Houston, Texas, USA) is submitting this GRAS Notice as an agent on behalf of Mushlabs GmbH (Humboldtstrasse. 59, 22083 Hamburg, Germany). Mushlabs claims that the use of Phoenix oyster mushroom mycelial biomass (POMM) as an ingredient in food, as described in Parts 2 through 7 of this GRAS Notice is not subject to the premarket approval requirements of section 409 of the Federal Food, Drug and Cosmetic (FD&C) Act based on its conclusion that POMM is GRAS under the conditions of its intended use.

1.A Name and Address of the Notifier

Mushlabs GmbH
Thibault Godard
Chief Science Officer
Humboldtstr. 59, 22083
Hamburg, Germany

1.B Name of GRAS Substance

Mycelial biomass of *Pleurotus pulmonarius* [Phoenix Oyster Mushroom Mycelial biomass] (POMM); the fruiting bodies of this mushroom are known as Phoenix Oyster Mushroom, Oyster Mushroom, Summer Oyster Mushroom, Italian Oyster Mushroom, or Indian Oyster Mushroom.

1.C Intended Conditions of Use

POMM is intended for use as a food ingredient[1] in processed food products including meat, seafood, and fish analogs, dairy analogs (milk, cream, yogurt), processed cheese, ice cream, and baked goods at a maximum level of 25% (dry mycelium weight/100 g product). Table 1.1 lists the intended uses and the maximum levels at which POMM could be added to those foods.

TABLE 1.1
Intended Uses of POMM as an Ingredient in Food Products

Food Category	Maximum Use Level of POMM (g/100g)
Milk analogs	10
Cream analogs	10
Ice cream analogs	15
Yogurt analogs	15
Processed cheese	25
Meat analogs	25
Fish analogs	25
Seafood analogs	25
Baked goods	25

POMM is not intended for use in infant formula or USDA-regulated products.

[1] POMM is not intended to comprise 100% of a food product.

1.D Basis for Conclusion of GRAS Status

Mushlabs GmbH (hereafter referred to as Mushlabs) concludes that the intended uses of POMM are Generally Recognized As Safe (GRAS) through Scientific Procedures. Mushlabs understands that although there is an extensive history of common use of the Phoenix oyster mushroom fruiting body in food prior to 1958, there is no documentation of marketing of products for human consumption derived from dikaryotic Phoenix oyster mushroom mycelial biomass (POMM) produced via submerged fermentation before 1958. Nonetheless, Mushlabs notes that the process of submerged fermentation of mushroom mycelium (including oyster mushrooms) and its potential for producing food has been described in scientific literature since 1953 (Block 1953; Sugihara and Humfeld 1954).

1.E Availability of Information

The data and information that form the basis of Mushlabs GmbH's conclusion that the use of POMM as an ingredient in food is GRAS are available for review and copying by FDA during customary business hours, at the location below, or will be sent to FDA upon request made to:

Mushlabs GmbH
Thibault Godard
Chief Science Officer
Humboldtstr. 59, 22083 Hamburg, Germany
thibault@infiniteroots.com

1.F Availability of FOIA Exemption

It is Mushlabs' view that the information contained in this GRAS notice is not exempt from disclosure under the Freedom of Information Act, 5 U.S.C. 552.

1.G Certification

To the best of my knowledge, this GRAS Notice is a complete, representative, and balanced submission that includes unfavorable and favorable information known to Mushlabs and pertinent to the evaluation of the safety and GRAS status of the intended uses of POMM.

Name, Position, and Signature of Authorized Individual and Date of Signature

[redacted] Chief Science Officer 6 September 2024

Thibault Godard **Position** **Date Signed**

PART 2. Identity, Method of Manufacture, Specifications, and Physical or Technical Effect

2.1 Identity

The subject of this notice is the dikaryotic mycelial biomass of *Pleurotus pulmonarius,* Phoenix oyster mushroom PXM1906, the fruiting bodies of which are commonly known as Phoenix oyster mushroom or oyster mushroom, hereafter referred to as "POMM". POMM is dikaryotic mycelium directly isolated as a piece from the fruiting body of *P. pulmonarius* PMX1906 and the dikaryotic mycelial biomass is subsequently produced by propagating the material in a fermenter under highly controlled conditions. The identity of PXM1906 was confirmed as an isolate of *P. pulmonarius* by genetic comparison of the isolate from the MycoBank database. Details can be found in Appendix A.[2]

Taxonomically, *P. pulmonarius* is categorized as *Pleurotus pulmonarius* (Fr.) Quél. The fruiting body of *P. pulmonarius* is commonly found in European, North American, and Asian diets.

Taxonomy
- Kingdom: Fungi
- Phylum: Basidiomycota[3]
- Class: Agaricomyces
- Order: Agaricales
- Family: Pleurotacaea
- Genus: Pleurotus
- Species: pulmonarius

2.1.1 Synonyms

Although there is a large diversity within the *Pleurotus* genus at a genetic level, they are generally referred to as "oyster mushrooms" and sold with little to no distinction of species at the market level (Pawlik *et al.* 2012; Zhao *et al.* 2016; Khan *et al.* 2017; Panek *et al.* 2019; Li *et al.* 2019; Lin *et al.* 2022). This can be attributed to the difficulty encountered in distinguishing one species from another based only on visualization of morphological characteristics. In particular, *P. ostreatus* and *P. pulmonarius* are often described in the literature as morphologically identical and as being seasonal variants of the same species (Anderson *et al.* 1973; Eger *et al.* 1979; Bresinsky *et al.* 1987; Guzman 2000; Lechner *et al.* 2004). Recent genetic evidence has shown that what previously had been considered to be a separate species, *P. sajor-caju*, is now considered to be within *P. pulmonarius*. The use of two different names derives from the independent description of the mushroom when found in different ecological niches before modern genetic analysis. The determination that both reside within a single species (Bao *et al.* 2004; Shnyreva and Schnyreva 2015) was due to the observed 97 to 100% concordance of the internal transcribed spacer (ITS)

[2] Mushlabs engaged a contract testing laboratory (Charles River Laboratories or CRL) to employ the "genetic barcode" of the internal transcribed spacer (ITS) region from the nuclear ribosomal RNA cistron (concluded to be a sequence that identifies a wide range of fungal species by Schoch *et al.* (2012)) to confirm the identity of its *P. pulmonarius* strain fungal species (Shnyreva and Shnyreva 2015; Rugolo *et al.* 2019).

[3] Basidiomycetes (the less formal version of Basidiomycota) are members of a subkingdom referred to as Dikarya.

reported by Zmitrovitch and Wasser (2016). Therefore, data on *P. sajor-caju* are directly relevant for the evaluation of the safety of POMM; these data are included in this GRAS Notice.

Due to their commercial importance, several studies have investigated the genetic relationships among various *Pleurotus* species (Pawlik *et al.* 2012; Zhao *et al.* 2016; Khan *et al.* 2017; Steenkamp *et al.* 2018; Li *et al.* 2019; Panek *et al.* 2019; Lin *et al.* 2022). Despite similar morphologies, there is enough genetic variability among the different species to prevent cross-species mating. Any differences that exist do not appear to represent significant differences in the biological processes that are involved in the process of forming the fruiting body (see discussion in Part 6) Further, differences important to evolutionary divergence with respect to taxonomy likely have minimal implications for food safety. Regardless, this notice is focused on *P. pulmonarius* dikaryotic mycelium.

2.2 Method of Manufacture

Mushlabs conducts all processing of the biomass in accordance with current Good Manufacturing Practice (cGMP), 21 CFR § 117, Subpart B.

2.2.1 Overview of Manufacture

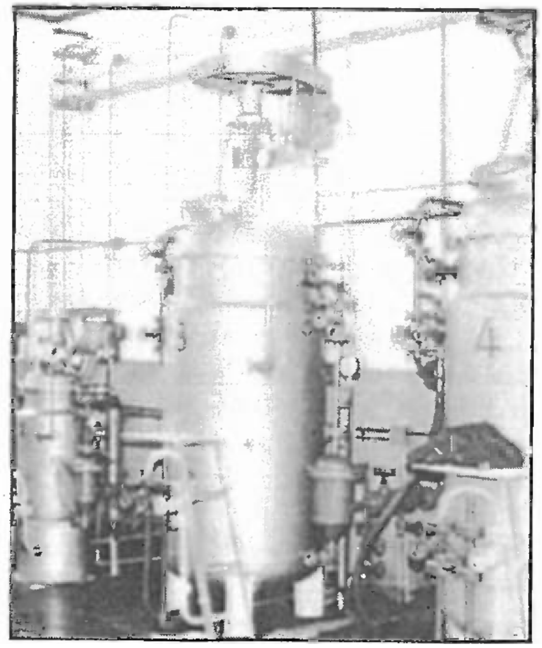
Mushlabs uses submerged fermentation to produce mycelial biomass from the edible mushroom *P. pulmonarius*. This process has been used in the mushroom industry since the 1960s to produce mycelium spawns that are then inoculated onto solid substrates from which fruiting bodies are collected after weeks of mycelial growth.

This method was first reported and patented by Humfeld (1948) for *Agaricus campestris*[4] (the button mushroom) and subsequently adopted by others for other edible mushroom species including oyster mushrooms (Sugihara and Humfeld 1954). In 1964, the French company Royal Champignon SA patented the process for species including oyster mushrooms.[5] Since then, submerged fermentation has been used commonly by mushroom growers and is discussed in manuals of mushroom production (*e.g.,* Zadrazil 1978). Stamets also described this method in detail including its applicability to *P. pulmonarius* (Stamets 1993; Figure 2.1).

[4] https://patents.google.com/patent/US2693665A/en

[5] https://patents.google.com/patent/FR1383075A/en; https://patents.google.com/patent/US3286399A/en

Figure 2.1 (from Stamets 1993). Pressurized vessel to produce mycelium spawns from spores or mycelium



The process developed by Mushlabs follows standard principles and can be separated into three stages: inoculum preparation and seed train, main fermentation, and separation of the mycelial biomass. Figure 2.2 provides a schematic description of the manufacturing process that includes critical control points.

The inoculum is prepared from dikaryotic mycelium[6] isolated from a fresh mushroom fruiting body of *P. pulmonarius* PMX1906 by methods described by Eger (1979), adapted specifically for *P. pulmonarius* (as *P. sajor-caju*) by Liu *et al.* (2016). These methods rely on the fact that fruiting bodies are solely made of totipotent dikaryotic hyphae, except for the spores, *i.e.* any fragment of a fruiting body can be used to start a mycelial culture that eventually will be assembled into dikaryotic mycelia, new fruiting bodies, or Mushlabs' mycelial biomass (Money 2002).

[6] Mushlabs does not use the vegetative monokaryotic mycelium found in the soil in nature as a source of mycelium in its production process.

Figure 2.2: Schematic of POMM Manufacturing Process

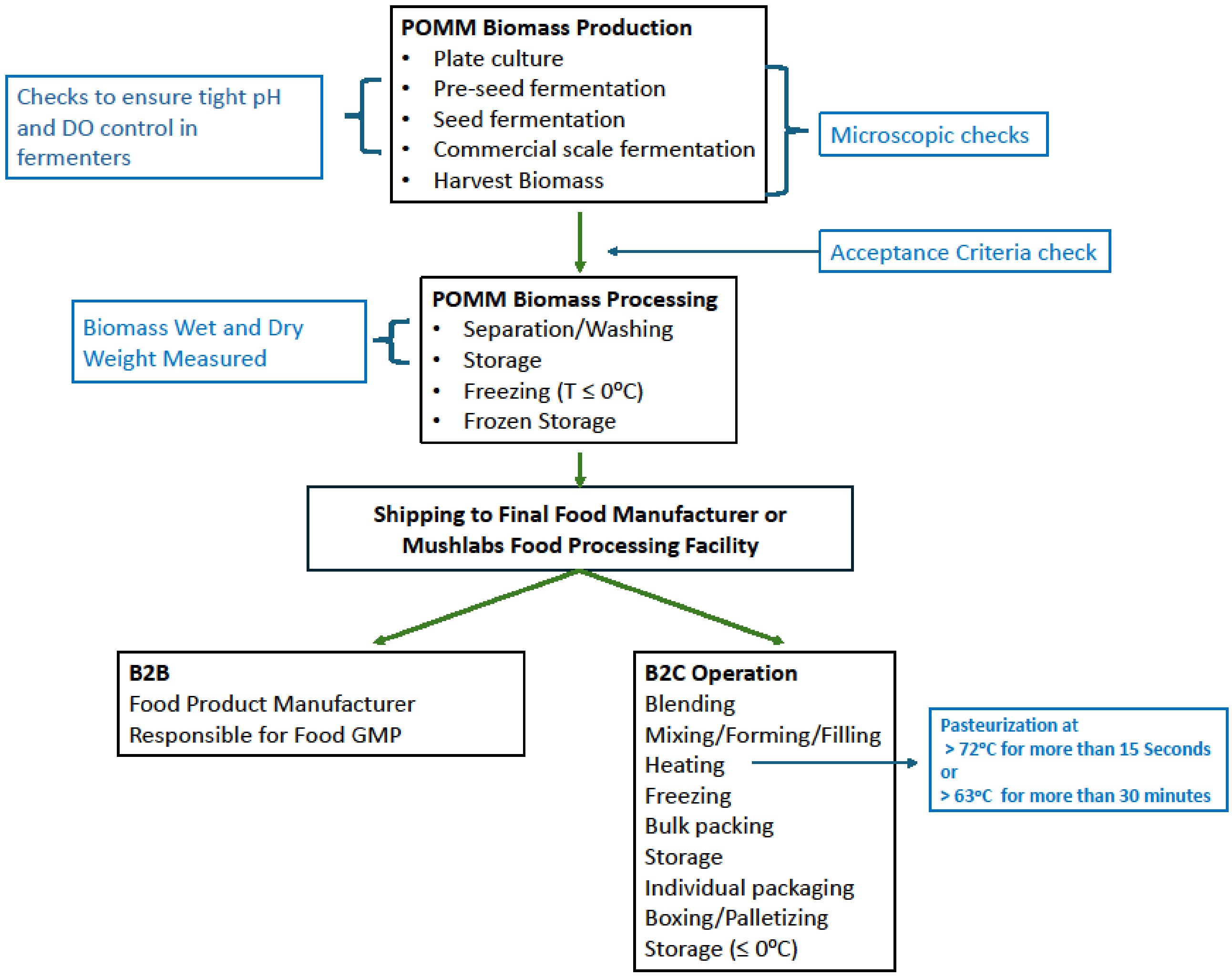


Figure 2.2 Legend: This figure represents the overall manufacturing process for POMM. Mushlabs can function as either a B2B (business to business) operation (*i.e.*, it manufactures a biomass (food ingredient) that is subsequently shipped to the ultimate food producer which is responsible for subsequent cGMP manufacture and safe handling) or a B2C (business to consumer operation) in which case it is responsible for the cGMP safe handling of the retail food products. In the B2B scenario, following the immediate cold storage of the biomass, Mushlabs will further separate and wash the biomass, store it at 4ºC, freeze it to 18ºC or dry it, and ship it to the ultimate food manufacturer in frozen or dry form. In the B2C scenario, in which Mushlabs operates as the ultimate producer of the food products, after the separation and washing of the POMM biomass, Mushlabs will blend POMM with other food-grade materials. Then, depending on the ultimate food product, it will mix, form, and fill processing containers; heat the resulting material at pasteurization conditions (for dairy substitutes) or baking temperatures, and freeze the resulting food products. When commercial need arises, Mushlabs will package the food product into retail sale form and prepare them for shipping to the retail establishment(s) for sale.

The pure isolate of POMM is first grown on agar plates containing appropriate commercially available culture media then transferred aseptically to liquid culture medium in small culture flasks with a sterile loop. In these flasks, the mycelium is submerged in nutritional medium and incubated (temperature between 25-35°C, pH between 3.5 and 6.5) until the culture has reached a biomass sufficient to inoculate larger culture vessels. The composition of all growth media used for production (described in Section 2.2) is simple, including only basic nutrients or substances typically found in food; all ingredients are food grade, and none are derived from the nine major allergens listed in the Federal Food, Drug, and Cosmetic Act (FFDCA) found in 201(qq)(1) *(i.e.,* milk, eggs, finfish, crustacean shellfish, tree nuts, peanuts, wheat, soybeans, and sesame). Table 2.2 lists the specific food grade materials used in the production of POMM, all of which are used within appropriate FDA regulations. Generally, the culture volume is increased by a factor between 10- to 100-fold in the new flasks and the process is repeated until sufficient mycelial biomass has been generated to inoculate a fully automated pre-fermenter.

As shown in Figure 2.2, all growth vessels are thoroughly cleaned, sterilized, and reused. Large fermenters are subjected to sterilization in place (SIP). These fermenters contain sterile culture medium composed of the food-grade materials listed in Table 2.2. Depending on the inoculum size and activity, the fermentation process in the pre-fermenter will take between 40 and 200 hours and its contents can then be used as inoculum for the next larger fermenter. Typical parameters that lead to transfer to larger fermenters include depletion of one of the key nutrients, reaching a specific pH value (*e.g.,* pH outside the 3.5-6.5 range that is optimal for growth), or high CO_2 content in exhaust air. For each transfer, the culture volume and fermenter size are increased 10-100-fold, and the process is repeated until the main fermenter (volume ≥ 200 L) can be inoculated with sufficient biomass. The final or main fermentation is performed at scales of up to 200 m^3 under controlled conditions (25 to 30°C, pH 3.5 to 6.9; controlled dissolved oxygen, controlled aeration rates, and controlled stirrer speed) using the medium described in Table 2.2. Fermentation continues for approximately 40-200 hours until one of the nutrient sources required for growth is depleted as determined by external measurement such as pH, exhaust gas analysis, or HPLC.

At final harvest, the biomass is separated mechanically using methods with long history of use in the food industry and relying mostly on centrifugal and gravitational forces (*e.g.,* decanting, sieving and/or filtration), alone or in combination, and the biomass is then washed with acidic water (pH between 3 and 6.9) to remove the culture broth and to minimize the potential growth of environmental microorganisms.[7] The biomass can then be washed again with water or salt solution and stored at 4°C before concentration using an additional separation step with the same methods to achieve 3-30% dry mass for the downstream process (production of final food products). The combination of an acid wash and optional water wash ensures that any media substances not depleted during growth have been removed. After harvest, the POMM biomass is frozen for storage before transfer to food production facilities for further processing. The freezing of POMM biomass (temperatures ≤ -18°C) effectively inactivates the biomass after harvest. Data shown in Table 2.1 confirms that after frozen storage for more than 24 hours, the product itself is non-viable and incapable of growth on an appropriate sterile agar plate. As indicated in Table 2.1, no yeast or mold growth was observed above the limit of detection in the final frozen product.

[7] This process decreases the microbial load of the harvested biomass to the levels defined as product acceptance criteria (Table 2.3) and product specifications (Table 2.7).

POMM is not a “ready to eat” ingredient and is intended for use by food manufacturers either in wet or dry forms[8] at fractional compositions of up to 25% (dry w/w) to formulate final food products. The final food products are intended to be further heat-processed by cooking or pasteurization, thus ensuring heat inactivation of POMM.

TABLE 2.1 Microbial Analysis of POMM*		
Organism	**Limit of Detection (CFU)**	**Frozen POMM (CFU)**
Total Viable Count**	1,000	<1,000
Yeasts	1,000	<1,000
Molds	1,000	<1,000
Enterobacteriaceae	100	<100
E. coli	10	<10
Coagulase (+) Staphylococci	100	<100
Bacillus cereus	100	<100
Clostridium perfringens	100	<100
Listeria monocytogenes	10	<10
Salmonella spp.	--	negative
* Measurements taken after storage at temperatures below -18°C for more than 24 Hours **Total Viable Count is equivalent to “Aerobic Plate Count” in Table 2.3 that describes microbiological acceptance criteria.		

2.2.2 Composition of Growth Medium

The composition of the growth medium used in POMM production is simple, including only basic cell nutrients or substances typically found in food; all ingredients are food grade, and none are derived from the nine major allergens according to US law (*i.e.,* milk, eggs, finfish, crustacean shellfish, tree nuts, peanuts, wheat, soybeans, and sesame). Table 2.2 lists the food grade materials used in the production of POMM. All raw materials are used in accordance with applicable U.S. regulations, including not exceeding allowable amounts for a particular use. Mushlabs’ review of each of the Certificates of Analysis (CoA) for the media components also indicates that none of them are major food allergens or are derived from the major food allergens[9].

TABLE 2.2 List of Substances Used in Various Formulations of Mushlabs’ Growth Media	
Category	**Substances Used**
Amino acids	20 common amino acids
Mineral salts	Ammonium sulfate, ammonium chloride, and ammonium carbonate Calcium carbonate and calcium chloride Copper sulfate Diammonium hydrogen phosphate

[8] The use of dry versus wet material is dependent on the application of the product. This difference does not pose a safety concern.

[9] US law considers milk, eggs, fin fish, crustacean shellfish, tree nuts, peanuts, wheat, soybeans, and sesame to be major allergens https://www.fda.gov/consumers/consumer-updates/allergic-sesame-food-labels-now-must-list-sesame-allergen

TABLE 2.2 List of Substances Used in Various Formulations of Mushlabs' Growth Media	
Category	**Substances Used**
	Disodium hydrogen phosphate Iron chloride and iron sulfate Magnesium chloride and magnesium sulfate Manganese chloride and manganese sulfate Potassium dihydrogen phosphate Sodium chloride Sodium phosphate monobasic hydrate Sodium selenate[10] Zinc sulfate
Vitamins	Biotin Choline chloride Folic acid[11] Myo-inositol Niacinamide Pantothenic acid Pyridoxal Riboflavin Thiamine Cobalamin Ascorbic acid
pH adjustment agents (Acids and bases)	Sodium hydroxide Potassium hydroxide Sulfuric acid Phosphoric acid Hydrochloric acid Citric acid Acetic acid
Nitrogen sources	Yeast extract Peptone Ammonia Urea
Carbon sources	Sucrose Glycerol Glucose Fructose Lactose Galactose Maltose Xylose Arabinose Starch Cellulose Malt extract Beet molasses

[10] The safety of sodium selenate for use in human food production is addressed in Section 6 of this Notice.

[11] Folic acid is added to the medium solely as required for mycelial growth; it is not added for the purpose of food fortification as is evident from the relatively low levels present in POMM (data provided in Section 6.1.1; selenium levels were not detected in POMM batches).

TABLE 2.2 List of Substances Used in Various Formulations of Mushlabs' Growth Media	
Category	**Substances Used**
	Cane molasses
Solid medium for isolation	Potato dextrose agar Malt extract agar Malt extract peptone agar
Additional reagents	Agar agar
Processing Aids (safe and suitable for food use)	Tween 20 Tween 80 Food grade antifoam agent

2.3 Specifications and Acceptance Criteria[12]

Unless otherwise indicated, Mushlabs has provided data and information in this notice on three non-consecutive batches of POMM labeled "A", "B", and "C". All data presented in this notice are derived from these three batches, which were produced under automated, commercial conditions. Mushlabs has set acceptance criteria used to determine whether a batch of POMM can be released for commercial sale (Table 2.3). Mushlabs also has established specifications that will accompany acceptable batches of POMM that are supplied to the ultimate food producer (Table 2.7).

Demonstration that Mushlabs can make POMM consistently over time is key to ensuring identity and quality. Once acceptance criteria and the safety of the qualifying batches have been established, demonstrating the consistency of batches helps ensure the food safety of all batches meeting those qualifications. The quality parameters used to set acceptance criteria for commercial batches have included data on the levels of proximates in POMM, limitations on levels of certain heavy metals, and limitations on levels of microbes in POMM, all three of which also are important to the demonstration of food safety. As a result, Mushlabs has tested three non-consecutive batches of POMM produced under commercial conditions taking each of these three quality parameters into account to establish appropriate acceptance criteria and specifications for POMM. The same three batches (A, B, and C) are referenced throughout this Notice.

Batch samples were freeze-dried prior to analysis to generate more accurate compositional analyses because high-water content can interfere with measurements. Freeze-drying ensures that nutritional components were conserved, and that sensitive compounds were not damaged during sample preparation. Table 2.3 summarizes the resulting acceptance criteria that must be met prior to release of POMM for commercial sale.

TABLE 2.3 Acceptance Criteria for POMM		
Parameter	**Acceptance Criteria**	**Method of Analysis§**
Composition (g/100 g dry weight)		
Protein	25-55	[1]ASU L 06.00-7 mod.[#]
Fiber	25-70	ASU L 6.00-18, mod.

[12] ICH considers "specifications" to refer to tests and/or references to analytical procedures, as well as appropriate acceptance criteria (*i.e.,* numerical limit ranges), or other criteria for tests described. In this document the terms are used with the assumption that the numerical limits referenced have been set by an accepted and validated test method. https://www.ich.org/page/quality-guidelines.

TABLE 2.3 Acceptance Criteria for POMM		
Parameter	**Acceptance Criteria**	**Method of Analysis§**
Fat	≤ 10	ASU L 6.00-6
Ash	≤ 10	ASU L 06.00-4
Heavy Metals (mg/kg)		
Lead	<0.1	[2]DIN [3]EN 15763 mod.
Metals (ppm)		
Cadmium[4]	<0.1	DIN EN 15763 mod.
Arsenic[4]	<0.1	DIN EN 15763 mod.
Mercury[4]	<0.1	DIN EN 15763 mod.
Microbiology (CFU/g)		
Aerobic plate count	<5000	DIN EN ISO 4833-1
Yeast and molds	<1000	ISO 21527-2
Enterobacteriaceae	<1000	DIN EN ISO 21528-2
E. coli	<10	DIN ISO 16649-2
Salmonella sp.	negative	DIN 10135 – PCR

[§] All methods of analysis are fit for purpose and have been internally validated by Mushlabs.
[1] ASU = Official collection of analysis methods according to § 64 of the German Food and Feed Code;
[2] DIN = Deutsches Institut für Normung
[3] EN = Europaische Norm
[4] These three metals will not be part of the routine specifications for POMM. This decision was made based on results in which all values were below the limit of detection (LOD) for the assays and the requirement that raw materials used in production runs of POMM will be verified to have levels of these three metals below the acceptance criteria values set forth here.
[#]mod. = modification

Batch data also were used to validate consistency under the conditions of production and confirm that safety had not been affected by virtue of changes in the composition of different batches. Proximate analysis is a well-established method for that purpose and relies on the description of the key physical characteristics of a food or food ingredient. Protein, fiber, fat, and ash are quantified in proximate analyses; carbohydrates in mushrooms are associated with fiber and are generally estimated by calculation. Mushlabs notes that variability in mushroom composition can result from the choice of growth substrates, substrate level and growth conditions during traditional mushroom farming (Wu *et al.* 2003; Cueva *et al.* 2017). Thus, POMM acceptance criteria and specifications also account for potential changes that might occur when the relative amounts of media components from Table 2.2, or the growth conditions, are adjusted within a controlled range but would not affect safety (as reported by Bakratsas *et al.* 2021).

Table 2.4 provides the results of three non-consecutive batches of POMM submitted for proximate analyses and adjusted for dry weight; these batches were produced using automated, commercial production methods. Little variability was observed across all compositional endpoints, indicating that the Mushlabs process is reproducible and confirms the ability of the process to consistently result in POMM biomass whose composition is within the ranges described in Table 2.3. For information purposes, Mushlabs has provided in Appendix B additional proximate data for three additional production batches (D, E, and F) produced with different relative amounts of media components (Table 2.2) and growth conditions[13]; these three additional production batches also met the acceptance criteria set by Mushlabs for POMM (Table 2.3). The data from the three non-consecutive batches alone (A, B, and C) confirm

[13] The differences are described in Appendix B.

Mushlabs' ability to produce POMM consistently, while the data in Appendix B provide further support for the reliability and consistency of its production method across the proximate ranges presented in Table 2.3.

TABLE 2.4 POMM Proximate Analyses (g/100g)*				
Batch	**A**	**B**	**C**	**Mean ± SD**
Protein	33.4	32.4	30.9	32.2 ± 1.03
Fiber	63.8	64.9	65.2	64.6 ± 0.6
Fat	0.9	1.1	1.1	1.0 ± 0.1
Ash	2.6	2.7	2.7	2.7 ± 0.05
Carbohydrates	-+	-+	0.1	0.1
*All values are based on dry weight. + -- = not quantifiable				

Table 2.5 provides the results of the heavy metals analyses for POMM batches A, B, and C. All reported values are based on dry weight. Levels of heavy metals including arsenic, cadmium, and mercury were consistently below levels of quantification[14]. Lead levels detected were just above the LOQ for the method employed and also were all below the acceptance criteria of 0.1 mg/kg. The results of these batch analyses demonstrate that the Mushlabs production process can yield a consistent product that conforms to the product quality acceptance criteria in Table 2.3, and that levels of heavy metals do not pose food consumption risks.

TABLE 2.5 POMM Heavy Metals Analyses				
Batch Number	**A**	**B**	**C**	**Method of Analysis**
Arsenic (mg/kg)	<0.04	<0.04	<0.04	DIN EN 15763 mod. LOQ = 0.04*
Cadmium (mg/kg)	<0.01	<0.01	<0.01	DIN EN 15763 mod. LOQ = 0.01*
Lead (mg/kg)	0.037	0.041	0.036	DIN EN 15763 mod. LOQ = 0.015*
Mercury (mg/kg)	<0.01	<0.01	<0.01	DIN EN 15763 mod. LOQ = 0.01*
* The laboratory that performed the analyses has stated that the LOD value is 1/3 of the LOQ value.				

Batches A, B, and C were analyzed for microbial levels as part of stability testing; analyses were conducted on POMM samples from the same three production batches that were then stored and refrigerated for up to 8 weeks after harvest. The results are provided below in Table 2.6. All microbial tests met the acceptance criteria (Table 2.3) at both time points of analysis (after 4 weeks of storage or after 8 weeks of storage). POMM is intended to be used as a food ingredient in food products that would require further processing steps by the ultimate food manufacturers or by Mushlabs (*i.e.,* heating as part of cooking or pasteurization), steps that would further reduce any risk of microbial contamination in final food products (see Figure 2.2).

[14] The analytical laboratories contracted by Mushlabs have stated that in general, the LOD is approximately one-third the LOQ.

TABLE 2.6 POMM Microbial Analyses After Four and Eight Weeks of Refrigerated Storage				
Batch Analyses Time Points	**A**	**B**	**C**	**Method of Analysis**
Total plate count (CFU/g) [Specification = <5000]				
Week 4	<1000	<1000	< 1000	DIN EN ISO 4833-1
Week 8	<1000	<1000	1000	DIN EN ISO 4833-1
Yeast and molds (CFU/g) [Specification = <1000]				
Week 4	<1000	<1000	< 1000	ISO 21527-2
Week 8	<1000	<1000	< 1000	ISO 21527-2
Enterobacteriaceae (CFU/g) [Specification= <1000]				
Week 4	<1000	<1000	< 1000	DIN EN ISO 21528-2
Week 8	<1000	<1000	< 1000	DIN EN ISO 21528-2
E. coli (CFU/g) [Specification = <10]				
Week 4	<10	<10	<10	DIN ISO 16649-2
Week 8	<10	<10	<10	DIN ISO 16649-2
Salmonella sp. (per 25 g) [Specification = negative]				
Week 4	Negative	Negative	Negative	DIN 10135-PCR
Week 8	Negative	Negative	Negative	DIN 10135-PCR

Table 2.7 lists the specifications for POMM that are to be used to describe the product to customers; the specifications selected for various categories were set to be the same as the acceptance criteria for that same parameter listed in Table 2.3. Because POMM is intended as a source of protein to replace protein in food products, protein was selected as the appropriate compositional parameter for purposes of a commercial sale specification. Mushlabs notes that its protein specification range (Table 2.7) reflects its batch data as well as the natural biological variability regarded as safe in widely consumed oyster mushroom fruiting bodies[15] a range that also is described for three additional POMM batches analyzed for proximates levels (D, E, and F), and provided for informational purposes only in Appendix B.

TABLE 2.7 Specifications for POMM		
Parameter	**Specification**	**Method of Analysis§**
Composition (g/100 g dry weight)		
Protein	25-55	[1]ASU L 06.00-7 mod.[#]
Heavy Metals (mg/kg)		
Lead	<0.1	[2]DIN [3]EN 15763 mod.
Cadmium[4]	<0.1	DIN EN 15763 mod.
Arsenic[4]	<0.1	DIN EN 15763 mod.
Mercury[4]	<0.1	DIN EN 15763 mod.
Microbiology (CFU/g)		
Aerobic plate count	<5000	DIN EN ISO 4833-1
Yeast and molds	<1000	ISO 21527-2

[15] References include: Bonatti *et al.* 2004; Goyal *et al.* 2006; Alam *et al.* 2008; Chirinang and Intarapichet 2009; Dunkwal and Jood 2009; Jonathan *et al.* 2011; Gupta *et al.* 2013; Patil 2013; Cogorni *et al.* 2014; Gogavekar *et al.* 2014; Goyal *et al.* 2015; Rana *et al.* 2015; Rashidi and Yang 2016; Hassan 2017; Islam *et al.* 2017; Nwoko *et al.* 2017; Adewoyin and Ayandele 2018; Familoni *et al.* 2018; Finimundy *et al.* 2018; Salehi 2019; Eke-Ejofor and Pollyn 2020; Goswami *et al.* 2020; Raman *et al.* 2021; Akyuz *et al.* 2022; Chilanti *et al.* 2022; Irshad *et al.* 2023.

TABLE 2.7
Specifications for POMM

Parameter	Specification	Method of Analysis§
Enterobacteriaceae	<1000	DIN EN ISO 21528-2
E. coli	<10	DIN ISO 16649-2
Salmonella sp.	negative	DIN 10135 – PCR

[§] All methods of analysis are fit for purpose and have been internally validated by Mushlabs.
[1] ASU = Official collection of analysis methods according to § 64 of the German Food and Feed Code;
[2] DIN = Deutsches Institut für Normung
[3] EN = Europaische Norm
[4] These three metals will not be part of the routine specifications for POMM. This decision was made based on results in which all values were below the limit of detection (LOD) for the assays and the requirement that raw materials used in production runs of POMM will be verified to have levels of these three metals below the LOD values set forth here.
[#]mod. = modification

2.4 Physical or Technical Effect

POMM is intended for use as a food ingredient that provides an alternative protein source in food products already found in the US diet; ultimate food manufacturers will add POMM during processing. Its intended applications are similar to those of textured vegetable protein, rice protein, or mycoprotein. POMM is not intended to be the sole source of protein in the human diet nor is it intended to comprise 100% of any food in any food category eaten daily by US consumers.

Consistent with information found in Section 1 (Table 1.1), it is intended to be incorporated at maximal levels of 25% (dry weight) in the food product, except for milk, where such a level of incorporation (25%) may alter texture.

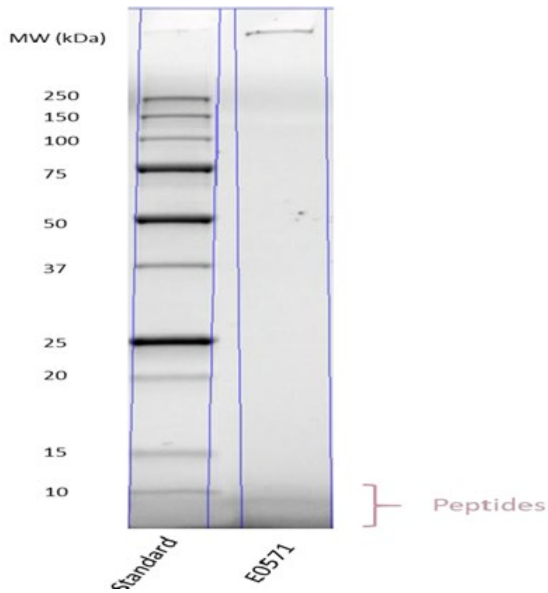
2.4.1 Protein and Amino Acids in POMM

As POMM is intended to provide a source of protein when added as an ingredient to food, studies have been performed to evaluate the digestibility and quality of the protein in POMM. Mushlabs contracted with Improve SAS[16], a firm specializing in protein production and analysis, to evaluate these two aspects of POMM.

Improve SAS simulated gastric digestion of POMM using a sequential enzyme hydrolysis of proteins. The first hydrolysis step used pepsin to mimic the gastric phase of digestion; and the second relied on pancreatin to mimic the intestinal phase. Details of the method used are found in the protocol provided on the Megazyme website.[17] Figure 2.3 indicates that the POMM sample was fully digested (100%), with no discrete bands in the SDS gel. Some residual peptides remained, all of which were below 10 kDa in size. As discussed more fully in Section 6, most allergenic proteins have molecular weights greater than 10 kDa (Huby *et al.* 2000). A discussion of the likelihood of POMM allergenicity is provided in Section 6.1.4.

[16] https://www.improve-innov.com/
[17] https://www.megazyme.com/documents/Assay_Protocol/K-PDCAAS_DATA.pdf

Figure 2.3: SDS PAGE of digested POMM
The left lane contains a run of the molecular weight markers, and the right lane is the POMM digestate.



Improve SAS also evaluated the amino acid content of the resulting POMM digestate, with particular emphasis on essential amino acids. Table 2.8 indicates that POMM is a high-quality protein, with all essential amino acids[18] present at levels that are above the 2002 WHO recommendations.[19] In addition, as it had been shown that POMM was fully digested (Figure 2.3), assays were performed to confirm that POMM is a high-quality protein by determining POMM's *in vitro* Protein Digestibility Adjusted Amino Acid Score (PDCAAS; Schaafsma 2000). The amino acid score (which is generally capped at 1.0 when performing a PDCAAS assessment) for POMM was 1.176, and the *in vitro* PDCAAS was 1.0. For context, sample PDCAAS values of commonly consumed protein sources include 1.00 for casein, 1.00 for egg, 0.92 for beef, and 0.65 for cooked kidney beans (GRN 904). GRN1117[20] provides PDCASS scores for other protein-containing foods (*e.g.,* 1.00 for whey protein isolate, 1.00 for whey protein concentrate, 0.91 for Rhiza mycoprotein, 0.91 for soy protein, 0.89 for pea protein concentrate, 0.67 for pea protein, 0.62 for cooked rice, 0.60 for cooked peas, 0.51 for roasted peanuts, and 0.45 for whole wheat).

Mushlabs has conducted an extensive analysis of its production batches (A, B, and C) to characterize the amino acid profile of POMM. Table 2.8 compares the amino acid composition of POMM with values reported [21] for *P. pulmonarius* fruiting bodies and other common protein replacement products (tofu and

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[18] The essential amino acids rows in Table 2.8 are shaded to emphasize their presence in POMM. All nine essential amino acids were present.
[19] WHO 2002 at https://iris.who.int/bitstream/handle/10665/43411/WHO_TRS_935_eng.pdf?sequence=1&isAllowed=y
[20] See Table 5 in GRN1117 at: www.fda.gov/media/175166/download?attachment
[21] from Chirinang and Intaqrapichet 2009; Gupta *et al.* 2013; Kayode *et al.* 2015; Adebayo and Oloke 2017; Fasoranti *et al.* 2019; Wang *et al.* 2022

rice protein. Inspection of the data in Table 2.8 indicates that the amino acid compositions are quite similar, with glutamic acid, aspartic acid, leucine, and arginine predominating.

TABLE 2.8
Amino Acid Content (g/100 g Dry Weight) of POMM Batches and Comparison with Published Values for *P. pulmonarius*, Tofu, and Rice Protein

Amino Acid	POMM* (Mean ± SD)	*P. pulmonarius* Fruiting Bodies*, 1 (Mean ± SD) [min/max]	Tofu Content*, 2	Rice Protein Based on GRN848*, 3
Alanine	6.9 ± 0.69	6.8 ± 2.21 [2.86/ 11.43]	2.0	5.8
Arginine	5.6 ± 0.96	7.5 ± 2.05 [3.64/ 12.54]	3.2	7.6
Aspartic acid	10.1± 1.17	9.8 ± 1.94 [6.32/ 12.52]	5.3	9.4
Cysteine/Cystine	3.7 ± 0.56	1.3 ± 0.36 [0.74/ 2.09]	0.7	1.2
Glutamic acid	13.7 ± 1.78	17.3 ± 3.59 [8.43/ 21.00]	8.3	20.4
Glycine	5.5 ± 0.73	5.2 ± 1.45 [2.02/ 7.29]	1.9	4.9
Histidine	2.8 ± 0.31	2.7 ± 0.80 [2.13/ 4.76]	1.4	2.5
Isoleucine	4.7 ± 0.21	4.1 ± 0.59 [3.34/ 5.43]	2.4	4.2
Leucine	8.3 ± 1.18	7.1 ± 0.78 [5.78/ 8.13]	3.6	8.3
Lysine	8.9 ± 1.31	5.8 ± 1.05 [3.88/ 6.91]	3.2	3.8
Methionine	0.9 ± 0.09	2.2 ± 0.64 [1.27/ 3.08]	0.6	2.3
Phenylalanine	4.9 ± 0.62	5.5 ± 0.99 [4.28/ 7.88]	2.3	5.2
Proline	4.7 ± 0.46	4.8 ± 1.95 [1.64/ 10.03]	2.6	4.7
Serine	4.8 ± 0.49	4.8 ± 0.75 [3.08/ 6.15]	2.3	5.2
Threonine	5.0 ± 0.59	4.8 ± 1.07 [1.82/ 6.58]	2.0	3.7
Tryptophan	0.8 ± 0.04	1.1 ± 1.08 [--/2.42]	0.7	1.3
Tyrosine	3.4 ± 0.50	3.9 ± 1.26 [2.83/ 6.69]	1.6	3.8
Valine	5.4 ± 0.58	5.7 ± 0.95 [4.88/ 8.56]	2.4	5.9

*All values have been rounded to one decimal for means, and two decimals for standard deviation (SD) around the mean.
[1] Values are extracted from Chirinang and Intaqrapichet 2009; Gupta *et al.* 2013; Kayode *et al.* 2015; Adebayo and Oloke 2017; Fasoranti *et al.* 2019; Wang *et al.* 2022
[2] Values reported in USDA FoodData Central (accessed 10 June 2024 at: https://fdc.nal.usda.gov/fdc-app.html#/food-details/172450/nutrients) reflect means, and did not include measures of variability.
[3] https://www.cfsanappsexternal.fda.gov/scripts/fdcc/?set=GRASNotices&id=848&sort=GRN_No&order=DESC&startrow=1&type=basic&search=848

2.4.2 Dietary Fiber in POMM

Although not intended to be a source of dietary fiber, fiber is present in POMM at levels that may be lower or higher than levels of protein. In the three non-consecutive batches of POMM (A, B, and C) analyzed, mean levels of total fiber were 64.6 g/100 g POMM (S.D. = 0.6). While as shown in Appendix B, under different conditions of production the average level of fiber in POMM batches D, E, and F was 27.70 g/100 g POMM (S.D.=1.36). As described in the peer-reviewed literature[22], *P. pulmonarius* fruiting bodies also

[22] Bonatti *et al.* 2004; Goyal *et al.* 2006; Alam *et al.* 2008; Chirinang and Intarapichet 2009; Dunkwal and Jood 2009; Jonathan *et al.* 2011; Gupta *et al.* 2013; Patil 2013; Cogorni *et al.* 2014; Gogavekar *et al.* 2014; Goyal *et al.* 2015;

contain fiber at relatively high levels. Total fiber levels reported in *P. pulmonarius* fruiting bodies ranged from 2.4 g/100g to as high as 63.6 g/100g (using ASU L 6.00-18, mod.). Thus, the POMM fiber levels detected indicate that total fiber levels in POMM batches within or near the range of total levels of fiber reported in the peer-reviewed literature on the product type that is widely marketed for human consumption, *P. pulmonarius* fruiting bodies.

In addition to analysis of POMM for total levels of fiber, Mushlabs also performed more detailed compositional comparisons of the monosaccharide levels within the fiber polysaccharides in POMM (Table 2.10). The values reported in Table 2.10 reflect analyses of levels in POMM Batches A, B, and C, as well as levels in five additional production batches that met the production acceptance criteria for POMM (a total of eight batches are reflected in the data in table 2.10). Comparisons of the average values for POMM with those reported in the published literature demonstrate that levels detected in POMM are within the range of biological variability for the dikaryotic mycelium present in the fruiting bodies of the species from which POMM has been extracted.

TABLE 2.10
Comparison of Mean Levels of Polysaccharides in POMM and Different Mushroom Structures (g/100 g protein)

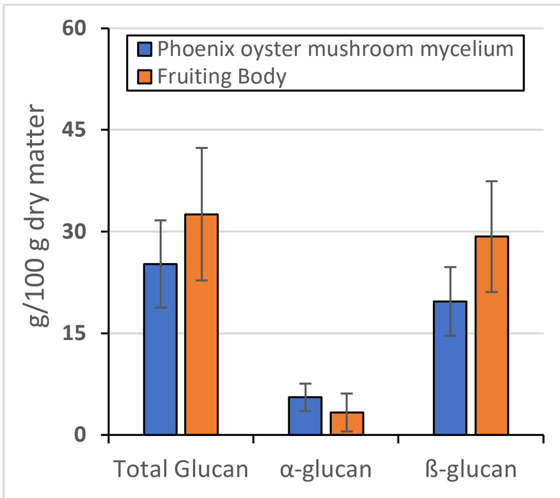
	Biological Source	Xylose	Mannose	Galactose	Glucose	Glucosamine	Uronic acid
Published Values for P. pulmonarius							
Cheung 1996	**Cap**	1.1	5.1	3.5	78.2	7.9	4.17
	Stalk	0.6	3.8	2.0	82.8	6.5	4.34
	Fruiting Body	0.9	4.5	2.7	80.5	7.2	4.255
	Mycelia	1.1	4.2	1.9	84.8	4.73	3.34
Guo *et al.* 2021	**Fruiting body**	2	5	3	90.0	Not measured	Not measured
Smiderle *et al.* 2012	**Mycelia**	2	9.8	3.2	84.1	Not measured	Not measured
POMM							
	Mycelial Biomass	1.1	3.5	1.6	84.9	5.9	3.1

In addition to the analysis of fiber polysaccharides, Table 2.11 and Figure 2.4 shows that levels of the soluble fibers α-glucan and β-glucan detected in POMM Batches A, B, and C are similar to data reported in the published literature for *P. pulmonarius* mycelia (Avni *et al.* 2017; Stastny *et al.* 2022; Sari *et al.* 2016).

Rana *et al.* 2015; Rashidi and Yang 2016; Hassan 2017; Islam *et al.* 2017; Nwoko *et al.* 2017; Adewoyin and Ayandele 2018; Familoni *et al.* 2018; Finimundy *et al.* 2018; Salehi *et al.* 2019; Eke-Ejofor and Pollyn 2020; Goswami *et al.* 2020; Raman *et al.* 2021; Akyuz *et al.* 2022; Chilanti *et al.* 2022; Irshad *et al.* 2023.

TABLE 2.11 Comparison of Soluble Fiber Found in POMM with Published Values [23] for *P. pulmonarius* Mycelia (Mean ± SD in g/100g)			
Source Material	**Total Glucan Content**	**α-Glucan Content**	**β-Glucan Content**
POMM	25.2 ± 6.4	5.5 ± 2.0	19.7 ± 5.1
Literature	32.6 ± 9.8	3.3 ± 2.8	29.3 ± 8.2

FIGURE 2.4: Comparison of Glucan Composition (per 100 g dry weight) in Phoenix Oyster Mushroom Mycelium Batches and Published Literature Values for *P. pulmonarius* Fruiting Bodies (Source Avni *et al.* 2017; Sari *et al.* 2017; Stastny *et al.* 2022)



Mushlabs measured the chitin composition in POMM; chitin is the key structural compound of the hyphal cell wall. Analyses show that POMM samples contain 9.8 ± 0.4 g chitin/100 g. Comparison with values reported in the published literature for mushroom caps and stalks of closely related oyster mushroom species indicates a high degree of similarity with reported values for oyster mushrooms (range between 8 and 11 g chitin/100g (Rahman *et al.* 2023; Ritota and Manzi 2023). The similarity in values between POMM and these related mushroom species confirms that the use of submerged fermentation to produce POMM does not fundamentally affect the nature and structure of the hyphal building block. Similarly, measurement of a characteristic fungal cell wall compound, ergosterol, resulted in a value of 3.25 g /100 g ± 1.66 for POMM samples, values that are again similar to the average value of 3.23 g /100 g ± 1.67 reported in literature (Banlangsawan *et al.* 2014; Huang *et al.* 2017; Wang *et al.* 2022)

2.4.3 Fatty Acids in POMM

Although POMM contains other macronutrients such as fatty acids, the amounts present are so low as to preclude having any substantive value in the characterization of POMM. Yet, the three non-consecutive batches of POMM (A, B, and C) were analyzed to determine the levels of fatty acids that can be found in POMM produced as a result of Mushlabs highly controlled submerged fermentation process. These data show that the fatty composition of POMM is a balance of saturated and unsaturated fatty acids with C18:2 (linoleic acid). C18:1 (oleic acid), and C16:0 (palmitic acid) fatty acids being the predominant components. Mushlabs notes that the fatty acid data for POMM are consistent with what has been reported in the

[23] Avni *et al.* 2017; Stastny *et al.* 2022; Sari *et al.* 2022

scientific literature for fruiting bodies of *P. pulmonarius* (Nair *et al.* 1989; Dimou *et al.* 2002; Goyal *et al.* 2015; Finimundy *et al.* 2018; Diamantis *et al.* 2022; Irshad *et al.* 2023).

2.4.4 Published Literature on Levels of Proximates in *P. pulmonarius*

To provide additional perspective on POMM's compositional analyses, Mushlabs surveyed the literature for data and information on the composition of *P. pulmonarius* mycelia and fruiting bodies, including published accounts of what was previously referred to as *P. sajor-caju*. Table 2.12 provides a summary of the proximates analyses (dry basis) for eight batches of POMM produced using the methods of manufacture described in Figure 2.2, and compares those values with published values for proximates in mycelia and fruiting bodies of *Pleurotus* spp. This comparison shows that the proximate levels measured in POMM are within the expected normal biological variability that is reported in the published literature for *P. pulmonarius* mycelia. Additionally, POMM compositional data are consistent with data found in the literature on *P. pulmonarius* fruiting bodies. Figure 2.5 depicts the data in Table 2.12 for ease of comparison.

Table 2.12
Proximates Data Comparison of POMM and
Published Values for *P. pulmonarius* Mycelia and Fruiting Bodies

POMM Values*	% Moisture [ASU L 0.600-3][a]	% Protein [ASU L 06.00-7 mod.]	% Fat [ASU L 06.00-6]	% Ash [ASU L 06.00-4]	% Carbohydrates [calculated]	% Fiber[b] [ASU L 00.00-18 mod.]	% Total Carbohydrates [calculated]
Mean	4.28	30.66	0.98	2.72	1.30	61.95	61.95
S.D.	0.43	1.97	0.07	0.21	1.59	2.22	2.22
Minimum	3.70	27.90	0.86	2.47	0.10	57.70	57.8
Maximum	4.80	34.00	1.03	3.13	3.10	64.90	68.0
Published Values for Mycelia of *P. pulmonarius*[24]							
Mean	--[c]	25.7	3.48	6.71	41.13[b]	22.29	63.42
S.D.	--	5.16	3.79	3.66	27.36	18.11	10.26
Minimum	--	23.09	1.69	3.91	4.10	2.53	50.55
Maximum	--	32.10	10.20	9.09	68.77	46.45	72.07
Values for *P. pulmonarius* Mycelia (Combined Literature Values and Phoenix Oyster Mushroom Mycelium Values)							
Mean	--	28.19	2.23	4.72	20.56	22.29	62.69
S.D.	--	3.5	1.77	2.82	29.08	28.04	1.04
Minimum	--	23.09	0.80	2.47	0.10	2.53	50.55
Maximum	--	34.00	10.20	9.09	68.77	64.90	72.07
Published Values for Fruiting Bodies of *P. pulmonarius*[25]							

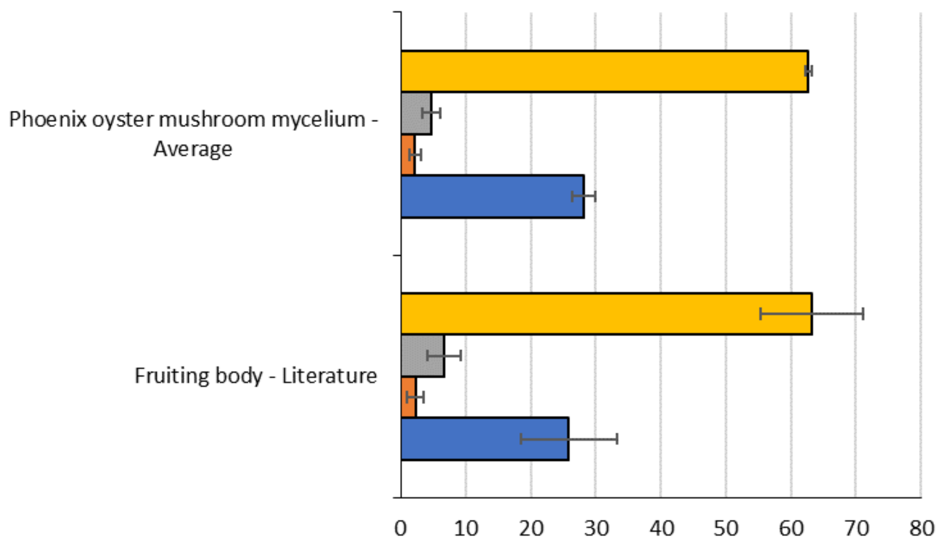
[24] Kausar *et al.* 2006; Confortin *et al.* 2008; Smiderle *et al.* 2012; Ojo *et al.* 2017; Yogachitra 2019

[25] Bonatti *et al.* 2004; Goyal *et al.* 2006; Alam *et al.* 2008; Chirinang and Intarapichet 2009; Jonathan *et al.* 2011; Gupta *et al.* 2013; Patil 2013; Cogorni *et al.* 2014; Gogavekar *et al.* 2014; Goyal *et al.* 2015; Rana *et al.* 2015; Rashidi and Yang 2016; Hassan 2017; Islam *et al.* 2017; Nwoko *et al.* 2017; Adewoyin and Ayandele 2018; Familoni

Table 2.12 Proximates Data Comparison of POMM and Published Values for *P. pulmonarius* Mycelia and Fruiting Bodies							
POMM Values*	**% Moisture [ASU L 0.600-3][a]**	**% Protein [ASU L 06.00-7 mod.]**	**% Fat [ASU L 06.00-6]**	**% Ash [ASU L 06.00-4]**	**% Carbohydrates [calculated]**	**% Fiber[b] [ASU L 00.00-18 mod.]**	**% Total Carbohydrates [calculated]**
Mean	--	25.89	2.28	6.69	35.85[b]	26.73	63.22
S.D.	--	7.45	1.33	2.52	17.60	19.11	7.85
Minimum	--	14.73	0.13	1.06	3.73	2.36	46.86
Maximum	--	42.90	6.62	10.35	65.41	63.61	83.29

* The data provided here are based on analysis of eight production of POMM, batches that included the batches referred to here as A, B, and C, as well as five additional batches produced at commercial scale.
[a] Information in [ ]'s indicates the method of analysis used by Mushlabs.
[b] There was a wide range of levels allocated between fiber and carbohydrates among the different publications, likely due to different assays used by various laboratories and noting that carbohydrates are usually calculated by difference.
[c] NA = not available or not reported

FIGURE 2.5: Proximates Data Comparison Depicting the Combined Mycelia Values (Eight Batches of Phoenix Oyster Mushroom Mycelium and Values from Published Data on *P. pulmonarius* Mycelia) and Literature Values for Fruiting Bodies



2.4.5 Conclusions Regarding POMM's Composition

POMM is intended to be added to food sources already on the market as an alternative source of protein. It is not intended to comprise 100% of the composition of any food. POMM has been well characterized with respect to composition: it contains all the amino acids essential to the human diet. Consistent with

et al. 2018; Finimundy *et al.* 2018; Salehi *et al.* 2019; Eke-Ejofor and Pollyn 2020; Goswami *et al.* 2020; Raman *et al.* 2021; Akyuz *et al.* 2022; Chilanti *et al.* 2022; Irshad *et al.* 2023 carbohydrate content of mushrooms found in the literature, POMM contains relatively high levels of carbohydrates including fiber. The acceptance criteria for release of POMM for incorporation into food are based on the detailed chemical and microbiological analyses of batches of POMM that have been produced using automated, commercial production methods. The data on the three non-consecutive batches of POMM that support this notice (POMM Batches A, B, and C) demonstrate a high degree of consistency among batches. Appendix B provides the data on three additional production batches of POMM that met acceptance criteria; the data are provided for informational purposes and provide additional support for the conclusion that Mushlabs produces POMM in a reliable and consistent manner consistent with its acceptance criteria (Table 2.3) as well as its specifications (Table 2.7).

PART 3. Dietary Exposure

3.1 Intended Use and Characteristics of the Dietary Exposure Assessment

Mushlabs intends to provide POMM to US food manufacturers as a GRAS food ingredient for incorporation into the food categories listed in Table 1.1. Although POMM contains both protein and fiber as its main constituents (Table 2.4), POMM is intended to provide an alternative source of protein.[26] Its uses are intended to be substitutional, *i.e.,* replacing some portion of the protein in any final food product into which it is incorporated. The ultimate food manufacturer, however, will determine the level of protein incorporated into its commercial products, provided the POMM levels do not exceed those listed in Table 1.1.

A food safety assessment includes the estimation of dietary exposures to POMM. This assessment includes a full description of the demographics of the US population likely to consume POMM (individuals over the age of two who eat foods with protein alternatives, *i.e.,* "eaters") and a description of the estimated consumption of POMM at the mean and 90th percentile levels. Mushlabs anticipates that POMM will be incorporated into foods at a maximum level of 25% of the final product (g dry mycelial biomass/100 g food) as indicated in Table 1.1.

Previous GRAS notices (GRN91, GRN904, and GRN945) have estimated daily dietary exposure to a protein source known as mycoprotein sourced from a different fungus (*Fusarium venenatum*). Additionally, a recent GRAS Notice (GRN1117) provides an estimate of daily dietary protein intake sourced from the fungal organism *Neurospora crassa*. Currently available US dietary surveys, such as the National Health and Nutrition Examination Survey (NHANES), however, have not collected data on intakes of mycoprotein or other alternative protein products that have been incorporated into all categories of foods relevant to the intended uses of POMM, *e.g.,* baked goods, ice cream, canned seafood, canned fish.

Mushlabs has estimated consumer dietary exposure based on the most recent relevant results from the 2017-2020 National Health and Nutrition Examination Survey (NHANES; Section 3.2.2). Levels of exposure from previous relevant GRAS Notices and published literature in which intended uses for the food ingredients are similar to the use for POMM, *i.e.,* alternative sources of protein for incorporation into food (Section 3.2.1) were used for comparison purposes only.

3.2 Estimated Dietary Exposure to POMM

3.2.1 Previous Relevant Exposure Estimates for Dietary Protein and Fiber

Previous GRAS notices have provided estimates of dietary intake of alternative sources of proteins derived from fungi (*e.g.,* GRN91; GRN904; GRN945; GRN1117), and there are estimates in the peer-reviewed literature on intake of foods incorporating alternative protein sources or plant-based products (*e.g.,* Fulgoni 2008; Gropper *et al.* 2019; Pikosky *et al.* 2022). For example, the FDA estimated by calculation a daily dietary intake level for mycoprotein of 28.8-55.2 g/person/day (90th percentile intake levels; GRN91). A previous analysis of data collected as part of the National Health and Nutrition Survey (NHANES) in 2011-2014 (Pikosky *et al.* 2022) indicated that daily intake of plant-based protein from all sources was 27.4

[26] As POMM contains fiber, an estimate of dietary fiber exposure resulting from its consumption has been provided in this dietary exposure assessment. Mushlabs does not intend to market POMM as a fiber source or to make any nutrient content claims with respect to fiber.

g/person/day (mean value in adults) with a maximum estimated intake level of 43.7 g/person/day (4th quartile in adults).

POMM is not intended to be used as a replacement source for fiber, and Mushlabs does not intend to make any nutrient content claims for fiber in POMM. Nonetheless, as fiber comprises a relatively large proportion of the composition of POMM, Mushlabs has estimated dietary exposure to fiber from the use of POMM under the conditions of use described earlier in this Notice (Table 1.1.).

One source of public information related to total daily fiber intakes, *"What We Eat in America"*[27], is based on information found in the 2017 to March 2020 pre-pandemic NHANES data. In that report, USDA lists mean total daily intake of fiber (all food sources) to be 16.0 g/person/day for all persons over two years of age with the estimated 90th percentile intake value of 32.0 g/person/day (calculated based on FDA guidance for a "pseudo-90th percentile"[28]). It should be noted that the recommended daily intake of fiber for adults in the US is somewhat higher than these intake values, in the range of 30-40 grams per day.[29]

Table 3.1 summarizes some publicly available information on daily US protein intake from food products containing alternative sources of protein. Since none of the previous dietary exposure assessments for alternative protein sources replicate the exact intended uses of POMM, the information in Table 3.1 was considered relevant only for purposes of providing context to a POMM-specific dietary exposure assessment.

TABLE 3.1
Estimates of Daily of Intake of Alternative Sources of Protein from Foods

Source of Data	Protein Intake (g/person/day)
GRN91 (FDA calculation)	28.8-55.2 (90th percentile intake; mycoprotein)
GRN904 (NHANES 2011-2012)	24.4 (mean intake; mycoprotein) 48.8 (pseudo 90th percentile intake)
GRN945 (NHANES 2003-2016)	72.06-73.7 (mean intake; mycoprotein)
GRN1117 (NHANES 2015-2018)	18.5 (mean intake) 40.2 (90th percentile intake)
Pikosky *et al.* 2022 (NHANES 2011-2014)	27.4 (mean intake; plant-based) 43.7 (4th quartile; plant-based)

[27] https://www.ars.usda.gov/ARSUserFiles/80400530/pdf/1720/Table_1_NIN_GEN_1720.pdf accessed 6 September 2024.
[28] https://www.fda.gov/regulatory-information/search-fda-guidance-documents/guidance-industry-estimating-dietary-intake-substances-food
[29] Found in the report by the Institute of Medicine in 2005 entitled *"Dietary Reference Intakes for Energy, Carbohydrate, Fiber, Fat, Fatty Acids, Cholesterol, Protein, and Amino Acids"* found at https://nap.nationalacademies.org/download/10490#

3.2.2. Estimated Dietary Exposure to POMM Based on NHANES Data

3.2.2.1 Data Extraction from NHANES

For purposes of estimating exposure to POMM under the conditions of use described in Table 1.1., Mushlabs extracted data from publicly available files related to the 2017-2020 NHANES survey conducted by the Centers for Disease Control and Prevention (CDC). The data sets accessed included the Dietary Interviews - Individual Foods, First Day and Second Day, the Dietary Interview Technical Support File - Food Codes, and the accompanying Demographic Variables and Sample Weights file. The original sample from this dataset consisted of 15,560 unique respondents (eaters only) across that three-year period, and included the following factors:

SEQN (unique respondent Identifier),
RIDAGEYR (age of respondent at the time of interview),
DRXFDCD (USDA numeric food code),
DRXFCSD (USDA food code short description),
DR1IGRMS & DR2IGRMS (grams per food item days 1 and 2)
DR1IPROT & DR2IPROT (grams of protein per food item on days 1 and 2)

The source codes used for foods employed in the dietary exposure assessment are provided in Appendix C.

From these datasets, Mushlabs first created a data subset to provide the exposure estimates for eaters only over two years of age (mean intake in g/day and 90th percentile intake in g/day) for the food categories previously indicated in Table 1.1. Table 3.2 presents these estimates per food category for eaters of foods in the selected categories for all US residents over the age of two covered by the survey.

To generate the information for Table 3.2, food codes listed in Appendix C were aligned with descriptors of the various food categories relevant to use of POMM as an alternative protein source. Data were extracted from NHANES on the basis of both grams of food intake per day and grams of protein intake per day for a particular food category, again identifying both mean intake levels as well as 90th percentile intake levels.

These data were used as a basis for further dietary exposure assessment work specific to POMM and its incorporation as a food ingredient (Table 3.3). It is important to note that the food categories of “baked goods (all)”, “ice cream”, “canned fish”, and “canned seafood” in the NHANES dataset do not provide data on products produced with alternative sources of protein.

TABLE 3.2 Data Extracted from NHANES 2017-2020 on Daily Dietary Exposure to Various Food Categories (Grams of Food and Protein Per Category) For Eaters Only (> 2 Years of Age to Adults)					
Food Category Listed in NHANES Data Files	**Number of Respondents (N=17598)**	**Mean Food Intake (g/day)**	**90th Percentile Food Intake (g/day)**	**Mean Protein Intake (g/day)**	**90th Percentile Protein Intake (g/day)**
Meat Substitutes	74	102.47	204.0	15.71	34.57
Baked Goods (All)	14553	79.54	181.16	5.76	11.54
Milk Non-Dairy	860	253.7	488.0	3.2	7.9
Yogurt Non-Dairy	8	131.1	178.7	2.0	6.3
Ice Cream (All Types)	2535	133.3	255	5.1	9.6
Canned Fish (All)	150	82.0	136.6	16.1	26.6
Canned Seafood (All)	1	113.4	113.4	23.2	23.2
Other (All)#	537	209.2	538.7	26.0	55.7
Imitation Coffee Creamer	52	43.6	117.0	0.4	1.2
Imitation Cheese	18	29.5	56.6	7.4	14.2
Imitation Crab	14	75.1	113.1	5.1	7.4
Misc. Protein Added Items	459	239.0	553.5	30.3	57.9
# Other (All) is a category that contains the combined NHANES entries for three products relevant to POMM intended uses (imitation cheese, imitation crab, and imitation coffee creamer) as well as one product not intended for POMM incorporation at this time (imitation sour cream), as well as a variety of "alternative protein types" such as textured vegetable protein. Thus, the "Other (All)" category estimates of intake are gross overestimates of alternative protein intake that would be representative of POMM exposure for its intended uses.					

To estimate daily dietary intake levels for POMM, values from Table 3.2 were used and corrected for the maximum level of incorporation provided for different food products, consistent with data on maximum levels of POMM incorporation into food that are found in Table 1.1. Because the NHANES data did not include categories for products containing alternative sources of protein for all of the categories relevant to the POMM dietary exposure assessment, the intake values for 'ICE CREAM', 'CANNED FISH', and 'CANNED SEAFOOD' in Table 3.2 were adjusted to take into account that alternative protein source food products do not hold large shares of the consumer market at this time. The adjustments for these three categories were therefore made based on market share data that was publicly available, as described below.

For the food categories where Mushlabs estimated POMM intake by consulting other data sources (Table 3.3), Mushlabs made conservative assumptions. In the case of 'ICE CREAM', non-dairy ice cream intake was assumed to be 30% of the dairy-based ice cream intake by employing statistics gathered by McKinsey in 2022 (Adams *et al.* 2022[30]). For fish and seafood analogs, which are also not given their own code in

[30] https://www.mckinsey.com/industries/consumer-packaged-goods/our-insights/how-to-stay-cool-as-competition-heats-up-in-ice-cream-and-yogurt last accessed 6 September 2024

NHANES, Mushlabs used data from The Good Food Institute[31] to adjust intakes where it has been estimated that the entire plant-based (alternative protein) meat and seafood market is 1.3% of total meat and seafood sales in the US. Mushlabs conservatively assumed 1% of the 'CANNED FISH' as well as 1% of the 'CANNED SEAFOOD' intake values in NHANES would represent the intake of analog fish products and analog seafood products. Even though the NHANES dataset did not include entries for 'BAKED GOODS' made with alternative sources of protein, no similar market information was available to allow for adjustment of the overall 'BAKED GOODS' daily intake value extracted from NHANES.

Mushlabs believes that adjustments of data from the NHANES dataset are an appropriate and data-driven method that allows for conservative and health protective POMM dietary exposure estimates. Mushlabs further believes that 100% market penetration of POMM is highly unlikely as market penetration is not anticipated to reach 100% of any one food category. In Table 3.3., Mushlabs therefore provided estimates of the mean and 90th percentile intakes of POMM assuming 20% market penetration, which is still anticipated to be a highly conservative overestimate.[32]

Estimates of POMM intake in Table 3.3 are provided based on both 100% market share (full substitution of all food intake with products with POMM incorporated at the maximum levels indicated in Table 1.1) and the more realistic market share of 20% (20% of marketed products would contain POMM as indicated in Table 1.1).

TABLE 3.3
Daily US Intakes (g/person/day) of POMM*

Food Category	NHANES US Mean Intake	NHANES 90th Percentile Intake	Fractional Composition of POMM in Food	100% Market Penetration		20% Market Penetration	
				Mean Daily POMM Intake	90th Percentile POMM Intake	Mean Daily POMM Intake	90th Percentile POMM Intake
Food Categories in NHANES with Data on Products Made with Alternative Sources of Protein							
Meat Analogs	102.5	204.0	25%	25.6	51.0	5.1	10.2
Milk analogs (Non-Dairy)	253.7	488.0	10%	25.4	48.8	5.1	9.8
Yogurt analogs (Non-Dairy)	131.1	178.7	15%	19.7	26.8	3.9	5.4
Other – All[a]	209.2	538.7	25%	52.3	134.7	10.5	26.9
Food categories in NHANES without Data on Products Made with Alternative Sources of Protein							
Baked Goods (All Categories)	79.5	161.2	25%	22.3	47.9	4.5	9.6
Ice Cream[b]	133.3 [40][b]	255.0 [76.5][b]	15%	6.0[b]	11.5[b]	1.2[b]	2.3[b]
Canned Fish[c]	92.1 [0.82][c]	166.6 [1.37][c]	25%	0.2[c]	0.34[c]	0.04[c]	0.07[c]
Canned Seafood[d]	113.4 [1.13][d]	113.4 [1.13][d]	25%	0.28[d]	0.28[d]	0.06[d]	0.06[d]

[31] gfi.org/wp-content/uploads/2023/09/COR23030_Plant-based-meat-and-seafood-one-pager.pdf
[32] See for example: gfi.org/wp-content/uploads/2023/09/COR23030_Plant-based-meat-and-seafood-one-pager.pdf

TABLE 3.3 Daily US Intakes (g/person/day) of POMM*							
Food Category	NHANES US Mean Intake	NHANES 90^{th} Percentile Intake	Fractional Composition of POMM in Food	100% Market Penetration		20% Market Penetration	
				Mean Daily POMM Intake	90^{th} Percentile POMM Intake	Mean Daily POMM Intake	90^{th} Percentile POMM Intake
Cumulative POMM INTAKE	--	--	--	**151.8**	**321.3**	**30.4**	**64.3**

* The intakes are for eaters only over 2 years of age for different foods into which POMM may be incorporated as a food ingredient.

[a] Encompasses the following entries found in NHANES 2017-2020 for intended uses of POMM including imitation cheese (representative of processed cheese), imitation crab (representative of seafood analogs), imitation sour cream, imitation coffee creamer (soy; liquid; representative of cream analogs), and protein[33]

b The number in brackets was used with the fractional composition value calculation since the value reported in NHANES is not representative of non-dairy ice cream intake. We will assume that the non-dairy intake is 30% of the dairy intake based on statistics reported by McKinsey in 2022 (Adams *et al.* 2022).

[C] In NHANES, 'CANNED FISH-ALL' is a category that combines intake data for the following specific types of canned fish products: canned anchovies, canned mackerel, canned salmon, canned sardines, and canned tuna. The number in brackets was used with the fractional composition value calculation since the NHANES reported value for canned fish are not plant-based alternative products. The Good Food Institute (gfi.org/wp-content/uploads/2023/09/COR23030_Plant-based-meat-and-seafood-one-pager.pdf) has stated that plant-based meat and seafood market is 1.3% of total meat and seafood sales in the US; we have conservatively assumed 1% of the canned tuna intake number would apply here.

[d] The number represents the estimate of market share of 1% for canned tuna analogs made with alternative protein as an ingredient.

Because the major dietary components in POMM are protein and fiber, Table 3.4 below is provided to present estimates of daily intakes of protein and fiber extracted from NHANES 2017-2020 dataset. Both the mean and 90^{th} percentile values for each food category relevant to POMM exposure, as well as the cumulative levels of intake, are provided. Table 3.4 also provides comparisons of the fiber and protein intakes for relevant food categories with estimated POMM daily intake (g/day from Table 3.3) as well as dietary reference values for protein and fiber.

The Food and Nutrition Board of the National Academies of Sciences Engineering, and Medicine has set acceptable daily intake levels for protein and dietary fiber.[34] The daily protein value is a Recommended Daily Allowance (RDA) while the fiber value is an Adequate Intake (AI) value. The Food and Nutrition Board defines the term RDA as the average daily level of intake sufficient to meet the nutrient requirements of nearly all (97–98%) healthy individuals. The term AI is defined as an intake level that is assumed to ensure nutritional adequacy; established when evidence is insufficient to develop an RDA.[35] Mushlabs believes these levels are appropriate comparators in the POMM dietary exposure assessment. In the last row of Table 3.4, the cumulative exposure values for POMM intake listed in brackets, "[ ]", were calculated based on the proximates data provided in Table 2.4 showing that each 100 g portion of POMM contains about one third protein and two thirds fiber.

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[34] https://nap.nationalacademies.org/catalog/10490/dietary-reference-intakes-for-energy-carbohydrate-fiber-fat-fatty-acids-cholesterol-protein-and-amino-acids last accessed 6 September 2024

[35] https://ods.od.nih.gov/HealthInformation/nutrientrecommendations.aspx last accessed 6 September 2024

TABLE 3.4 Cumulative Daily Intake (g/day) for Protein and Fiber in Foods (NHANES 2017-2020) into Which POMM Could Be Incorporated as Compared to US Dietary Reference Values (g/day)								
Food Category	**Mean Protein Intake (g/day)**	**90th Percentile Protein Intake (g/day)**	**Mean Fiber Intake (g/day)**	**90th Percentile Fiber Intake (g/day)**	**Mean POMM Intake Assuming 20% Market Penetration (g/day)**	**90th Percentile POMM Intake Assuming 20% Market Penetration (g/day)**	**US Daily Protein RDA[a] (g/day)**	**US Daily Fiber AI[a] (g/day)**
Food Categories in NHANES with Data on Products Made with Alternative Sources of Protein								
Meat Analogs	16.1	35.7	3.4	7.5	5.1	10.2	Not Applicable	Not Applicable
Milk analogs (Non-Dairy)	3.0	6.7	0.54	1.2	5.1	9.8	Not Applicable	Not Applicable
Yogurt analogs (Non-Dairy)	2.9	7.3	0.16	0.42	3.9	5.4	Not Applicable	Not Applicable
Other – All[b] (imitation cheese, imitation coffee creamer, imitation crab, imitation sour cream, proteins)	11.0	20.3	1.93	4.9	10.5	26.9	Not Applicable	Not Applicable
Food Categories in NHANES without Data on Products Made with Alternative Sources of Protein								
Baked Goods (All Categories)	5.8	11.5	2.2	4.4	4.5	9.6	Not Applicable	Not Applicable
Ice Cream[c]	5.1 [0.51][c]	9.6 [0.96][c]	1.2 [0.4][c]	2.4 [0.7][c]	1.2[b]	2.3[b]	Not Applicable	Not Applicable
Canned Fish[d]	19.2 [0.19][d]	31.5 [0.32][d]	0	0	0.04[c]	0.07[c]	Not Applicable	Not Applicable
Canned Seafood[e]	23.16 [0.23][e]	23.16 [0.23][e]	0	0	0.06[d]	0.06[d]	Not Applicable	Not Applicable
Cumulative INTAKE	**39.7**	**83.01**	**8.63**	**19.1**	**30.4 [9.12 g protein] [20.4 g fiber]**	**64.3 [19.3 g protein] [43.1 g fiber]**	**34.7**	**27.0**

[a] https://ods.od.nih.gov/HealthInformation/nutrientrecommendations.aspx

[b] This category encompasses the following entries found in NHANES 2017-2020 that relate to intended uses of POMM and includes: imitation cheese (representative of processed cheese), imitation crab (representative of seafood analogs), imitation sour cream, imitation coffee creamer (soy; liquid; representative of cream analogs), and protein[36]

[36] These inclusion terms were identified during the process of data extraction of the NHANES database. They were chosen either through exclusion terms from the other categories (such was the case for IMITATION CHEESE not being part of the "Meat Substitute" category), and a snowball sample of terms based on skimming the entries and collecting

TABLE 3.4 Cumulative Daily Intake (g/day) for Protein and Fiber in Foods (NHANES 2017-2020) into Which POMM Could Be Incorporated as Compared to US Dietary Reference Values (g/day)								
Food Category	**Mean Protein Intake (g/day)**	**90th Percentile Protein Intake (g/day)**	**Mean Fiber Intake (g/day)**	**90th Percentile Fiber Intake (g/day)**	**Mean POMM Intake Assuming 20% Market Penetration (g/day)**	**90th Percentile POMM Intake Assuming 20% Market Penetration (g/day)**	**US Daily Protein RDA[a] (g/day)**	**US Daily Fiber AI[a] (g/day)**

[c] The numbers in brackets were used since the value reported in NHANES is not representative of non-dairy ice cream intake. We will assume that the non-dairy intake is 30% of the dairy ice cream intake based on statistics reported by McKinsey in 2022 (Adams *et al.* 2022[37]).
[d] In NHANES, 'CANNED FISH-ALL' is a category that combines intake data for the following specific types of canned fish products: canned anchovies, canned mackerel, canned salmon, canned sardines, and canned tuna. The number in brackets was used with the fractional composition value calculation since the NHANES reported value for canned fish are not plant-based alternative products. The Good Food Institute (gfi.org/wp-content/uploads/2023/09/COR23030_Plant-based-meat-and-seafood-one-pager.pdf) has stated that plant-based meat and seafood market is 1.3% of total meat and seafood sales in the US; we have conservatively assumed 1% of the canned fish intake number would apply here.
[e] The number represents the estimate of market share of 1% for canned seafood analogs made with alternative protein as an ingredient.

3.3 Conclusions Regarding Estimates of Dietary Exposure to POMM

Table 3.5 summarizes the dietary exposure assessment findings regarding POMM when used as intended as an alternative source of protein for food products.

Comparison of the POMM intake values with intakes calculated for mycoprotein (GRN945; mean intake of 24.4 g/day and 90th percentile intake of 48.8 g/day) shows that the estimates of US dietary exposure to POMM assuming a market penetration of 20% (30.4 g/day mean intake and 64.3 g/day 90th percentile) are somewhat similar.

A realistic dietary exposure to POMM would be expected to be lower than the 20% market penetration estimates. The Good Food Institute published a plant-based market overview[38] in 2023 indicating that overall market penetration (share) of plant-based food products was 1% of the total value of all sales in the total food and beverage market in the US. As a result, the assumption that one food ingredient, POMM, when used as an alternative source of protein in foods, would capture 20% of the market for the food categories considered in this exposure assessment is highly unlikely.

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a repository of terms. The term "Protein" encompassed several search terms of interest including products like "Textured Vegetable Protein" (often added as a meat replacement), various granola and protein snack bars, and powdered protein mixes used as supplements to smoothies and other beverages. Including "Protein" in this category results in some double counting because things like textured vegetable protein are also used in meat analogs, for example.

[37] https://www.mckinsey.com/industries/consumer-packaged-goods/our-insights/how-to-stay-cool-as-competition-heats-up-in-ice-cream-and-yogurt

[38] gfi.org/wp-content/uploads/2023/09/COR23030_Plant-based-meat-and-seafood-one-pager.pdf

TABLE 3.5 Summary of US Dietary Exposure to POMM Based on NHANES 2027-2020 Data		
Intake Category	**POMM Intake Assuming 100% Market Penetration (g/day)**	**POMM Intake Assuming 20% Market Penetration (g/day)**
Cumulative POMM Exposure (All Relevant Product Categories)		
Mean Intake	151.8	30.4
90th Percentile Intake	321.3	64.3
Cumulative Protein Exposure (All Relevant Product Categories)		
Mean Intake	39.7	9.12
90th Percentile Intake	83.01	19.3

PART 4. Self-Limiting Levels of Use

In general, no self-limiting levels of use for the product have been identified. One possible exception is the incorporation of mycelium product into fluid milk, as amounts in excess of 15% may affect its texture. Mushlabs limits incorporation of POMM as a food ingredient in various food products to no more than 25% (w/w on a dry basis). For those products in which substitutional use of POMM could result in increased daily fiber intake for any individual, onset of gastrointestinal (GI) symptoms would serve as an individual's self-limiting use.

PART 5. Experience Based on Common Use in Food Before 1958

Although there is a history of common use of the fruiting body in food prior to 1958, there is no formal documentation of *P. pulmonarius* mycelia produced via submerged fermentation being used as food before 1958. Nonetheless, Mushlabs also notes that the process of submerged fermentation of mushroom mycelium (including oyster mushrooms) and the use of the potential for use of the process to produce food has been publicly available since 1953 (Block 1953; Sugihara and Humfeld 1954).

PART 6. Safety Narrative

This safety assessment is premised on the observation that Mushlabs' dikaryotic POMM is derived directly from the dikaryotic fruiting body of *P. pulmonarius* PXM1906. It therefore carries the exact same genome as a fruiting body of the species, which has been a common constituent of US and world-wide diets. Production of *Pleurotus spp*. has been reported to account for about 25% of mushrooms produced (Raman *et al.* 2021). They are the second most produced and consumed mushrooms worldwide after button mushrooms (Rajarathnam *et al.* 1987; Chang 1999; Cohen *et al.* 2002; Sánchez 2010; Lee *et al.* 2021). In the United States, *P. pulmonarius* fruiting bodies are commonly marketed as "oyster mushrooms" without any indication of species. In 2020-2021, more than 72 million pounds of oyster mushrooms (*Pleurotus* spp.) were produced in the US.[39] In addition to their uses in food, many traditional medicines employ mushrooms for therapeutic uses, purportedly because of their antioxidant properties[40] (Jose 2002).

In order to demonstrate general recognition of safety by scientific procedures, Mushlabs has divided the safety assessment into two parts.

Traditional Safety Assessment. Section 6.1 focuses on more "traditional" (as opposed to molecular) approaches to demonstrate the safety of POMM. It provides discussions of data and information on POMM that relate to compositional analyses not previously covered in Section 2.4 (*i.e.,* proximates; amino acids and proteins; carbohydrates including fiber; and fatty acids). It also addresses some general food safety issues related to anti-nutrients in food and their relevance to POMM, taking into account that POMM will be used as an ingredient in food that undergoes cooking or other heat treatment (*e.g.,* pasteurization). It also summarizes the literature on the allergenic potential of *P. pulmonarius* with emphasis on immune-mediated allergic responses after oral consumption. It further discusses the body of literature that addresses toxic responses associated with oral exposure to *P. pulmonarius*, including the potential for the presence of toxins in POMM. Additional POMM data is presented in Section 6.2, data that directly addresses toxicity and allergenicity of POMM based on genomic and proteomic analyses of POMM. Section 6.1.2 also presents measurements of residual nucleic acids in POMM.

Molecular Confirmation of the Lack of Presence of Hazards That May Pose Food Consumption Risks. The primary goal of Section 6.2 is to provide Mushlabs' extensive searches for potential hazards that may be found in the mycelia of *P. pulmonarius* (and thus POMM). It begins with basic information on the life cycle of *P. pulmonarius* to provide some context on the similarities between the fruiting body and mycelium, with particular emphasis on morphology and karyology, and the expected small differences in gene expression between the fruiting body and mycelia characterized by Joh *et al.* 2007. Then, in order to demonstrate that potential toxins or allergens that may have been reported as being present in the fruiting body (an edible mushroom) are not present in POMM, Mushlabs has fully characterized the genome of POMM, completed a transcriptomic analysis, and concluded with a proteomic analysis. The results of these molecular studies demonstrate that no proteins present in POMM could pose a food consumption risk.

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[39] https://www.nass.usda.gov/Publications/Todays_Reports/reports/mush0821.pdf last accessed 6 September 2024

[40] https://www.mskcc.org/cancer-care/integrative-medicine/herbs/oyster-mushroom

6.1 Traditional Components of a Safety Assessment for POMM As a Food Ingredient

In Section 2.4 of this GRAS Notice, Mushlabs provided results of its analysis of the major constituents of POMM, including a proximates analysis on three non-consecutive batches of POMM (Batches A, B, and C). Section 2.4 also presented information on protein digestibility. The results of these assays showed that the food ingredient:

- Has a macronutrient profile that is consistent with the macronutrient profile expected to be present in a food ingredient produced from dikaryotic mycelia isolated from the fruiting bodies of *P. pulmonarius*
- Consists principally of protein and fiber
- Is highly digestible with no proteins > 10 kDa after *in vitro* digestion
- Comprises all the essential amino acids in adequate amounts

Mushlabs also provided data from three non-consecutive batches of POMM demonstrating that Mushlabs' method of production was not contaminated with microbes, that POMM was non-viable after freezing, and that POMM had no levels of heavy metals that would pose a food safety concern. These traditional food safety consideration endpoints contribute to the overall finding by Mushlabs that POMM is safe to use as intended as a food ingredient providing an alternative protein source for human consumption.

6.1.1 Vitamins and Minerals in POMM

As described in Section 2.2, the media used in the submerged fermentation of POMM contains vitamins and minerals necessary for the mycelia to grow and are therefore incorporated into the biomass that comprises POMM. In fact, like all living organisms, mushrooms incorporate constituents from their environments as they grow. These include minerals, vitamins, and heavy metals; several reports have demonstrated that the content of minerals, vitamins and heavy metals detected in mushroom fruiting bodies is directly related to the concentration of those constituents in the substrate upon which they are grown (Bressa *et al.* 1988; Favero *et al.* 1990a, 1990b; Sales-Campos *et al.* 2011). As a result, variation in mineral and vitamin profiles is expected in the fruiting bodies of oyster mushrooms depending on where they were grown. All of these are considered safe-to-eat. Mushlabs has demonstrated through its analyses of three non-consecutive batches of POMM (Batches A, B, and C), that POMM does not contain levels of heavy metals that would pose a food safety concern (Table 2.5).

Experts in nutrition have set levels needed in the diet of humans to maintain health, referred to collectively as dietary reference values.[41] One commonly referred to value is known as a "Recommended Dietary Allowance" or RDA and is the average daily dietary intake level that is sufficient to meet the nutrient requirement of nearly all (97 to 98 percent) healthy individuals in a particular gender and life stage group (life stage considers age and, when applicable, pregnancy or lactation). Because certain vitamins and minerals can have toxic effects when levels of exposure greatly exceed recommended daily levels of intake, particularly if exceeded on a repeated basis, scientists have established "upper limits" on dietary reference values for some vitamins and minerals, referred to a "tolerable upper intake level" or

[41] https://www.ncbi.nlm.nih.gov/books/NBK45182/#:~:text=The%20Recommended%20Dietary%20Allowance%20(RDA,applicable%2C%20pregnancy%20or%20lactation) last accessed 6 September 2024

"UL".[42] These values represent the highest level of daily nutrient intake that is likely to pose no significant risk of adverse health effects to almost all individuals in the general population.

Levels of vitamins and minerals that were detected at measurable levels in POMM are provided in Table 6.1. All methods used in analyses were fit for purpose and had been validated. These data show a high degree of consistency across Mushlabs' batches in the detected levels of vitamins or minerals. Moreover, the levels detected are consistent with levels reported by the OECD[43] in 2013 for oyster mushrooms.

Other vitamins and minerals that were assayed for but were not detected at levels above the limit of quantification (LOQ[44]) for the methods used, and as such, are not listed in Table 6.1. These vitamins and minerals and the corresponding LOQ and LOD values included vitamin A (LOQ=0.5 mg/kg; LOD=0.167 mg/kg), pyridoxal (LOQ=0.01 mg/kg; LOD=0.003 mg/kg), pyridoxine (LOQ=0.01 mg/kg; LOD=0.003 mg/kg), vitamin D2 (LOQ=0.15 mg/kg; LOD=0.05 mg/kg), β-tocopherol (LOQ=0.01 mg/kg; LOD=0.003 mg/kg), Δ-tocopherol (LOQ=0.086 mg/kg; LOD=0.029 mg/kg), γ-tocopherol (LOQ=0.066 mg/kg; LOD=0.022 mg/kg), α-carotene (LOQ=0.01 mg/kg; LOD=0.003 mg/kg), β-carotene (LOQ=0.01 mg/kg; LOD=0.003 mg/kg), vitamin K1 (LOQ=0.1 mg/kg; LOD=0.033 mg/kg), selenium (LOQ=0.2 mg/kg; LOD=0.07 mg/kg), and cobalt (LOQ=0.05 mg/kg; LOD=0.017 mg/kg). The absence of cobalt is notable as it is not allowed in human food in the US (21 CFR 189.120).

TABLE 6.1
Vitamin and Mineral Levels Detected in POMM

Parameter[a]	Batch A	Batch B	Batch C	Mean Value	Method
Vitamins					
Vitamin B1 (mg/kg)	0.42	0.45	0.46	0.44	DIN EN 14122, HPLC/Fl
Vitamin B2 (mg/kg)	0.89	0.93	0.89	0.76	DIN EN 14152, HPLC/Fl
Pyridoxamine (mg/kg)	0.20	0.22	0.18	0.20	DIN EN 14663, HPLC/Fl
Vitamin B6 (total) (mg/kg)	0.20	0.22	0.18	0.20	DIN EN 14663, HPLC/Fl
Niacin (mg/kg)	8.82	9.56	9.07	9.15	AOAC 944.13
Pantothenic acid (mg/kg)	1.80	1.67	1.69	1.72	AOAC 945.74
Biotin (µg/kg)	4.12	4.56	4.82	4.50	SOP M 3552, LC-MS-MS
Folic acid (µg/kg)	143	133	141	139	SOP M 2885, LCMS-MS
Vitamin B12 (µg/kg)	0.83	0.91	1.13	0.96	AOAC 952.20/986.23
Vitamin C (mg/kg)	1.43	2.84	1.52	1.93	SOP M2885, LC/MS-MS
Vitamin D3(µg/kg)	18.7	1.48	<0.05[b]	6.74	SOP M2885, LC/MS-MS
Vitamin E (mg/kg)	0.27	0.34	0.29	0.30	DIN EN 12822, HPLC/Fl
α-Tocopherol (mg/kg)	0.04	0.11	0.08	0.08	DIN EN 12822, HPLC/Fl
Minerals					
Sodium (mg/kg)	2,310	2,140	2,370	2273	DIN EN 15621, mod.
Calcium (mg/kg)	3,520	3,930	3,370	3607	DIN EN 15621, mod.

42 https://www.ncbi.nlm.nih.gov/books/NBK45182/#:~:text=The%20Recommended%20Dietary%20Allowance%20(RDA,applicable%2C%20pregnancy%20or%20lactation)

43 OECD ENV/JM/MONO(2013)23 "Consensus document on compositional considerations for new varieties of oyster mushroom (*Pleurotus ostreatus*): key food and feed nutrients, antinutrients and toxicants" from November 2013. https://one.oecd.org/document/env/jm/mono(2013)23/en/pdf

44 The testing laboratories used by Mushlabs have stated that in general, the LOD is approximately one third the LOQ.

TABLE 6.1 Vitamin and Mineral Levels Detected in POMM					
Parameter[a]	Batch A	Batch B	Batch C	Mean Value	Method
Potassium (mg/kg)	1,460	1,470	1,540	1490	DIN EN 15621, mod.
Phosphorus(mg/kg)	4,910	5,000	5,270	5060	DIN EN 15621, mod.
Magnesium (mg/kg)	705	727	784	739	DIN EN 15621, mod.
Iron (mg/kg)	89.2	83.6	75.2	82.7	DIN EN 15621, mod.
Copper (mg/kg)	9.87	10.7	10.9	10.5	DIN EN 15763, mod.
Zinc (mg/kg)	30.1	32.4	32.2	31.6	DIN EN 15621, mod.
Manganese (mg/kg)	6.57	6.87	6.96	6.80	DIN EN 15763, mod[45]
[a] Values listed for each vitamin or mineral are based on dry weight. [b] Values were below the limit of quantification.					

Because some vitamins and minerals have UL values set, *i.e.,* maximum intake levels expected to be free of any adverse health effects, Mushlabs estimated margins of exposure (MOEs) between the ULs for those substances and the levels detected in POMM batches (Table 6.2). MOEs are commonly used in food safety assessment to determine whether a level of exposure would raise a food safety concern compared to some already established physiological or regulatory level. MOEs were calculated only for those vitamins and minerals with established ULs that were detected at quantifiable levels in the POMM batches (Batches A, B, and C). For purposes of calculation of vitamin and mineral MOEs relative to the UL values, Mushlabs used the 90th percentile daily exposure values from the POMM dietary exposure assessment of 64.3 g POMM/person/day. As previously noted in Part 3 (Dietary Exposure Assessment), that value was calculated assuming a 20% market penetration of products containing POMM (cumulative); also provided in Part 3 was the justification for why it is neither realistic nor reasonable to expect 100% market penetration. Inspection of the data in Table 6.2 indicates that the MOEs for components in POMM are at appropriate levels for safe consumption in food under the intended conditions of use.

TABLE 6.2 Margin of Exposure (MOE) Between Measured Levels of Vitamins and Minerals in POMM and Dietary Reference Values for Individual Vitamins and Minerals with Established UL Values (UL of a RDA[a] or RDI[b])				
Vitamin	Mean Level Detected[c] (mg/kg)	Level Based on the Amount of Daily Exposure to POMM[d]	Adult UL of RDA [RDI] Level	MOE US UL Value: Daily Serving Amount
Vitamins				
Folic acid	0.139	8.9 µg	1000 µg/d	112
Pantothenic acid	1.72	110.6 µg	[5 mg/d]	45
Vitamin B6	0.20	12.9 µg	100 mg/d	7,752
Vitamin C	1.93	124.1 µg	2000 mg/d	16,116
Vitamin E	0.30	19.3 µg	1000 mg/d	51,813
Vitamin B12	0.001	0.064 µg	[2.4 µg/d]	38
Minerals				
Sodium	2273	146 mg	2.3 g/d	16
Calcium	3607	232 mg	2.5 g/d	11
Phosphorus	5060	325 mg	4 g/d	12

[45] The method referenced uses inductively coupled plasma mass spectrometry (ICP-MS) suitable for the detection of manganese and selenium.

TABLE 6.2 Margin of Exposure (MOE) Between Measured Levels of Vitamins and Minerals in POMM and Dietary Reference Values for Individual Vitamins and Minerals with Established UL Values (UL of a RDA[a] or RDI[b])				
Vitamin	**Mean Level Detected[c] (mg/kg)**	**Level Based on the Amount of Daily Exposure to POMM[d]**	**Adult UL of RDA [RDI] Level**	**MOE US UL Value: Daily Serving Amount**
Iron	82.7	5.32 mg	40 mg/d	8
Copper	10.5	0.68 mg	10 mg/d[e]	15
Zinc	31.6	2.03 mg	40 mg/d	20
Manganese	6.80	0.44 mg	11 mg/d	25

[a] RDA = Recommended Daily Allowance
[b] RDI = Recommended Daily Intake
[c] Levels listed in this column reflect the average (mean) levels detected in three non-consecutive batches of POMM.
[d] The "amount of daily exposure" uses the estimated US 90th percentile daily POMM intake of 64.3 g POMM/person (20% market share) for all products into which POMM is incorporated) as estimated in Section 3.2.2.
[e] The UL for copper intake in children is 1 mg/day and is based on a single acute intake of copper (one time exposure).

6.1.2 Levels of Residual Nucleic Acids in POMM

Another safety endpoint relevant to POMM production is the level of residual nucleic acids in the food ingredient. Mushlabs has measured levels of RNA in three non-consecutive POMM batches and found that the average value was 1.67 % w/w, well within the limits suggested by FAO of 2% (Filho *et al.* 2019; Coelho *et al.* 2022). Table 6.3 indicates the individual values.

Table 6.3 Measured Amounts of RNA in POMM				
	Sample Batch A	**Sample Batch B**	**Sample Batch C**	**Average Of All Samples**
RNA Content# (% w/w)	1.67	1.41	1.93	1.67

RNA levels were measured by NovoCIB (https://www.novocib.com/nucleotide-analysis-service). An HPLC-UV analysis was performed. Each sample was analyzed by ion-paired HPLC twice, before and after enzymatic RNA hydrolysis to NMP.

Mushlabs notes that POMM is intended to be an ingredient in food and thus the actual amount of RNA will likely be lower in the final food product. Further, POMM would be heated/ cooked when incorporated as part of a final food product, processes that would denature RNA[46] even further.

[46] Although DNA is fairly stable at high temperatures, most RNA degrades under high temperatures. The half-life of an mRNA molecule of 1000 or more nucleotides is approximately 1 minute at temperatures of 100°C; tRNAs, which are much smaller, have a half-life of ~ 10 minutes at the same temperature (Becskei and Rahaman 2022).

6.1.3 Antinutrients and POMM

Antinutrients are chemical compounds produced by organisms used as food, primarily plants, that can protect the organism from predators, usually insects and nematodes, but their presence in food can diminish the bioavailability of specific nutrients and cause other deleterious effects. As discussed in the publicly available literature, such compounds are primarily associated with plant foods (Popova and Mihaylova 2019). For products produced from certain plants, the presence of antinutrients has been considered as a nutritional concern because of their tendency to lower the bioavailability of certain nutrients.

Recent reviews (Petroski and Minich 2020; Nath *et al.* 2022) found that antinutrient substances can affect nutrition in raw unprocessed human and animal food. In almost all instances, however, food processing (including soaking, cooking, fermentation, or thermal processing) either inactivates or otherwise ameliorates the effects of these substances. Mushlabs notes that POMM is a fungal product and not a plant product. Furthermore, POMM products are intended to be subjected to further processing, including heat (cooking); such treatments would be expected to inactivate any such substances that might be present in POMM (Petroski and Minich 2020; Nath *et al.* 2022). Many plants in the US diet also contain significant levels of these or very similar compounds such as flavonoids, alkaloids, tannins, phytate and saponins (Table 6.4).

Alkaloids are a wide family of basic nitrogen-containing natural products that have been studied for decades as bioactive compounds of pharmacological interest, with most attention focused on plants as sources. The most familiar plant-derived alkaloids include caffeine, nicotine, and morphine. In their review of the available data on specific alkaloid compounds identified in various fungal species (Zorrilla and Evidente 2022) identified only two indole alkaloids, terpendole N and terpendole O in the edible mushroom *P. ostreatus*. Hou *et al.* (2022), in their recent review of the topic, describe these two compounds as being found in some edible plants (*e.g.,* grains) and mushrooms.

Saponins are glycosidic molecules that arise as secondary metabolites, primarily in plants; they are heat-stable and amphiphilic (exhibit both hydrophilic and lipophilic properties). They are naturally found in many plants (*e.g.,* tomatine in tomatoes; avenascoside in oats; soyasaponin in soy) where they may serve as defense molecules against disease and herbivores (Mugford and Osbourn 2013). Because of their natural bitter or sweet flavors, they can be used in food (*i.e.,* glycyrrhizin is found in licorice roots). Saponins can, however, reduce the uptake of nutrients such as glucose and cholesterol, and decrease the activity of enzymes including trypsin and chymotrypsin (Gemede and Retta 2014). For example, oxalic acid can reduce bioavailability of calcium, magnesium, and zinc; tannins can affect taste and palatability; and phytate can affect the bioavailability of divalent cations. Nonetheless, saponins historically have been used in human medicine (*e.g.,* ginsenosides are extensively used in Chinese medicines; digitalis has been used in heart disease), and are currently being further investigated for pharmaceutical uses, particularly as drug delivery vehicles due to their amphiphilic properties (Suresh *et al.* 2021).

Flavonoids are molecules that contain various phenolic structures and are found in fruits, vegetables, grains, roots, stems, flowers, tea, and wine (Panche *et al.* 2016). In nature, flavonoids can contribute to the plant's survival by providing pigments that attract pollinators, or they can serve as natural defense molecules by acting as UV-protectants or they can be used to combat natural predators. Flavonoids have been linked to positive human health effects because of their antioxidant, anti-inflammatory, and anti-mutagenic or anti-carcinogenic properties. As discussed in Appendix D studies discussed in the published literature indicate that extracts of *P. pulmonarius* fruiting bodies and mycelia possess these functional characteristics (*e.g.,* antioxidant, tumoricidal, anti-diabetic, and anti-inflammatory activities), characteristics that have been summarized in the context of benefits to human health.

Although there has been discussion of antinutrients such as tannins, saponins, alkaloids, and flavonoids in plants, data on these components in mushrooms in general, and more specifically in *Pleurotus spp*., show that levels of the purported antinutrient components in the fruiting bodies of species grown under different conditions are lower than in plants found in the US diet (Table 6.4).

Only recently have a very few investigators in Africa and India published reports of levels of antinutrients in mushrooms. As can be seen in Table 6.4, the reported levels of these compounds in *Pleurotus spp*. are generally far lower than levels in plants. These values were measured in the fruiting bodies, which are subject to threats by pests in the environment that mycelia such as POMM would not be subject to when cultured in an aseptic submerged fermentation process. There is, therefore, no *a priori* reason to expect that the levels of these substances would be significantly higher in POMM than in the fruiting bodies. This conclusion is consistent with the information identified in the peer-reviewed literature (Table 6.4)

TABLE 6.4
Reported Levels of Antinutrients in *Pleurotus spp*. or Oyster Mushrooms Compared to Levels in Plants

Species studied	Phytates (mg/g)	Tannins (mg/g)	Oxalate (mg/g)	Trypsin inhibitor (mg/g)	Saponins (mg/g)	Citation
Pleurotus ostreatus	1.56 ± 0.12	0.048 ± 0.014	ND	ND	ND	Woldegiorgis *et al.* 2015
Pleurotus tuber – regium	0.001	0.0004	ND	ND	ND	Blessing *et al.* 2015
"Oyster mushroom"	0.01 ± 0.003	ND	2.4 ± 0.05	1.9 ± 0.68	0.67 ± 0.18	Oly-Alawuba and Obiakor – Okeke 2014
Pleurotus ostreatus	0.009	0.0011	0.0041	0.067	0.016	Kelechii *et al.* 2019
Legumes: Soy, lentils, chickpeas, peanuts, beans	3.86 to 7.14	1.8 to 18	0.008	Not reported	1.06-1.70	Popova and Mihaylova 2019
Grains: Wheat, barley, rye, oat. Millet, corn, spelt, kamut, sorghum	50 to 74	Not reported	0.35 to 2.70	Not reported	Not reported	Popova and Mihaylova 2019

6.1.4 Potential Allergenicity of POMM

Two types of allergenicity are relevant to the discussion of POMM: food allergy associated with ingestion of allergenic mushroom species, and respiratory allergy associated with inhalation of allergenic spores of mushrooms. In this part of the POMM safety assessment, Mushlabs addresses the peer-reviewed literature on the historical exposure to *P. pulmonarius* and closely related species. To be complete in its assessment, however, Mushlabs has further evaluated this issue from a molecular perspective in Section 6.2.

As with other cultivated mushroom species, inhalation-based allergies have been documented especially among workers who process mushrooms. It is well-known that some antigens present on spores are allergenic, resulting in allergic reactions to inhaled antigens. Branicka *et al.* (2021) reported on experience from a family mushroom farm in Poland, in which an otherwise healthy, 32-year-old non-atopic woman exhibited respiratory symptoms consistent with respiratory allergy or allergic asthma following processing and packaging *P. ostreatus*, a related species of edible mushroom. In an earlier work, Fischer *et al.* (2002) described delayed hypersensitivity-type reactions in some individuals upon intra-dermal, but not oral, challenge of hypersensitive subjects with *P. pulmonarius* spore extract. Characteristic symptoms of *Pleurotus* spp. allergy have included allergic rhino-conjunctivitis, asthma, and hypersensitivity pneumonitis (Lehrer *et al.* 1994; Helbling *et al.* 1998; Mori *et al.* 1998; Saikai *et al.* 2002). In addition, allergic contact dermatitis after exposure to *P. ostreatus* has been described (Rosina *et al.* 1995).

Because food products consisting of mycelial biomass propagated by submerged fermentation do not contain spores, inhalation allergenicity would not pose a food safety concern for use of POMM as a food ingredient.

With respect to undesired gastrointestinal effects having been reported following the ingestion of *Pleurotus* spp., Beug *et al.* (2006) identified nine cases of gastrointestinal symptoms in individuals ingesting *P. ostreatus* fruiting body. None of these reactions to *P. ostreatus* or undefined *Pleurotus* spp. were identified specifically as food allergy, but rather characterized as dietary intolerance to the fiber or sugars in the mushroom.

One recent case report of an anaphylactic reaction following ingestion of P. *ostreatus* has been reported (Rangkakulnuwat *et al.* 2023). A 12-year-old boy developed hives, abdominal pain, and vomiting following ingestion of a soup described as a "home-made oyster mushroom soup". Although previously healthy, the boy had exhibited an anaphylactic reaction following consumption of a mixed-mushroom tempura two years earlier, and according to the medical history presented in the report, had been "itchy" after eating mushrooms at school. A standard immunological work-up including skin prick- and prick-to-prick skin tests. In addition, an oral food challenge with six different kinds of cooked mushrooms in separate sessions was performed. A positive result was observed after ingestion of a sample purported to be oyster mushroom of the species *P. ostreatus,* but not the closely related king oyster mushroom (*P. eryngii*; >95% sequence identity with oyster mushroom). SDS PAGE and immunoblotting on samples of raw and cooked *P. ostreatus* (oyster mushrooms) were also performed; these identified a protein of MW 150-250 kD as a potential allergen. Mass spectroscopy of this protein and subsequent search against a UniProt data based identified this protein as trehalose phosphorylase (UniProt identifier A6yRN9). Further amino acid alignment with known allergens in the Comprehensive Protein Allergen Resource (COMPARE) database indicated that the primary amino acid sequence of TPase did not align with any other known allergen. The authors speculate that TPase "*presents as heat-stable homodimer in cooked [P. ostreatus]*[47]" but cautioned against concluding that "*trehalose phosphorylase is a novel allergen and allergenic protein [until] further experiments using recombinant or purified protein [are] conducted for further validation*."

The relevance of this case report for the food safety of POMM is unclear. First, it is a single case report purportedly linked to exposure to *P. ostreatus* and not to another *Pleurotus* species that was also tested. *P. pulmonarius* and other oyster mushroom fruiting bodies have been consumed historically without such

[47] Mushlabs notes that the amino acid sequence cited by Rangkakulnuwat *et al.* is from an accession number in UniProt for *P. pulmonarius*, not *P. ostreatus*. It is not clear whether the mushroom was misidentified by the authors, or they assumed that the proteins would be identical between the two species.

reported reactions. Trehalose phosphorylase is a protein involved in one of the two biosynthesis pathways of trehalose and is commonly found in fungi, plants, and other microbes that are widely consumed by humans. In fungi, trehalose is generally found in spores, fruiting bodies, and vegetative cells and is produced as a protective measure in response to environmental stress (*e.g.* heat, osmotic stress, *etc*.; Iordachescu and Imai 2008; Yan *et al.* 2021). Trehalose is not produced by mammals (Thammahong *et al.* 2017) but it is a sugar found in different kingdoms and phyla and may be found in microorganisms in the mammalian microbiome. As a result, trehalose-degrading enzymes are also found in detoxifying organs (*i.e.,* kidney and the brush border of the small intestine) in humans and other mammals. Further, data presented in the Rangkakulnuwat paper indicate that thermal stability of the protein may be limited at temperatures above 50°C. Given the child's previous milder symptoms, it is thus likely that this single report is of a child previously sensitized by ingestion of unspecified mushrooms who then developed a severe reaction. Except for the Rangkakulnuwat *et al.* case report, no other reports exist describing immune-modulated allergic reactions to the oral consumption of this protein.

Nonetheless, Mushlabs has pursued the question of allergenicity at an empirical molecular level (genomics and proteomics), described in Section 6.2. The results of those analyses *do not* provide evidence to support a finding that allergenicity following ingestion of POMM would be a food safety concern.

6.1.5 POMM and Potential Toxins

Although some fungi are known to produce toxins, edible mushrooms generally do not. It is common, however, to screen food ingredients from natural sources for the presence of commonly known toxic substances that include mycotoxins and certain environmental contaminants. Thus, POMM was screened for these commonly encountered substances, even though *P. pulmonarius* has not been linked to the production of the commonly known mycotoxins. This is particularly the case as the controlled conditions of POMM production would be expected to produce a food ingredient that was free of environmental contaminants. A search of the peer-reviewed literature was performed as well to determine if there were other less well-known toxins that have been linked to *Pleurotus spp*., including *P. pulmonarius*.

6.1.5.1 Mycotoxins and POMM

As part of the traditional safety assessment for POMM, batches were screened for common mycotoxins found in agricultural food products using liquid chromatography with tandem mass spectrometry ([LC-MS/MS]; SOP M 3650); the list of toxins is provided in Table 6.5. The levels of these toxins were all below the Limit of Detection, and thus pose no food safety concerns (Table 6.5). Further, as these mycotoxins are only produced by *Aspergillus* and *Fusarium* species, they would only be present if POMM were contaminated by those species; routine testing for the toxins is therefore not warranted due to the highly controlled submerged fermentation method of manufacture under GMP conditions.

TABLE 6.5
Analyzed Mycotoxins with Associated Limits of Detection

Toxin	Limit of Detection (µg/kg)
Aflatoxin B1	< 0.2
Aflatoxin B2	< 0.2
Aflatoxin G1	< 0.2
Aflatoxin G2	< 0.2
Ochratoxin A	< 0.5

TABLE 6.5 Analyzed Mycotoxins with Associated Limits of Detection	
Toxin	**Limit of Detection (µg/kg)**
Deoxynivalenol (DON)	< 10
Zearalenone	< 5.0
Deoxynivalenol (DON)3-Acetyl-Deoxynivalenol	< 10
15-Acetyl-Deoxynivalenol	< 20
Nivalenol	< 10
T-2 Toxin	< 2.0
HT-2 Toxin	< 2.0
4,15-Diacetoxyscirpenol	< 10
Fusarenon-X	< 10
Fumonisin B1	< 50
Fumonisin B2	< 50

6.1.5.2 Environmental Contaminants and POMM

Assays for chemical contaminants sometimes associated with agricultural production were conducted on batches of POMM using GC/MS[48] (Table 6.6). None of the substances included in the analysis were at levels above the limit of detection for the assays used. These results confirm that POMM source material was free of common environmental contaminants and thus there should be no food safety concerns for the source material. Moreover, because POMM is produced under controlled conditions, further analyses for these substances are not warranted.

TABLE 6.6 Analyses of Chemical Contaminants with Associated Limits of Detection	
Chemical	**Limit of Detection (µg/kg)**
Benzo(a)anthracene	< 0.2
Benzo(c)fluorene	< 0.5
Chrysene	< 0.2
Cyclopenta(c,d)pyrene	< 0.2
5-Methylchrysene	< 0.2
Benzo(b)fluoranthene	< 0.2
Benzo(k)fluoranthene	< 0.2
Benzo(j)fluoranthene	< 0.2
Benzo(a)pyrene	< 0.2
Indeno(1,2,3-c,d)pyrene	< 0.2
Dibenzo(ah)anthracene	< 0.2
Benzo(ghi)perylene	< 0.2
Dibenzo(a,l)pyrene	< 0.5
Dibenzo(a,e)pyrene	< 0.5
Dibenzo(a,i)pyrene	< 0.5
Dibenzo(a,h)pyrene	< 0.5

[48] SOP M 2920, GC/MS has an LOQ of 0.2 and 0.5 µg depending on the chemical being detected.

6.1.5.3 *Pleurotus* spp. and Potential Toxins

Oyster mushrooms are considered edible; this implies that like other edible mushrooms, they do not produce toxins that could pose a food safety concern. Nonetheless, fungi (including edible mushrooms) have evolved chemical means to protect themselves. These take the form of various peptides or proteins that can harm predators. For example, the most consumed mushrooms in the US, *Agarius bisporus*, a species that includes the white button, cremini, and portobello mushrooms, contain agaritine, a compound implicated in mutagenesis. Because agartine is degraded upon cooking, only mushrooms consumed crude or uncooked (Lagrange and Vernoux 2022) may pose a food consumption risk.

There are reports that members of the *Pleurotus* genus also have evolved chemoprotective mechanisms. *P. ostreatus* has been reported to contain a cytolytic protein, ostreolysin, primarily as a defense against nematodes (Zuzek *et al.* 2006; Frangez *et al.* 2017). This pore-forming protein occurs as a dimer[49] and both proteins that form the dimer are necessary for bioactivity (Frangez *et al.* 2017). Ostreolysins are heat-labile and sensitive to pH, losing activity at temperatures above 25°C and pH below 5 (Berne *et al.* 2005). The genus also contains the closely related pleurotolysins A and B.

Even though there are papers describing the association of ostreolysin protein with *Pleurotus* mushroom fruiting bodies, Vidic *et al.* (2005) confirmed the findings of Berne *et al*. (2002) and reported that ostreolysin is not found in the mycelia of basidiomycetes (of which *Pleurotus spp.* is one). Ostreolysin expression is turned on during the stage of early primordia and increases during the maturation of the fruiting bodies, resulting in the presence of this protein in young fruiting bodies. Ostreolysin was measured both by immunofluorescence and hemolytic activity (the toxin forms pores in lipid-rich vesicles such as erythrocytes) in the Vidic study. No immunofluorescence or hemolytic activity was detected in the mycelia, leading the investigators to hypothesize that the lysin participates in hyphal differentiation and the formation of basidia and basidiospores.

Section 6.2 describes the molecular analyses performed for Mushlabs to rule out the presence and/or expression of similar genes and proteins.

6.1.6 Overall Summary of Published Literature on *P. pulmonarius* and Toxicity

In traditional safety assessments, a source of useful data is often found in the peer-reviewed literature investigating the pharmacology and toxicology of a substance. In the case of a food ingredient such as POMM, oral feeding studies are not typically considered to be useful for identifying human food safety concerns in the US. Regardless, an extensive literature search was performed to identify data or information that could inform assessments of the safety of *P. pulmonarius* as a food ingredient. Searches were performed in 2022, 2023, and 2024 and used the U.S. National Library of Medicine databases known as PubMed and TOXLINE. Also employed was a search using the online ProQuest Dialog subscription database. No date limitations were applied in the searches and search terms used were directed to information on mycelia from *P. pulmonarius* but used names also associated with the mushroom such as "oyster mushroom". Although the focus was on mycelia, literature on extracts of the whole mushroom as well as those of the fruiting body were considered relevant because, like POMM, the fruiting body of oyster mushrooms consists of entangled dikaryotic hyphae (mycelia). Other keywords applied in the searches included "toxic", "toxicity", "safety", "allergy", "toxin", "adverse effect", and "oral". Studies

[49] The dimer proteins are ostreolysin A and pleurotolysin B which are both needed to form the active pore protein (Frangez *et al.* 2017).

conducted in both animals and humans, as well as in cells and tissues, were considered relevant to the safety assessment.

Much of the identified literature provides reviews of the use of *P. pulmonarius* for its medicinal properties (*e.g.,* Dalonso *et al.* 2015; Golak-Siwulska *et al.* 2018; Krakowska *et al.* 2020; Tung *et al.* 2020). Although these studies were not designed to be traditional toxicological evaluations, they did contain useful information on the pharmacological or biological activity of either the *P. pulmonarius* mushroom itself (fruiting bodies) or extracts of the mushroom (both fruiting bodies and mycelia) relevant to a food safety assessment.

The key observations from these reports relevant to the safety of *P. pulmonarius* as a food ingredient are summarized in Appendix D and are all focused on the bioactivity of different *P. pulmonarius* preparations (*e.g.,* dried mushroom; aqueous and/or organic extracts of fruiting bodies of *P. pulmonarius*; aqueous and/or organic extracts of mycelia of *P. pulmonarius*). Overall, these studies report that extracts of *P. pulmonarius* appear to have beneficial effects on the endpoints of human health studied. As stated above, although these studies were not designed to be safety or toxicology evaluations, the data can serve as indicators of the lack of toxicity (or its inverse, safety) of *P. pulmonarius* after oral exposure (*e.g.,* no change in body weight; no histopathological changes in organs after oral dosing; *etc.*).

The *in vivo* studies included in this safety assessment were selected because they met one or more of the following criteria:

- Published in a peer-reviewed journal (relevant to study quality)
- Used a common laboratory animal species (relevant to study quality and interpretation of any findings/ observations)
- Involved systemic exposure of animals to preparations of *P. pulmonarius*, either after oral dosing or intraperitoneal injection (routes most relevant to food safety assessment)
- Included control groups in the study design (aids in interpretation of any reported effects/ observations)
- Included more than one exposure or dose level in the study design (dose-response evaluation)
- Investigated duration of exposure (relevant to food safety assessment)
- Provided details on the nature of the *P. pulmonarius* substance being administered (relevant to food safety assessment)

A review of the key features and outcomes of these studies (Appendix D) shows that regardless of the type of *P. pulmonarius* preparation/extract studied, there were no reports that the mushroom or its extracts exhibited any adverse effects at pharmacologically active (*i.e.,* bioactive) doses. In this context, “bioactive” doses refer to levels of exposure affecting basic cellular physiology in disease models (*e.g.,* animal models of diabetes or cancer in the case of *P. pulmonarius* and its extracts) but does not imply toxicity. Appendix D includes studies (*e.g.,* Xu *et al.* 2014, Balaji *et al.* 2020), in which data were collected from control groups *i.e.,* those in which the only intervention was administration of a *P. pulmonarius* preparation to non-disease induced or otherwise untreated animals. No toxicity was reported in the outcomes of such studies. The following provides a brief narrative of some the studies that are particularly relevant to the safety assessment of POMM after oral exposure.

6.1.6.1 Published Studies Relevant to Safety Assessment of Mycelia of Oyster Mushrooms

6.1.6.1.1 Published *In Vitro* Studies

Of the *in vitro* studies described in Appendix D, two involved uses of aqueous extracts of *P. pulmonarius* mycelial biomass (Jeong *et al.* 2013; Milovanovic *et al.* 2014). Although not *in vivo* evaluations of *P. pulmonarius*, these *in vitro* studies are relevant to safety assessment when considered as part of weight-of-the-evidence approach to safety assessment (*e.g.,* Blaauboer *et al.* 2014; Schilter *et al.* 1996). Such studies can provide information on underlying mechanisms by which toxic responses may occur. A common example of use of such data is in cancer risk assessment in which *in vitro* studies of genotoxicity are included in weight-of-the-evidence assessments. Therefore, the *in vitro* studies identified in the published literature were considered relevant to the food safety assessment for POMM.

Milovanovic and colleagues (2014) studied the biological activity of aqueous extracts of the mycelia from three Pleurotus species: *P. pulmonarius*, *P. ostreatus,* and *P. eryngii* to follow up on reports that genus *Pleurotus* had been shown to have significant nutritional and medicinal activity (Gunde-Cimerman 1999; Stamets 2000). The positive effects noted included anti-hypercholesterolemic, anticancer, antiviral, antimicrobial and immunomodulating activities (Gunde-Cimerman 1999). All of the mycelial extracts were tested for anti-fungal effects against a variety of common pathogenic fungal organisms, for antioxidant activity, and for cytotoxic effects in human cervical adenocarcinoma HeLa and human colon carcinoma LS174 cell lines. The range of exposure levels (doses) to test for cytotoxicity included 200, 100, 50, 25 and 12.5 µg/mL. All the aqueous mycelial extracts were reported as showing "low cytotoxic activity." In particular, the aqueous *P. pulmonarius* mycelial extract was reported to have an IC_{50} value of >400 µg/mL in cultured human cells. The authors also reported that an antioxidant effect (radical scavenging) was correlated with the phenol/flavonoid content of all three extracts. This report supports the safety assessment of POMM as it does not indicate the presence of toxins in the mycelia of these *Pleurotus* species, even when concentrated in extracts.

Jeong and colleagues (2013) also tested aqueous extracts of *P. pulmonarius in vitro*. They focused on antioxidant and immunomodulatory effects of exo- and endo-polysaccharides produced in submerged cultures of mushroom mycelia. The extracts studied were from seven different mushroom species, including *P. pulmonarius* (other species from which extract were derived included *P. citrinopileatus*, *P. australis*, *P. pulmonarius*, *Tremella mesenterica*, *Cryptoporus volvatus*, *Cordyceps militaris* and *Cordyceps sinensis;* all are edible mushroom species). As in the study by Milovanovic *et al.* 2014, flavonoids were identified as major components of the polysaccharide extracts from the *P. pulmonarius* mycelia as well as the other six mycelia extracts. As expected, the *P. pulmonarius* polysaccharide mycelial extracts showed significant antioxidant activity as well as immunomodulatory activity. The authors stated that the data presented were consistent with a general suppressive effect on cytokine production and therefore "*are more likely to have any beneficial effects through reducing the magnitude of any immune/inflammatory response*." These data also are generally supportive of the safety of *P. pulmonarius* extracts and provide a mechanistic basis for the bioactivity of *P. pulmonarius* mycelia that is described in review papers.

6.1.6.1.2 Published *In Vivo* Animal Studies

Although Appendix D contains a description of studies that involved exposure to extracts of *P. pulmonariu*s fruiting bodies, only the studies that describe use of extracts of the whole mushroom (not solely fruiting bodies) or mycelial extracts are presented in this section. Regardless of the test substance's origins, however, review of the studies in which *P. pulmonarius* mycelial extracts have been tested indicate that under the conditions of those testing protocols, no food safety concerns were identified, nor was any toxicity observed with *P. pulmonarius* extracts. These data are consistent with the data to be discussed in Section 6.2 related to the absence of known toxins in POMM molecular testing.

The *in vivo* micronucleus test is often used to assess genotoxicity (OECD guideline 474). Akanni *et al.* (2010) reported that rats were exposed orally to extracts of *P. pulmonarius* using a protocol consistent with that outlined by the Organization for Economic Cooperation and Development (OECD) 474.[50] Animals were injected intraperitoneally to bypass the first pass liver metabolism making the circulating concentration of the extract a "worst-case" exposure level (no immediate metabolism). All treatments with *P. pulmonarius extracts* were negative (the positive control compound, cyclophosphamide, demonstrated clastogenicity). These results suggest that *P. pulmonarius* extract does not appear to be genotoxic *in vivo*, thereby providing additional evidence to the weight of evidence supporting the safety of POMM. These results are consistent with findings reported for other edible mushroom extracts (*e.g.,* GRN848 for Shiitake mycelia; Emsen *et al.* 2020 for *P. ostreatus* extracts; Boulaka *et al.* 2020 for *P. eryngii* and *P. ostreatus).*

Wasonga *et al.* (2008) reported that aqueous extracts of *P. pulmonarius* (whole mushrooms samples, dried) delayed progression of liver cancer in a mouse model of the disease. Mice were fed daily for 12 weeks with food enriched in aqueous extracts of *P. pulmonarius* (one part extract with four parts normal diet). Histopathological examination of the liver of animals exposed to the extracts as well as the carcinogen being tested showed that tissue appeared normal. Based on these and similar observations, the authors supported elevating extracts of *P. pulmonarius* to clinical studies.

Lavi *et al.* (2010; 2012) contain data relevant to the bioactivity and safety of aqueous extracts of both mycelia and fruiting bodies of *P. pulmonarius*. In the 2012 study, a mouse model of colitis was used to examine *in vivo* anti-inflammatory activity of the extracts at doses that approximated dietary fiber intakes in humans (Lavi *et al.* 2012). Earlier *in vitro* studies by the research group had linked *P. pulmonarius* extracts with anti-inflammatory properties likely due to the presence of glucans in the extracts (Lavi *et al.* 2010). In the *in vivo* mouse study, animals were administered extracts for 10 days (either 2 or 20 mg/day orally in food) before treatment to induce colitis, with extract exposure continuing for an additional 18 days. One group of mice only received 20 mg/day in diet (high dose extract groups, both mycelial and fruiting body extracts were tested this way) with no induction of colitis and served as a control group for examining the potential adverse effects of the extract itself in the mice. The authors reported that there were no adverse histopathological changes in colon tissue when mice were treated for 29 days with the two extracts in the diet. In the mice where colitis was induced, the authors reported that both extracts (fruiting bodies and mycelia) were able to reverse the histological and biochemical alterations in the experimentally induced colitis mouse model. The lack of effects in the control animals with repeated dose exposure (diet) to extracts of both fruiting body and mycelium extracts supports the safety of the *P. pulmonarius* mycelia specifically.

Olufemi *et al.* (2012) examined the bioactivity of extracts of mycelia of selected oyster mushrooms (*P. pulmonarius* and *P. ostreatus*) found in Nigeria. The mycelia were cultured on a novel medium of yeast extract supplemented with an ethanolic extract of *Annona senegalensis*, and the anti-leukemic potential of their metabolites was studied. The study included one group of rats administered the extracts alone (no carcinogen) by oral gavage. These rats appeared healthy and *"showed no significant alterations in erythrocyte counts, leukocyte counts, and hematological indices, when compared with the healthy animals*

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[50] https://www.oecd-ilibrary.org/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects_20745788 last accessed 6 September 2024

in group F who received commercial feed and water only." These data are supportive of the oral safety of oyster mushroom mycelial extracts.

Balaji *et al*. (2020) evaluated the *in vivo* bioactivity of aqueous, hot water (HWE) and organic (acetone) (AE) extracts of *P. pulmonarius* mycelia in a diabetic rat model. Included in the study was an evaluation of the acute oral toxicity of the extract in the rats; a standard acute oral toxicity study design was used. The authors reported*: "The results of the acute toxicity study of HWE and AE showed no mortality even after 72 hours at 2000 mg/kg. The AE showed mild sedation at 2000 mg/kg but however, HWE didn't alter any of the general behavior. No fatality or noxious reactions were found during and after the treatment period with both extracts."* Additionally, the HWE and AE mycelial extracts reversed the adverse histopathological effects in the diabetic rat model. Again, these data are supportive of the safety of orally administered mycelial extracts from *P. pulmonarius*.

Notably, none of the published studies report adverse effects in animals exposed to *P. pulmonarius* extracts.

6.2. Molecular Confirmation of the Lack Hazards That May Pose Food Consumption Risks

As context for the molecular characterization of POMM, this section provides some underlying information on the life cycle of *P. pulmonarius*, with particular emphasis on morphology and karyology, and the expected, but small differences between the fruiting body and mycelia characterized by Joh *et al.* (2007) and Wang *et al.* (2017). In order to demonstrate that any potential toxin or allergens would not be present in POMM, Mushlabs fully characterized the genome of POMM, completed a transcriptomic analysis, and concluded with a proteomic analysis. The results of these studies demonstrated that there were no proteins in POMM that could be linked to potential allergenicity or toxicity, and that the molecular analyses support the conclusions of information provided herein that POMM does not pose a food safety concern.

6.2.1 Differences Between Fungi and Plants

Plants and fungi constitute two distinct biological kingdoms of eukaryotes, each of which has evolved a distinct biology. Although historical and superficial structural analogies once led to the classification of fungi within the plant kingdom (*e.g.,* stationary nature and presence of cell walls), key advancements in understanding their structures, functions, and genetics have shown that fungi differ fundamentally from plants in several respects. Key distinguishing traits include karyology, heterotrophic nutrition (obtaining nutrients by digesting and absorbing organic material), the absence of chlorophyll, a chitinous cell wall as opposed to the cellulosic cell walls, relatively undifferentiated cellular structure in the form of hyphae, and very different modes of reproduction. This contrasts with plants, which have well developed distinct cellular structures comprising different tissues of the plant (*e.g.,* roots, stems, leaves, flowers, and their component parts) that have specific functions. These different structures may have significant differences in composition as well; for example, rhubarb leaves are toxic, but stems are edible. In contrast, basidiomycetes produce dikaryotic hyphae that form a continuum over the organism, except for the haploid spores (see Figure 6.1). These considerations led to the introduction of a new system (Whittaker 1969; Hagen *et al.* 2012) in which fungi are classified as a separate kingdom based on their unique characteristics. As a result, biology would dictate that care should be taken when attempting to draw conclusions from observations in one kingdom to another, including from the plant kingdom to the fungal kingdom.

As an example, the historical nomenclature referring to the reproductive form of the mushroom as the "fruiting body" appears to be derived from an analogy to the more commonly perceived "fruit" of a plant that houses the seeds and may have occurred at a time when the differences between the fungal and plant kingdoms were not fully articulated. The net result is often a confusion among non-mycologists who may improperly characterize the "fruiting body" as a "fruit" that harbors "mushroom seeds". The only analogy of the haploid precursors to plant seeds is the formation of haploid nuclei that become basidiospores; these are dispersed in nature where they germinate as monokaryotic mycelium again and the life cycle repeats. Although these may be analogous to true seeds in plants, they have no homology to them (Martinez-Carrera 1998).

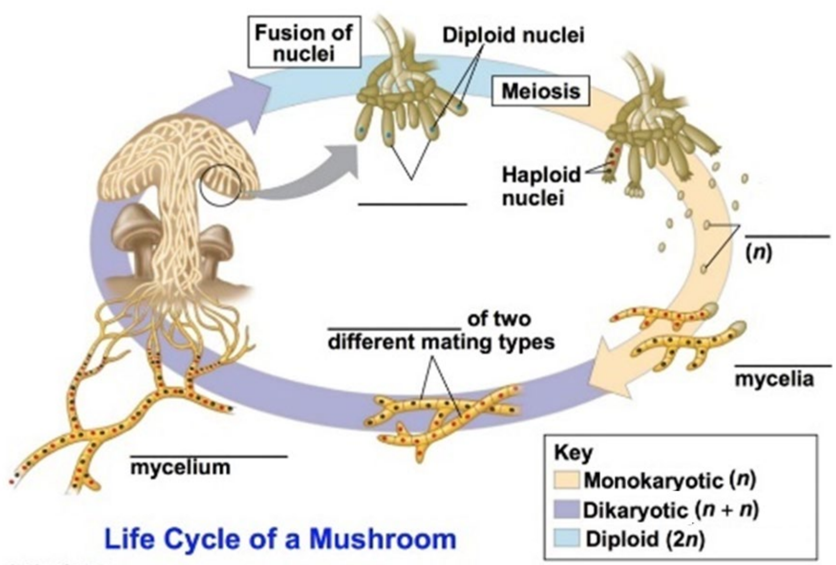
6.2.2 The Life Cycle of *P. pulmonarius* in Nature

The following section provides a brief overview of the life cycle of Basidiomycetes, including *P. pulmonarius*, in nature. It is provided as background to provide context for the similarities, and perhaps more importantly, the lack of food-safety related differences between the edible, and therefore safe, fruiting body of *P. pulmonarius* and POMM propagated by Mushlabs using submerged fermentation, both being dikaryotic mycelial forms of the Phoenix oyster mushroom.

6.2.3 Life Phases and Karyology of *P. pulmonarius*

The genus *Pleurotus*, including *P. pulmonarius*, has two distinct "life phases". These differ from those exhibited by more differentiated eukaryotes such as most edible plants and non-colonial animals in which gametogenesis is limited to a few highly specialized structures that produce haploid gametes, usually within the greater part of the organism that is diploid. Basidiomycetes, particularly *Pleurotus spp.,* alternate between monokaryotic phases (one nucleus or haploid) and dikaryotic or pseudo-diploid states, which arise following the fusion of two monokaryons (mating). This is depicted in Figure 6.1.

Figure 6.1 Mushroom Life Cycle derived from https://quizlet.com/291268807/life-cycle-of-a-mushroom-diagram/



The monokaryon results from the maturation of a basidiospore, which develops into a haploid mycelium composed of hyphae as their simplest building unit. When compatible monokaryotic mycelia encounter each other in the soil, they are able to "mate" by fusing hyphae in a process referred to as "plasmogamy." The resulting hyphae become binucleate, containing two genetically different nuclei in each hyphal compartment of the mycelium and are therefrom described as dikaryotic (Martinez-Carrera 1998). Under specific conditions, the dikaryotic mycelium can morphose into basidiocarp, a structure colloquially referred to as the fruiting body.

As seen below in Figure 6.2, morphological mycelium (dikaryotic hyphae) is often found (and therefore consumed) as "hairy" structures on the caps of oyster mushroom fruiting bodies at the market level.

Figure 6.2 Presence of white mycelium on the cap of consumed oyster mushrooms (outlined in red)[83]

6.2.4 Fruiting Body and Mycelial Structure of *P. pulmonarius*

The form of *P. pulmonarius* commonly available to consumers is the dikaryotic fruiting body. The basic structural unit of the dikaryotic fruiting body is referred to as a hypha (pl. *hyphae*), which is also the basic structural unit of the mycelium. What is now known as the hyphal systems of mushrooms *i.e.,* how the hyphae within the fruiting bodies ensure its shape and structure, was first described in 1932 and then later reprised by Van der Westhuizen (1971), Pegler (1996), and Porter and Naleway (2022). The British Mycological Society summarizes this as "*The main body of the fungus is made up of fine threads (hyphae) that group together to make a mycelium….the fine threads that make up the mycelium that lies hidden from view can also form the fruit body of the fungus that we see*."[51]

The hyphal system of oyster mushrooms is described as "monomitic" or having only one type of hypha. Hence, all dikaryotic hyphae in *P. pulmonarius* are considered to be "generative" (*i.e.,* totipotent) and structurally similar. These generative (dikaryotic) hyphae can be assembled into threads, which can then be referred to as dikaryotic generative mycelium or fruiting bodies, depending on their structural forms. This means that fruiting bodies of *P. pulmonarius* are solely composed of a bundle of dikaryotic generative hyphae (except for the spore) and "unidimensional" dikaryotic mycelia is also observed at market level as a fluffy white layer expending on the cap of oyster mushrooms during their storage and sale (Figure 6.2).

[51] https://www.britmycolsoc.org.uk/mycokids/mycokids-how-are-fruit-bodies-made last accessed 6 September 2024

POMM originates from dikaryotic generative hyphae directly isolated from the fruiting body of *P. pulmonarius* and is propagated through submerged fermentation (as described in Section 2.2.1). Hence, it also consists of dikaryotic hyphae assembled as dikaryotic mycelia, just as are the widely consumed fruiting bodies (Ginterova and Janotkova 1975; Eger 1979; Zervakis and Balis 1996; Petersen and Hughes 1997; Bao *et al.* 2004; Li *et al.* 2019; Porter and Naleway 2022).

6.2.5 Gene Expression in *P. pulmonarius*

Little information is available on the specific patterns of gene expression in *P. pulmonarius* (or any mushroom) found at the retail level. Once identified as being "edible", with the exception of microbial degradation or exposure to inappropriate storage conditions, these mushrooms are generally considered safe to eat.[52] Thus, specific differences in gene expression related to different growth conditions do not affect the safety of oyster mushrooms currently available on the markets and consumed worldwide. In fact, multiple vendors sell oyster mushroom growing kits to the general public[53] on the assumption that the resulting mushrooms (and attached white fluffy mycelia) will be safe to eat, regardless of changes in gene expression that may occur as the result of different growth media or substrates.

Phenotypes of any living organism intrinsically rely on changing gene expression, which in turn reflect the internal and external factors such as development phase, age, and environmental stimuli encountered throughout the growth of these mushrooms. In particular, different gene expression patterns are expected from oyster mushrooms grown on different substrates or under different conditions. As previously stated in Section 2.2.1, Mushlabs controls the growth of POMM stringently to avoid any differentiation of the mycelia into the fruiting body stage.

The effects on gene expression that may be attributed to internal and external factors have been described in literature:

- Wu *et al.* (2021) demonstrated that the protein and metabolite profiles of consumed fruiting bodies is not constant during their storage after harvest but instead constantly evolve over time.

- Wang *et al.* (2020) and Yu *et al.* (2023) showed that even within the same fruiting body, the different parts (pileus, stem and gills) have different nutritional, transcriptome and proteome profiles.

- Xiao *et al.* (2017), Angelini *et al.* (2021), Zeng *et al.* (2022), and Lee *et al.* (2023) highlight that fruiting bodies obtained from different growth substrates show very different profiles and levels of metabolites and proteins.

[52] A key exception is the morel mushroom, and the "false morel" with which it is often confused. Morels may contain the highly toxic hydrazine and must be cooked thoroughly prior to eating. https://www.fda.gov/food/outbreaks-foodborne-illness/investigation-illnesses-morel-mushrooms-may-2023 last accessed 6 September 2023

[53] https://learn.freshcap.com/growing/how-to-grow-oyster-mushrooms/ (last accessed 6 September 2024) which lists such substrates a hardwood sawdust, soy hulls, wheat straw, sugar cane, and coffee grounds among others; https://www.motherearthnews.com/organic-gardening/gardening-techniques/how-to-grow-oyster-mushrooms-at-home-zm0z23zatro/ (last accessed 6 September 2024) which sells oyster mushroom kits to generate fruiting bodies with a proprietary medium; https://www.fieldforest.net/product/587/BLOG (last accessed 6 September 2024) which sells kits to grow oyster mushrooms on straw or (non-specified) grain.

As a result, there is no "reference" mushroom fruiting body gene expression library that can serve as a comparator for food safety assessment. Nonetheless, as previously indicated, fruiting bodies of *Pleurotus spp.* are considered "edible mushrooms" and have been consumed safely regardless of any differences in gene expression, which may manifest as changes in shape, maturity, composition, or proteome and metabolite profiles.

As previously discussed in Section 2.2, Mushlabs stringently controls the growth of POMM in its fermenters to favor the propagation of dikaryotic mycelia as threads and avoid transition to the fruiting body phenotype and the potential for spore formation. Wang *et al.* (2017) reported that the mature fruiting body of *P. pulmonarius* is always the developmental stage with the highest variation of expressed proteins (+50%), while little to no protein is differentially expressed in the mycelium of *P. pulmonarius*.

Gene expression must differ among the different morphologies and developmental phases of all living organisms. Joh *et al.* (2007) identified genes expressed uniquely at different developmental stages of *P. ostreatus* and found that under his experimental conditions, among 4,060 genes, only eight (0.2%), thirteen (0.3%) and two (0.05%) were expressed only in mycelia, fruiting body, and basidiospore, respectively. The remaining genes are expressed in all stages although their expression may differ under different growth conditions, age, specific portion of the fruiting body (stem or cap), and, most importantly, how their sources of nutrition and environmental conditions may vary (*e.g.,* substrate, temperature, light, humidity; Sales-Campos *et al.* 2011; Wang *et al.* 2020; Berger and Ersoy 2022; Xu *et al.* 2022; Lee *et al.* 2021). Wang *et al.* (2017) specifically investigated the proteome of different morphological forms of *P. pulmonarius* and came to similar conclusions, showing that among 1698 detected proteins only 11 (0.6%) were solely expressed in mycelium and these were mainly belonging to known central metabolic pathways (carbohydrate, lipid and ATP synthesis).

A review by Berger and Ersoy (2022), described studies of gene expression in three edible mushroom species, *Agaricus bisporus*, *Pleurotus ostreatus*, and *Lentinus edodes.* They noted that all three species exhibited differences in levels of gene expression at different stages of the fungal life cycle. Ye *et al.* (2021) reported a detailed analysis of differential expression in vegetative mycelia and primordia (developing fruiting bodies) in a closely related species P. *eryngii.* A number of genes related to cellular functions such as transcription, carbohydrate utilization, and increased chitin production for the energy-consuming process of building the fruiting body were up- or down-regulated. Several genes related to trehalose metabolism were also identified as being up-regulated; trehalose is a stress protectant for cells. Virtually all the differences identified were in the expression of enzymes generally attributed to typical housekeeping functions present in all life stages, although their levels may differ.

Differences identified in the literature were related to changes in levels of expression rather than in the presence of additional proteins in the vegetative mycelia compared to the fruiting body. Wang *et al.* (2017) has demonstrated that approximately 65% of the proteins (and their isoforms) expressed across the different life stages of *P. pulmonarius* likely have housekeeping functions

In summary, the gene expression differences noted in mycelia compared to fruiting bodies are likely responsible for the differences in assembly and relative compositions of the major life forms of *P. pulmonarius*. Gene expression changes are necessary for the efficient function of any organism. Edible mushrooms have been consumed safely without consideration (or knowledge) of such differences in levels of gene expression and changes in protein structure; we have simply referred to them as "oyster mushrooms".

As discussed here, the actual results from Mushlabs' molecular analyses of POMM genomes and gene expression levels are consistent with the published literature and provide specific instances of differential gene expression. The conclusions of these analyses for food safety are discussed in Section 6.3. As elaborated more fully in the following section, Mushlabs has reviewed differences in gene expression between POMM and *P. pulmonarius* fruiting bodies that hypothetically may have an impact on food safety at the molecular level. No genes with potential food safety concerns have been identified that could affect the food safety of POMM.

6.2.6. Mushlabs Molecular Analyses with POMM

As discussed earlier in this section, *P. pulmonarius* has been reported to contain the dimer-forming proteins (ostreolysin A and pleurotolysin B) in the early germinating tissue. These proteins, when present together, can form a pore-forming complex that can lyse cells. Berne *et al.* (2005) have demonstrated by immunofluorescence that these proteins are present in the early germinal tissue of the fruiting body, but not in the mycelia. The latter finding, the absence of lytic proteins in mycelia of *Pleurotus* spp. was confirmed by Vidic *et al.* (2005). In order to determine whether POMM contained this or any potential toxic proteins, Mushlabs conducted a comprehensive series of molecular analyses. The analysis specifically addressed the *Pleurotus* toxin genes that encode the proteins ostreolysin A and pleurotolysin B[54], as well as detailed comparisons of the molecular data from POMM to determine whether any other potential known toxins from other fungi or allergens were present.

Mushlabs undertook a three-pronged approach to demonstrate that ostreolysin-related proteins, or other proteins with toxic properties were not found in POMM

1. Whole Genome Sequence (WGS) analysis using DIAMOND BLASTX conducted by Charles River Laboratories (CRL)[55]
2. Transcriptomic Analysis conducted by Novogene[56]
3. Sequence specific analysis of the proteome of POMM conducted by the University of Griefswald (located in Griefswald, Germany)

These analyses were performed by laboratories with the specific expertise needed and were selected by Mushlabs based on their qualifications and experience in molecular analyses of genomes and proteomes in biological materials, with particular emphasis on food safety. The results of these molecular studies are internally consistent: none of the studies were able to predict or discern the presence of any of the hemolysin proteins or any other known mushroom toxin in POMM, as described in more detail in the following sections.

6.2.6.1. Genomic Analyses

Charles River Laboratories (CRL) performed two analyses of the genome of POMM. Because of the literature identifying ostreolysin and related hemolysins as potential toxins in *Pleurotus* mushrooms, the emphasis was put on ostreolysin A and the two known isoforms of pleurotolysin (pleurotolysin A and pleurotolysin B). It is important to remember that because the source of POMM was dikaryotic hyphae

[54] These proteins are occasionally referred to as "hemolysins" as their activity was first detected by the lysis of sheep blood erythrocytes (Simpson and Gozzo 1978).
[55] https://www.criver.com/
[56] https://novogene.com

material isolated from a *P. pulmonarius* fruiting body, the genome of POMM is identical to the genome of the widely consumed fruiting body of *P. pulmonarius.* That genome has the ability to produce different transcripts and their resulting translation products depending on the environmental conditions to which they are exposed and will determine the morphology they assume. As Mushlabs isolated its POMM starting material, dikaryotic mycelium from the fruiting body, from the fruiting body of PMX1906, that material should be regarded as a subset of the fruiting body> As a result, it would be counter-intuitive to expect that new compounds are produced that have not been detected in fruiting bodies of the species grown under very diverse conditions (*e.g.* substrate, temperature, *etc*).

Database for comparison/annotation

CRL first performed a thorough search from GenBank for all fungal peptides that are closely or distantly related to “toxins” and “defensins”. Defensins are frequently short peptides that are not well conserved and are used to protect the organism from bacterial and eukaryote attack. They are often overlooked with standard annotation pipelines because they are typically only 18-45 amino acids in length.

After building this library, CRL used the DIAMOND BLASTX (Diamond) method to analyze and query the POMM genome. This approach uses the same NCBI public database (NR and NT) as the commonly used BlastX. Diamond is routinely used by CRL and other laboratories because it utilizes a more efficient algorithm to allow for a more rapid analysis with a comparable detection sensitivity to NCBI BlastX (Buchfink *et al.* 2015). Potential loci of interest were annotated using Diamond match to non-redundant peptides and fungal toxins previously identified during the construction of the toxin and defensin library. Each of these loci were examined individually including referring to the original fungal toxin entry in Genbank. In some cases, NCBI web blast was also used to identify conserved domains and match these peptides with their function.

Stringency Cutoffs

One concern regarding genomic matches is that of stringency, often expressed as “expect values”, *i.e.,* the number of “hits” one could expect to see by chance for a search of a database of some finite size; these are often expressed as “e-values or e-value thresholds”. CRL noted that with the standard/default online e-value threshold used by NIH of 0.05 (5e-2), a larger list with more low-scoring (sensitive, not specific) hits can be reported. This is often thought of as a convenient significant threshold for reporting results. Nonetheless, CRL routinely employs the more stringent 1e-10 as the best trade-off for general users in that it provides some sensitivity while winnowing out many, but not all, false positives. CRL notes that the usual practice in the literature is 1e-3 or smaller, which removes most false positives.

To help ensure that conservative (*i.e.,* health protective) concerns are met, CRL first employed a cut-off of 1e-10 to remove rare false positives that can occur from repetitive sequence matches. Then, to demonstrate that lower stringency searches would not yield more relevant information for food safety, CRL re-ran its genomic analysis for potential sequences in the genome that would possibly code for known toxins (and/or allergens) using the far less stringent cut-off of 1e-2.

TABLE 6.7 **Annotation of Potential Fungal Toxins in *P. pulmonarius* Genome**	
Putative Peptide ID	**Descriptor***
VW098972.1	Dual O-methyltransferase/FAD dependent monooxygenase CTB3 (Cercosporin toxin biosynthesis cluster protein 3) [*Ganoderma boninense*]
VW097593.1	O-methyltransferase CTB2 (EC (Cercosporin toxin biosynthesis cluster protein 2) [*Ganoderma boninense*]
VWO98972.1	Dual O-methyltransferase/FAD-dependent monooxygenase CTB3(Cercosporin toxin biosynthesis cluster protein 3) [*Ganoderma boninense*]
VWO97593.1	O-methyltransferase CTB2 (EC (Cercosporin toxin biosynthesis cluster protein 2) [*Ganoderma Boninense*]
VW097386.1	Cercosporin MFS transporter CTB4 ICercosporin toxin biosynthesis cluster protein 4), partial [*Ganoderma boninense*]
VW096292.1	Ketoreductase CTB6 (EC Cercosporin toxin biosynthesis cluster protein 6)
VW095044.1	Cercosporin MFS transporter CTB4 ICercosporin toxin biosynthesis cluster protein 4), [*Ganoderma boninense*]
spA0A384XHA3,1STR11_STRTC	Full = Phenylanaline ammonia-lyase str11; Full=Stobilurin biosynthesis cluster protein r11
UFQ31923.1	Ergot alkaloid biosynthetic protein A partial [*Claviceps Africana*]
QIM40768.1	Prolyl oligopeptidase [*Amanita pallidorosea*]
*These descriptors are of proteins putatively encoded by genes identified by peptide ID numbers in this table.	

This new analysis did not significantly increase the sensitivity of the assay, yielding only 98 additional new hits; all these corresponded to cellular pathways that have not been implicated in toxin production. Overall, seven loci were identified with an 80% match to loci for known *Pleurotus* hemolysins, Pleurotolysin B and Ostreolysin A6, and 26 loci were found with a 30-50% match to genes encoding proteins potentially involved in known fungal toxin biosynthetic pathways.

The first seven loci that matched the loci from known *Pleurotus* hemolysins Pleurotolysin B and Ostreolysin A6 with high identity (>80%) were deemed not of safety concern as both of these genes are also found in other edible *Pleurotus* species regularly consumed worldwide.

The Organization for Economic Co-operation and Development (OECD) has addressed this topic in its consensus document on oyster mushrooms (OECD 2006). They concluded that:

> "*there is no data on the toxicity of P. ostreatus hemolysins in oyster mushroom consumers. Oyster mushroom is considered a non-toxic mushroom and the presence of hemolysins does not influence this conclusion. Hemolysins are thermo-labile, and potential toxicity would be considered only for raw mushrooms. Furthermore, proteins are usually degraded in the gastrointestinal tract when ingested. Administration of aqueous extracts of P. ostreatus to mice demonstrated no acute*

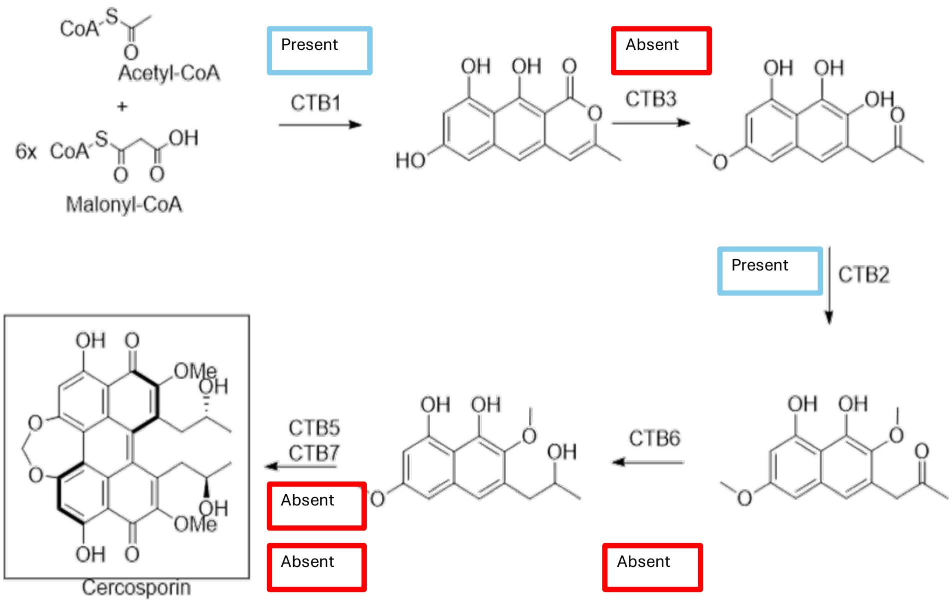
effects and repeated oral feeding of the mushroom to rodents revealed no histopathological changes of cardiac or hepatic tissue".

More importantly, the presence of hemolysin protein dimers in the biomass of *P. pulmonarius* was later excluded after review of the transcriptomic and proteomic data for POMM. The protein known as Pleurotolysin B was not found in either dataset; the simultaneous presence of both Pleurotolysin B and Ostreolysin A6 is required for an active hemolysin dimer to form. Thus, the presence of just one is not indicative of a potential food consumption risk.

With respect to the remaining 26 loci initially identified by CRL in POMM, the loci overlapped with genes encoding the eight peptides listed in Table 6.7. Each of these peptides was reviewed further by CRL, who concluded that the proteins encoded by the identified sequences pose no food safety concern: they are not allergens or toxin themselves, but are only distantly related to one step of an incomplete toxin pathway, while having other key functions in the central metabolism of *Pleurotus spp*. For instance, the first entry (VWO98972.1) encodes an RNA-helicase, an enzyme involved in 40S ribosomal subunit biogenesis, while the fourth entry (VWO96292.1) encodes a glucose-6-isomerase, a central enzyme involved in energy generation in the glycolysis pathway and in converting glucose-6-phosophate to fructose-6-phospate as precursors of fungal cell wall biosynthesis.

Although some of the peptides in Table 6.7 are somewhat similar to enzymes involved in the cercosporin pathway, CRL has determined that this pathway is incomplete in *P. pulmonarius* as several key enzymes are absent from the POMM genome, transcriptome and/or proteome. For example, peptide VWO97386.1 Cercosporin MFS transporter encodes one protein on the biosynthetic pathway for cercosporin; other proteins are missing. This is illustrated in Figure 6.3, which shows the key biosynthetic steps in the cercosporin pathway (adapted from Newman and Townsend 2016). In this figure, Mushlabs has noted which genes were encoded in the WGS sequencing of POMM and which were not. If the enzyme that catalyzes the synthesis of any one intermediate is missing, the final product, in this case, cercosporin, cannot be produced. Mushlabs further notes that laccases that are found in *Pleurotus*. degrade cercosporin, thus highlighting again that the presence of this pathway in *Pleurotus spp.* is rather unlikely (Caesar-Thonthat *et al.* 2009; US Patent No. 6,872,388).

FIGURE 6.3 Biosynthetic pathway for Cercosporin Indicating Which Enzymatic Steps Are Present or Absent in POMM



6.2.6.2 Transcriptome Analyses

Although gene transcripts are less relevant in terms of food safety assessment than translated proteins, Mushlabs commissioned Novogene[57] to conduct a full analysis of the transcripts present in POMM. In particular, Novogene was charged to determine whether the genomic sequences highlighted by CRL in its whole genome analysis resulted in transcripts. Of the 26 loci identified in the initial whole genome analysis conducted by CRL, Novogene found only eight transcripts. These referred to putative presence of peptide IDs VWO98972.1, QIM40768.1, VWO97593.1, UFQ31923.1, VWO96292.1, VWO95044.1 and spA0A384XHA3,1STR11_STRTC (listed in Table 6.7). Of these, only four (not including ostreolysin A6) were found in the subsequent POMM proteomic analysis conducted by the University of Griefswald described in Section 6.2.6.3.

6.2.6.3 Proteome Analysis

Mushlabs conducted two types of studies related to the proteome of POMM. First, POMM biomass underwent tryptic digestion to enable the characterization of its complete proteome (the set of proteins present in POMM) and assess whether potential “toxic” or “allergenic” proteins were actually present in the biomass. Second, a series of digestion studies were performed with POMM to evaluate the potential allergenicity and toxicity of POMM based on the presence of any residual larger peptides or undigested proteins following simulated gastric digestion

[57] https://www.novogene.com/us-en/services/research-services/transcriptome-sequencing/?keyword=rna%20sequencing%20company&campaign=11002997209&gclid=CjwKCAjwp4m0BhBAEiwAsdc4aMWl6-hjvKCeTPK0iZZOpuWhgaPNngmvfNFFnfavm0cqi7dDWuBvNxoCwxUQAvD_BwE&gad_source=1 last accessed 6 September 2024

For the first study, Mushlabs contracted with the Department of Microbial Physiology and Molecular Biology of the Institute of Microbiology at the University of Greifswald (Greifswald Germany) to perform a proteomic study based on mass spectroscopy to determine whether any potentially toxic or allergenic proteins could be detected in POMM; the CRL genomics report was used as an initial guide. As previously stated, the POMM samples that were analyzed were taken from the same fermenter batches (Batches A, B, and C) that were part of the analyses described in Section 2. Proteins were extracted from the mycelial biomass and the quality of the protein extracts (no degradation, correct determination of protein concentration) was checked by 1D SDS PAGE prior to mass spectrometry sample preparation. Subsequently, the proteins were digested into tryptic peptides using S-Traps according to the manufacturer's instruction[58] to ensure efficient denaturation of proteins and maximal efficiency of digestion. Visual inspection of total ion chromatograms was done to confirm the quality of peptide mixture. Peptides were separated and analyzed by HPLC-MS/MS prior to identification using the CRL strain-specific protein sequence database that predicts potential proteins.

Although the entire proteome was evaluated (n >1000 proteins), the Mushlabs analysis focused on the peptides initially identified by CRL as of interest with respect to food safety and their degree of homology with proteins potentially involved in toxin biosynthesis pathways. Mushlabs notes that CRL had already determined that there was no risk of relevant toxin production in POMM in the course of their whole genome analysis. Therefore, the additional proteome analysis only added further support in terms of POMM safety evaluation.

Table 6.8 below summarizes the list of proteins of interest identified in POMM.

TABLE 6.8
Proteomic Analysis of POMM from the University of Greifswald

Gene ID	Peptide ID	Potential Function[a]	Number of Peptide Identifications[b]	Sequence Coverage (%)[c]	Molecular Weight (kDa)[d]
FUN_013526	KAF4577128.1	Ostreolysin A6	4	34.1	15.079
FUN_000224	VWO98972.1	ATP-dependent RNA helicase HAS1	9	20.3	88.64
FUN_005155	QIM40768.1	Prolyl oligopeptidase A	11	19.4	87.383
FUN_008881	VW096292.1	Glucose-6-phosphate isomerase	19	39.9	62.542
FUN_015464	spA0A384XHA3,1STR11_STRTC	Phenylalanine ammonia-lyase	8	7.5	125.4

a = Potential function was inferred based on UniProt (https://www.uniprot.org/) data base, which provides functional annotations of proteins based on their sequences.
b = The number of identified peptides that could be assigned to a given protein.
c = indicates the percent of the protein length that is covered by the identified proteins (previous column). Coverage is influenced by a number of factors and should not per se be considered as a proxy for the reliability of protein identification. A protein should be identified with two different peptides.
d = the anticipated molecular weight of the putatively identified protein.

[58] https://protifi.com/pages/s-trap last accessed 6 September 2024

With respect to the peptides related to hemolysins, only the ostreolysin A6 protein sequence was identified in the POMM samples. No Pleurotolysin B was detected in the digested POMM. Thus, there is no potential to form the active dimer in POMM, which requires both ostreolysin A6 and pleurotolysin B expression.

Apart from ostreolysin A6, only four other proteins from the list of peptides were found in the POMM proteome that had even minimal homology to proteins with potential roles in toxin synthesis. The four peptides are listed in Table 6.8. These proteins have been investigated in more depth using current literature and public databases (NCBI, SwissProt, PubMed). The results showed that they are not toxin biosynthesis genes. Their deduced functions were identified as follows:

- **Protein 1**: ATP-dependent RNA helicase is an enzyme involved in 40S ribosomal subunit biogenesis.
- **Protein 2**: Prolyl oligopeptidase is a serine protease that hydrolyzes proline-containing peptides < 30 amino acids long.
- **Protein 3**: Glucose-6-Isomerase is a central enzyme in the glycolytic pathway that, among other functions, converts glucose-6-phosophate to fructose-6-phospate (a precursor of fungal cell wall biosynthesis).
- **Protein 4**: Phenylalanine ammonia-lyase catalyzes the non-oxidative deamination of L-phenylalanine to form trans-cinnamic acid, a commitment step in the pathway that produces secondary metabolites such as lignins, coumarins, and flavonoids.

Proteins 1 through 4 above occur in most living organisms including mammals; they are involved in cellular functions key to energy production, RNA production, and carbon and nitrogen metabolism. Although it is possible that some toxin biosynthesis pathways may contain proteins similar to the four proteins detected in POMM, the other key components of such toxin pathways are not present in the genome of *P. pulmonarius*. This supports the conclusion that POMM does not contain proteins that would pose a food safety concern. The example of the cercosporin biosynthesis pathway has been provided in Figure 6.4 as an example of this concept because most of the loci (22 out of 26) listed in the initial whole genome anlaysis search from CRL were related to the cercosporin pathway. It is noted as well that the other four loci in the initial CRL whole genome analysis were related to Protein 3 and Protein 4, again supporting the conclusion that POMM does not contain proteins that would pose a food safety concern.

Important to the food safety considerations in the case of POMM is the fact that the food ingredient is incorporated into products that would result in heating, cooking and eventual digestion. This would be expected to lead to degradation of any proteins or peptides present in POMM, consistent with conclusions in the OECD consensus document (OECD 2006) on *Pleurotus spp: "There is [sic] no data on the toxicity of P. ostreatus hemolysins in oyster mushroom consumers. Oyster mushroom is considered a non-toxic mushroom, and the presence of hemolysins does not influence this conclusion. Hemolysins are thermo-labile, and potential toxicity would be considered only for raw mushrooms. Furthermore, proteins are usually degraded in the gastrointestinal tract when ingested"* [59]

The proteome data analysis results are supportive of the digestion data for POMM provided by Improve SAS that were presented in Section 2 of this notice. Improve SAS specializes in protein production and

[59] OECD (2006), "Section 11 - Oyster Mushroom (PLEUROTUS SPP.)", in *Safety Assessment of Transgenic Organisms, Volume 1: OECD Consensus Documents*, OECD Publishing, Paris, https://doi.org/10.1787/9789264095380-14-en. page 255.

analysis, and they were asked to evaluate the protein digestibility of POMM and the extent to which the protein present in the biomass would be digested under simulated gastric conditions. The results of these studies showed that POMM was fully (100%) digested, and any remaining residual peptides were all below 10 kDa. As already discussed, published literature suggests that most allergenic proteins have molecular weights greater than 10 kDa (Huby *et al.* 2000). Thus, in concordance with the scientific literature review on allergenicity (Section 6.1.4), the lack of proteins in POMM that can pose an allergenic risk if POMM were consumed as part of the human diet support the conclusion that POMM does not pose a food allergenicity concern.

6.3 Overall Conclusions Regarding POMM Food Safety

Sections 6.1 and 6.2 provide a detailed discussion of the information and data that support the finding that when used as a food ingredient at the incorporation levels indicated in Table 1.1, POMM is safe to eat. The key issues and information described in the notice are summarized below.

1. **Confirmation of Identity:** Mushlabs has thoroughly described the provenance of the species of *P. pulmonarius* from which POMM is derived, *i.e.*, it is cultured from a dikaryotic fruiting body to yield dikaryotic mycelia (typical fruiting body tissue).

2. **Highly Controlled Manufacturing:** All manufacture of POMM is conducted under food GLP conditions consistent with US Good Manufacturing Practices for foods. POMM is produced by submerged fermentation that precludes the further differentiation of the dikaryotic mycelia into fruiting bodies. POMM is manufactured. Further, given the tightly controlled conditions employed by Mushlabs when producing POMM compared to the less prescribed conditions for the cultivation of mushroom fruiting bodies on various lignocellulosic substrates in traditional mushroom farming, POMM would be less likely to contain substances that could be injurious to health that may be produced in response to changes to the environment of the fruiting body.

3. **Establishment of Acceptance Criteria and Specifications:** Mushlabs has established acceptance criteria based on multiple production batches. Data from three non-consecutive batches are provided herein. Acceptance criteria and specifications have been established for protein levels in POMM as well as levels of heavy metals and microbes. Any batches that do not meet the stringent acceptance criteria will be discarded. The data from the POMM batches analyzed show that Mushlabs can reliably produce POMM that meets both the acceptance criteria and the specifications.

4. **Characterization of Growth Media:** The medium in which POMM is produced is composed of simple ingredients, does not contain any of the 9 major allergens, and none of the components are derived from the nine allergens listed by FDA.

5. **Description of Intended Uses of POMM and the Conditions of Its Use:** POMM is intended to be an ingredient in food that is incorporated as an alternative source of protein, and not to comprise 100% of any food. POMM is intended to be cooked prior to consumption.

6. **Residual Nucleic Acids in POMM:** Nucleic acid levels in POMM are below 2% as recommended by FAO.

7. **Completion of Dietary Exposure Assessment:** Mushlabs performed a POMM-specific dietary exposure assessment based on use of the most recent US data (NHANES data). Daily intakes of POMM for eaters-only over two years of age were estimated to be 30.4 g/day (mean intake) and 64.3 g/day (90th percentile intake).

8. **Safety Narrative:**
    a. A safety assessment conducted using more traditional methods did not indicate the presence of any toxins, allergens, or other substances that could pose food safety concerns.
    b. To confirm that there are no substances in POMM that could be toxic or allergenic, Mushlabs conducted a series of comprehensive analyses of the genome, transcriptome, and proteome of POMM. There is no evidence that POMM contains any hazards that could pose a food safety concern.

Based on a weight-of-the-evidence of the data and information, as well as the intended uses of POMM, Mushlabs has concluded that POMM is generally recognized as safe by scientific methods when present as an ingredient in food at the maximum levels indicated in Table 1.1.

PART 7. List of Supporting Data and Information

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7.2 APPENDIX A: Data Demonstrating Identity of Phoenix Oyster Mushroom Mycelium Source Material

charles river

Processing Lab:
Detlev-Karsten-Rohwedder-Straße 7
41564 Kaarst Germany
www.criver.com/accugenix
Accugenix.labEU@crl.com

Accugenix®
AccuGENX-ID® Report
NEW-SOP-00130

Customer:	Mushlabs GmbH	Account:	605530 (MUH1)
Address:	Rosenthaler Str. 13, Berlin, berlin, 10119, Germany		
Accugenix®C# / Run Date:	C5206557-20220225127 / 2022-02-24 16:40:28	ID Request Form#:	538070
Customer Sample ID:	ML_ITS_22	Due Date:	2022-02-25

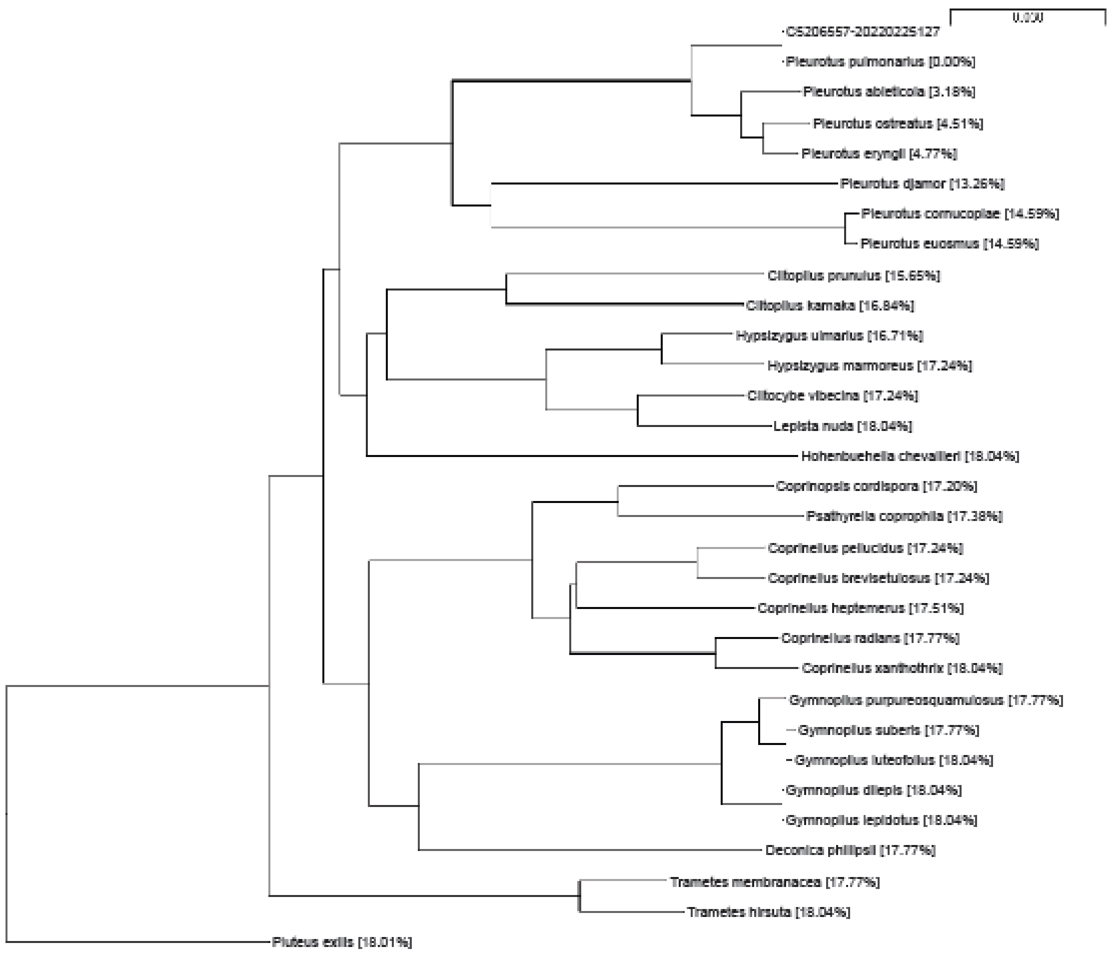
AccuGENX-ID® Database Search Result - Fungal ITS Library

Identification: *Pleurotus pulmonarius*

Confidence Level: **Species**

•Based on published literature, the above identified species has been described as a filamentous fungi. Click HERE for more information on this organism.

Neighbor Joining Tree



The value shown in the bracket represents the percent difference in the sequence alignment between the unknown and each individual library entry.

Not intended for *in vitro* diagnostic use. The results apply to the sample as received.

Rev. 08Feb2022 JDB

Prepared By Michael D. Testa at Newark,DE, United States on 2022-02-25 10:31:06
Reviewed By Robert Stevens at Newark, DE, United States on 2022-02-25 10:32:06
QA Approved By Alyssa K Ohara at Newark, DE, United States on 2022-02-25 16:14:07

This document has been signed electronically by the individual(s) shown above, in compliance with FDA regulation CFR 21 Part 11.

7.3 APPENDIX B: Proximate data for 3 additional POMM batches (D, E, and F)

The tables below provide additional data that supports the safety of POMM. These three additional POMM baches (D, E, and F) are batches from large scale automated production produced with the same production method as described in Section 2 of this notice, with the same media constituents as listed in Table 2.2 but with use of some different levels of individual media constituents. As seen below in Table E.1, changes in levels of individual media constituents can affect overall protein levels as evidenced by mean protein levels in batches D, E, and F of 50.62 g/100 g POMM as compared to a mean level of 32.2 g/100 g POMM in batches A, B, and C (see Table 2.4). As noted, all production batches met the specification set by Mushlabs for protein of 25 to 55 g protein/100 g POMM. Tables E.2 and E.3 provide evidence that the three additional batches met Mushlabs' acceptance criteria for heavy metals and microbiology.

TABLE E.1
POMM Proximate Analyses (g/100g)*

Batch	D	E	F	Mean ± SD
Protein	53.14	48.47	50.26	50.62 ± 2.36
Fiber	28.74	26.16	28.19	27.70 ± 1.36
Fat	5.07	4.80	4.80	4.89 ± 0.16
Ash	4.32	4.55	5.21	4.69 ± 0.46
*All values are based on dry weight.				

TABLE E.2
POMM Heavy Metals Analyses (mg/kg)*

Batch	D	E	F	Method of Analysis
Arsenic	<0.04	<0.04	<0.04	DIN EN 15763 mod.
Cadmium	<0.01	<0.01	<0.01	DIN EN 15763 mod.
Lead	0.03	0.00	0.02	DIN EN 15763 mod.
Mercury	<0.01	<0.01	<0.01	DIN EN 15763 mod.
*All values are based on dry weight.				

TABLE E.3
POMM Microbiology Analyses (mg/kg)*

Batch	D	E	F	Method of Analysis
Aerobic plate count	< 1000	< 1000	< 1000	DIN EN ISO 4833-1
Yeast and molds	< 1000	< 1000	< 1000	ISO 21527-2
Enterobacteriaceae	< 100	< 100	< 100	DIN EN ISO 21528-2
E.coli	< 10	< 10	< 10	DIN ISO 16649-2
Salmonella sp.	Negative	Negative	Negative	DIN 10135 – PCR
*All values are based on dry weight.				

7.4 APPENDIX C: Source Food Codes Used in the POMM Dietary Exposure Assessment

FDCD	DRXFCSD	CATEGORY
51180010	BAGEL	BAKED GOODS - BREAD
54440010	BAGEL CHIPS	BAKED GOODS - BREAD
51630000	BAGEL, MULTIGRAIN	BAKED GOODS - BREAD
51630100	BAGEL, MULTIGRAIN, WITH RAISINS	BAKED GOODS - BREAD
51501080	BAGEL, OAT BRAN	BAKED GOODS - BREAD
51404500	BAGEL, PUMPERNICKEL	BAKED GOODS - BREAD
51301700	BAGEL, WHEAT	BAKED GOODS - BREAD
51301900	BAGEL, WHEAT BRAN	BAKED GOODS - BREAD
51301800	BAGEL, WHEAT, WITH RAISINS	BAKED GOODS - BREAD
51300100	BAGEL, WHOLE GRAIN WHITE	BAKED GOODS - BREAD
51301750	BAGEL, WHOLE WHEAT	BAKED GOODS - BREAD
51301805	BAGEL, WHOLE WHEAT, WITH RAISINS	BAKED GOODS - BREAD
51180080	BAGEL, WITH FRUIT OTHER THAN RAISINS	BAKED GOODS - BREAD
51180030	BAGEL, WITH RAISINS	BAKED GOODS - BREAD
51182010	BREAD STUFFING	BAKED GOODS - BREAD
51182020	BREAD STUFFING MADE WITH EGG	BAKED GOODS - BREAD
51801010	BREAD, BARLEY	BAKED GOODS - BREAD
51801020	BREAD, BARLEY, TOASTED	BAKED GOODS - BREAD
51407010	BREAD, BLACK	BAKED GOODS - BREAD
52401000	BREAD, BOSTON BROWN	BAKED GOODS - BREAD
51300175	BREAD, CHAPPATTI OR ROTI	BAKED GOODS - BREAD
51111010	BREAD, CHEESE	BAKED GOODS - BREAD
51111040	BREAD, CHEESE, TOASTED	BAKED GOODS - BREAD
51113010	BREAD, CINNAMON	BAKED GOODS - BREAD
51113100	BREAD, CINNAMON, TOASTED	BAKED GOODS - BREAD
51105010	BREAD, CUBAN	BAKED GOODS - BREAD
51140100	BREAD, DOUGH, FRIED	BAKED GOODS - BREAD
51119010	BREAD, EGG, CHALLAH	BAKED GOODS - BREAD
51119040	BREAD, EGG, CHALLAH, TOASTED	BAKED GOODS - BREAD
51107010	BREAD, FRENCH OR VIENNA	BAKED GOODS - BREAD
51107040	BREAD, FRENCH OR VIENNA, TOASTED	BAKED GOODS - BREAD
51301540	BREAD, FRENCH OR VIENNA, WHOLE WHEAT	BAKED GOODS - BREAD
51301550	BREAD, FRENCH OR VIENNA, WHOLE WHEAT, TOASTED	BAKED GOODS - BREAD
52405010	BREAD, FRUIT	BAKED GOODS - BREAD
51808000	BREAD, GLUTEN FREE	BAKED GOODS - BREAD
51808010	BREAD, GLUTEN FREE, TOASTED	BAKED GOODS - BREAD
52408000	BREAD, IRISH SODA	BAKED GOODS - BREAD
51109010	BREAD, ITALIAN, GRECIAN, ARMENIAN	BAKED GOODS - BREAD

FDCD	DRXFCSD	CATEGORY
51109040	BREAD, ITALIAN, GRECIAN, ARMENIAN, TOASTED	BAKED GOODS - BREAD
51000180	BREAD, MADE FROM HOME RECIPE OR PURCHASED AT A BAKERY, NS AS	BAKED GOODS - BREAD
51401030	BREAD, MARBLE RYE AND PUMPERNICKEL	BAKED GOODS - BREAD
51601020	BREAD, MULTIGRAIN	BAKED GOODS - BREAD
51602020	BREAD, MULTIGRAIN, REDUCED CALORIE AND/OR HIGH FIBER, TOASTE	BAKED GOODS - BREAD
51601010	BREAD, MULTIGRAIN, TOASTED	BAKED GOODS - BREAD
51601210	BREAD, MULTIGRAIN, WITH RAISINS	BAKED GOODS - BREAD
51108100	BREAD, NAAN	BAKED GOODS - BREAD
51106010	BREAD, NATIVE, WATER, PUERTO RICAN STYLE	BAKED GOODS - BREAD
51000100	BREAD, NS AS TO MAJOR FLOUR	BAKED GOODS - BREAD
51000110	BREAD, NS AS TO MAJOR FLOUR, TOASTED	BAKED GOODS - BREAD
52403000	BREAD, NUT	BAKED GOODS - BREAD
51501040	BREAD, OAT BRAN	BAKED GOODS - BREAD
51501050	BREAD, OAT BRAN, TOASTED	BAKED GOODS - BREAD
51501010	BREAD, OATMEAL	BAKED GOODS - BREAD
51501020	BREAD, OATMEAL, TOASTED	BAKED GOODS - BREAD
51121110	BREAD, ONION	BAKED GOODS - BREAD
51300185	BREAD, PARATHA	BAKED GOODS - BREAD
51109100	BREAD, PITA	BAKED GOODS - BREAD
51301620	BREAD, PITA, WHEAT OR CRACKED WHEAT	BAKED GOODS - BREAD
51301600	BREAD, PITA, WHOLE WHEAT	BAKED GOODS - BREAD
51127010	BREAD, POTATO	BAKED GOODS - BREAD
51127020	BREAD, POTATO, TOASTED	BAKED GOODS - BREAD
51404010	BREAD, PUMPERNICKEL	BAKED GOODS - BREAD
51404020	BREAD, PUMPERNICKEL, TOASTED	BAKED GOODS - BREAD
52404060	BREAD, PUMPKIN	BAKED GOODS - BREAD
51300180	BREAD, PURI	BAKED GOODS - BREAD
51129010	BREAD, RAISIN	BAKED GOODS - BREAD
51129020	BREAD, RAISIN, TOASTED	BAKED GOODS - BREAD
51122000	BREAD, REDUCED CALORIE AND/OR HIGH FIBER, WHITE OR NFS	BAKED GOODS - BREAD
51806010	BREAD, RICE	BAKED GOODS - BREAD
51401010	BREAD, RYE	BAKED GOODS - BREAD
51401020	BREAD, RYE, TOASTED	BAKED GOODS - BREAD
51133010	BREAD, SOUR DOUGH	BAKED GOODS - BREAD
51133020	BREAD, SOUR DOUGH, TOASTED	BAKED GOODS - BREAD
51300300	BREAD, SPROUTED WHEAT	BAKED GOODS - BREAD
51300310	BREAD, SPROUTED WHEAT, TOASTED	BAKED GOODS - BREAD
51134000	BREAD, SWEET POTATO	BAKED GOODS - BREAD

FDCD	DRXFCSD	CATEGORY
51135000	BREAD, VEGETABLE	BAKED GOODS - BREAD
51301010	BREAD, WHEAT OR CRACKED WHEAT	BAKED GOODS - BREAD
51301040	BREAD, WHEAT OR CRACKED WHEAT, MADE FROM HOME RECIPE OR PURC	BAKED GOODS - BREAD
51301050	BREAD, WHEAT OR CRACKED WHEAT, MADE FROM HOME RECIPE OR PURC	BAKED GOODS - BREAD
51301510	BREAD, WHEAT OR CRACKED WHEAT, REDUCED CALORIE AND/OR HIGH F	BAKED GOODS - BREAD
51301520	BREAD, WHEAT OR CRACKED WHEAT, REDUCED CALORIE AND/OR HIGH F	BAKED GOODS - BREAD
51301020	BREAD, WHEAT OR CRACKED WHEAT, TOASTED	BAKED GOODS - BREAD
51301120	BREAD, WHEAT OR CRACKED WHEAT, WITH RAISINS	BAKED GOODS - BREAD
51301130	BREAD, WHEAT OR CRACKED WHEAT, WITH RAISINS, TOASTED	BAKED GOODS - BREAD
51101000	BREAD, WHITE	BAKED GOODS - BREAD
51102010	BREAD, WHITE WITH WHOLE WHEAT SWIRL	BAKED GOODS - BREAD
51102020	BREAD, WHITE WITH WHOLE WHEAT SWIRL, TOASTED	BAKED GOODS - BREAD
51101050	BREAD, WHITE, MADE FROM HOME RECIPE OR PURCHASED AT A BAKERY	BAKED GOODS - BREAD
51101060	BREAD, WHITE, MADE FROM HOME RECIPE OR PURCHASED AT A BAKERY	BAKED GOODS - BREAD
51101010	BREAD, WHITE, TOASTED	BAKED GOODS - BREAD
51300050	BREAD, WHOLE GRAIN WHITE	BAKED GOODS - BREAD
51300060	BREAD, WHOLE GRAIN WHITE, TOASTED	BAKED GOODS - BREAD
51300110	BREAD, WHOLE WHEAT	BAKED GOODS - BREAD
51300140	BREAD, WHOLE WHEAT, MADE FROM HOME RECIPE OR PURCHASED AT BA	BAKED GOODS - BREAD
51300150	BREAD, WHOLE WHEAT, MADE FROM HOME RECIPE OR PURCHASED AT BA	BAKED GOODS - BREAD
51300120	BREAD, WHOLE WHEAT, TOASTED	BAKED GOODS - BREAD
51300210	BREAD, WHOLE WHEAT, WITH RAISINS	BAKED GOODS - BREAD
51300220	BREAD, WHOLE WHEAT, WITH RAISINS, TOASTED	BAKED GOODS - BREAD
52407000	BREAD, ZUCCHINI	BAKED GOODS - BREAD
51808050	BREADSTICKS, HARD, GLUTEN FREE	BAKED GOODS - BREAD
51184000	BREADSTICKS, HARD, NFS	BAKED GOODS - BREAD
51184100	BREADSTICKS, HARD, REDUCED SODIUM	BAKED GOODS - BREAD
51306000	BREADSTICKS, HARD, WHOLE WHEAT	BAKED GOODS - BREAD
51183990	BREADSTICKS, NFS	BAKED GOODS - BREAD
51184210	BREADSTICKS, SOFT, FAST FOOD / RESTAURANT	BAKED GOODS - BREAD
51184220	BREADSTICKS, SOFT, FROZEN	BAKED GOODS - BREAD
51184200	BREADSTICKS, SOFT, NFS	BAKED GOODS - BREAD

FDCD	DRXFCSD	CATEGORY
51184260	BREADSTICKS, SOFT, STUFFED OR TOPPED WITH MELTED CHEESE	BAKED GOODS - BREAD
51184250	BREADSTICKS, SOFT, TOPPED WITH MELTED CHEESE	BAKED GOODS - BREAD
51184230	BREADSTICKS, SOFT, WITH PARMESAN CHEESE, FAST FOOD / RESTAUR	BAKED GOODS - BREAD
51184240	BREADSTICKS, SOFT, WITH PARMESAN CHEESE, FROM FROZEN	BAKED GOODS - BREAD
53111000	CAKE OR CUPCAKE, GINGERBREAD	BAKED GOODS - BREAD
53113000	CAKE, JELLY ROLL	BAKED GOODS - BREAD
91703010	CARAMEL, CHOCOLATE-FLAVORED ROLL	BAKED GOODS - BREAD
71930200	CASABE, CASSAVA BREAD	BAKED GOODS - BREAD
53610100	COFFEE CAKE, CRUMB OR QUICK-BREAD TYPE	BAKED GOODS - BREAD
53610170	COFFEE CAKE, CRUMB OR QUICK-BREAD TYPE, WITH FRUIT	BAKED GOODS - BREAD
53239000	COOKIE, SHORTBREAD	BAKED GOODS - BREAD
53239010	COOKIE, SHORTBREAD, REDUCED FAT	BAKED GOODS - BREAD
53239050	COOKIE, SHORTBREAD, WITH ICING OR FILLING	BAKED GOODS - BREAD
52206010	CORNBREAD MUFFIN, STICK, ROUND	BAKED GOODS - BREAD
52206060	CORNBREAD MUFFIN, STICK, ROUND, MADE FROM HOME RECIPE	BAKED GOODS - BREAD
52204000	CORNBREAD STUFFING	BAKED GOODS - BREAD
52202060	CORNBREAD, MADE FROM HOME RECIPE	BAKED GOODS - BREAD
52201000	CORNBREAD, PREPARED FROM MIX	BAKED GOODS - BREAD
54305010	CRACKERS, CRISPBREAD	BAKED GOODS - BREAD
54305020	CRACKERS, FLATBREAD	BAKED GOODS - BREAD
51108010	FOCACCIA, ITALIAN, PLAIN	BAKED GOODS - BREAD
51121025	GARLIC BREAD, FROM FAST FOOD / RESTAURANT	BAKED GOODS - BREAD
51121035	GARLIC BREAD, FROM FROZEN	BAKED GOODS - BREAD
51121015	GARLIC BREAD, NFS	BAKED GOODS - BREAD
51121065	GARLIC BREAD, WITH MELTED CHEESE, FROM FAST FOOD / RESTAURAN	BAKED GOODS - BREAD
51121075	GARLIC BREAD, WITH MELTED CHEESE, FROM FROZEN	BAKED GOODS - BREAD
51121045	GARLIC BREAD, WITH PARMESAN CHEESE, FROM FAST FOOD / RESTAUR	BAKED GOODS - BREAD
51121055	GARLIC BREAD, WITH PARMESAN CHEESE, FROM FROZEN	BAKED GOODS - BREAD
51807000	INJERA, ETHIOPIAN BREAD	BAKED GOODS - BREAD
13210110	PUDDING, BREAD	BAKED GOODS - BREAD
51000400	ROLL, BRAN, NS AS TO TYPE OF BRAN	BAKED GOODS - BREAD
51154600	ROLL, CHEESE	BAKED GOODS - BREAD

FDCD	DRXFCSD	CATEGORY
51154510	ROLL, DIET	BAKED GOODS - BREAD
51154550	ROLL, EGG BREAD	BAKED GOODS - BREAD
51155000	ROLL, FRENCH OR VIENNA	BAKED GOODS - BREAD
51156500	ROLL, GARLIC	BAKED GOODS - BREAD
51808100	ROLL, GLUTEN FREE	BAKED GOODS - BREAD
51000300	ROLL, HARD, NS AS TO MAJOR FLOUR	BAKED GOODS - BREAD
51158100	ROLL, MEXICAN, BOLILLO	BAKED GOODS - BREAD
51620000	ROLL, MULTIGRAIN	BAKED GOODS - BREAD
51000200	ROLL, NS AS TO MAJOR FLOUR	BAKED GOODS - BREAD
51420000	ROLL, RYE	BAKED GOODS - BREAD
51159000	ROLL, SOUR DOUGH	BAKED GOODS - BREAD
51160110	ROLL, SWEET, CINNAMON BUN, FROSTED	BAKED GOODS - BREAD
51160100	ROLL, SWEET, CINNAMON BUN, NO FROSTING	BAKED GOODS - BREAD
51161050	ROLL, SWEET, FROSTED	BAKED GOODS - BREAD
51160000	ROLL, SWEET, NO FROSTING	BAKED GOODS - BREAD
51161020	ROLL, SWEET, WITH FRUIT, FROSTED	BAKED GOODS - BREAD
51161030	ROLL, SWEET, WITH FRUIT, FROSTED, DIET	BAKED GOODS - BREAD
51320010	ROLL, WHEAT OR CRACKED WHEAT	BAKED GOODS - BREAD
51320060	ROLL, WHEAT OR CRACKED WHEAT, HOT DOG BUN	BAKED GOODS - BREAD
51153000	ROLL, WHITE, HARD	BAKED GOODS - BREAD
51157000	ROLL, WHITE, HOAGIE, SUBMARINE	BAKED GOODS - BREAD
51154010	ROLL, WHITE, HOT DOG BUN	BAKED GOODS - BREAD
51150000	ROLL, WHITE, SOFT	BAKED GOODS - BREAD
51320700	ROLL, WHOLE GRAIN WHITE	BAKED GOODS - BREAD
51320710	ROLL, WHOLE GRAIN WHITE, HOT DOG BUN	BAKED GOODS - BREAD
51320500	ROLL, WHOLE WHEAT	BAKED GOODS - BREAD
51320550	ROLL, WHOLE WHEAT, HOT DOG BUN	BAKED GOODS - BREAD
41310900	BEAN CHIPS	BAKED GOODS - SNACKS
54401026	CORN CHIPS, FLAVORED	BAKED GOODS - SNACKS
54401035	CORN CHIPS, FLAVORED (FRITOS)	BAKED GOODS - SNACKS
54401021	CORN CHIPS, PLAIN	BAKED GOODS - SNACKS
54401031	CORN CHIPS, PLAIN (FRITOS)	BAKED GOODS - SNACKS
54401090	CORN CHIPS, REDUCED SODIUM	BAKED GOODS - SNACKS
54440020	CRACKER CHIPS	BAKED GOODS - SNACKS
54420210	MULTIGRAIN CHIPS (SUN CHIPS)	BAKED GOODS - SNACKS
54402700	PITA CHIPS	BAKED GOODS - SNACKS
54408110	PRETZEL CHIPS, HARD, FLAVORED	BAKED GOODS - SNACKS
54408115	PRETZEL CHIPS, HARD, GLUTEN FREE	BAKED GOODS - SNACKS
54408105	PRETZEL CHIPS, HARD, PLAIN	BAKED GOODS - SNACKS

FDCD	DRXFCSD	CATEGORY
54408300	PRETZELS, HARD, CHEESE FILLED	BAKED GOODS - SNACKS
54408200	PRETZELS, HARD, CHOCOLATE COATED	BAKED GOODS - SNACKS
54408260	PRETZELS, HARD, COATED, GLUTEN FREE	BAKED GOODS - SNACKS
54408190	PRETZELS, HARD, COATED, NFS	BAKED GOODS - SNACKS
54408035	PRETZELS, HARD, FLAVORED	BAKED GOODS - SNACKS
54408082	PRETZELS, HARD, FLAVORED, GLUTEN FREE	BAKED GOODS - SNACKS
54408070	PRETZELS, HARD, MULTIGRAIN	BAKED GOODS - SNACKS
54408015	PRETZELS, HARD, NFS	BAKED GOODS - SNACKS
54408310	PRETZELS, HARD, PEANUT BUTTER FILLED	BAKED GOODS - SNACKS
54408081	PRETZELS, HARD, PLAIN, GLUTEN FREE	BAKED GOODS - SNACKS
54408017	PRETZELS, HARD, PLAIN, LIGHTLY SALTED	BAKED GOODS - SNACKS
54408016	PRETZELS, HARD, PLAIN, SALTED	BAKED GOODS - SNACKS
54408030	PRETZELS, HARD, PLAIN, UNSALTED	BAKED GOODS - SNACKS
54408210	PRETZELS, HARD, WHITE CHOCOLATE COATED	BAKED GOODS - SNACKS
54408250	PRETZELS, HARD, YOGURT COATED	BAKED GOODS - SNACKS
54408000	PRETZELS, NFS	BAKED GOODS - SNACKS
54408470	PRETZELS, SOFT, FILLED WITH CHEESE	BAKED GOODS - SNACKS
54408455	PRETZELS, SOFT, FROM FROZEN, SALTED	BAKED GOODS - SNACKS
54408456	PRETZELS, SOFT, FROM FROZEN, UNSALTED	BAKED GOODS - SNACKS
54408475	PRETZELS, SOFT, FROM SCHOOL LUNCH	BAKED GOODS - SNACKS
54408485	PRETZELS, SOFT, GLUTEN FREE	BAKED GOODS - SNACKS
54408480	PRETZELS, SOFT, MULTIGRAIN	BAKED GOODS - SNACKS
54408400	PRETZELS, SOFT, NFS	BAKED GOODS - SNACKS
54408420	PRETZELS, SOFT, READY-TO-EAT, CINNAMON SUGAR COATED	BAKED GOODS - SNACKS
54408422	PRETZELS, SOFT, READY-TO-EAT, COATED OR FLAVORED	BAKED GOODS - SNACKS
54408405	PRETZELS, SOFT, READY-TO-EAT, NFS	BAKED GOODS - SNACKS
54408410	PRETZELS, SOFT, READY-TO-EAT, SALTED, BUTTERED	BAKED GOODS - SNACKS
54408415	PRETZELS, SOFT, READY-TO-EAT, SALTED, NO BUTTER	BAKED GOODS - SNACKS
54408432	PRETZELS, SOFT, READY-TO-EAT, TOPPED WITH CHEESE	BAKED GOODS - SNACKS
54408430	PRETZELS, SOFT, READY-TO-EAT, TOPPED WITH MEAT	BAKED GOODS - SNACKS
54408411	PRETZELS, SOFT, READY-TO-EAT, UNSALTED, BUTTERED	BAKED GOODS - SNACKS
54408416	PRETZELS, SOFT, READY-TO-EAT, UNSALTED, NO BUTTER	BAKED GOODS - SNACKS
54401111	TORTILLA CHIPS, COOL RANCH FLAVOR (DORITOS)	BAKED GOODS - SNACKS
54401085	TORTILLA CHIPS, FLAVORED	BAKED GOODS - SNACKS

FDCD	DRXFCSD	CATEGORY
54401170	TORTILLA CHIPS, LOW FAT, UNSALTED	BAKED GOODS - SNACKS
54401110	TORTILLA CHIPS, NACHO CHEESE FLAVOR (DORITOS)	BAKED GOODS - SNACKS
54401112	TORTILLA CHIPS, OTHER FLAVORS (DORITOS)	BAKED GOODS - SNACKS
54401075	TORTILLA CHIPS, PLAIN	BAKED GOODS - SNACKS
54401122	TORTILLA CHIPS, REDUCED FAT, FLAVORED	BAKED GOODS - SNACKS
54401121	TORTILLA CHIPS, REDUCED FAT, PLAIN	BAKED GOODS - SNACKS
54402080	TORTILLA CHIPS, REDUCED SODIUM	BAKED GOODS - SNACKS
71220000	VEGETABLE CHIPS	BAKED GOODS - SNACKS
53803300	BABY TODDLER BISCUIT	BAKED GOODS - SWEETS
53803100	BABY TODDLER COOKIE	BAKED GOODS - SWEETS
52104200	BISCUIT WITH FRUIT	BAKED GOODS - SWEETS
52104100	BISCUIT, CHEESE	BAKED GOODS - SWEETS
52103000	BISCUIT, FROM FAST FOOD / RESTAURANT	BAKED GOODS - SWEETS
52102040	BISCUIT, FROM REFRIGERATED DOUGH	BAKED GOODS - SWEETS
52104010	BISCUIT, HOME RECIPE	BAKED GOODS - SWEETS
52101000	BISCUIT, NFS	BAKED GOODS - SWEETS
52104040	BISCUIT, WHEAT	BAKED GOODS - SWEETS
55701000	CAKE MADE WITH GLUTINOUS RICE	BAKED GOODS - SWEETS
55703000	CAKE MADE WITH GLUTINOUS RICE AND DRIED BEANS	BAKED GOODS - SWEETS
53102200	CAKE OR CUPCAKE, APPLE	BAKED GOODS - SWEETS
53102100	CAKE OR CUPCAKE, APPLESAUCE, WITHOUT ICING OR FILLING	BAKED GOODS - SWEETS
53102700	CAKE OR CUPCAKE, BANANA	BAKED GOODS - SWEETS
53102800	CAKE OR CUPCAKE, BLACK FOREST	BAKED GOODS - SWEETS
53104260	CAKE OR CUPCAKE, CARROT	BAKED GOODS - SWEETS
53104100	CAKE OR CUPCAKE, CARROT, WITHOUT ICING OR FILLING	BAKED GOODS - SWEETS
53105270	CAKE OR CUPCAKE, CHOCOLATE WITH CHOCOLATE ICING, BAKERY	BAKED GOODS - SWEETS
53105272	CAKE OR CUPCAKE, CHOCOLATE WITH CHOCOLATE ICING, FROM MIX	BAKED GOODS - SWEETS
53105262	CAKE OR CUPCAKE, CHOCOLATE WITH WHITE ICING, BAKERY	BAKED GOODS - SWEETS
53105264	CAKE OR CUPCAKE, CHOCOLATE WITH WHITE ICING, FROM MIX	BAKED GOODS - SWEETS
53105275	CAKE OR CUPCAKE, CHOCOLATE, NO ICING	BAKED GOODS - SWEETS
53104400	CAKE OR CUPCAKE, COCONUT	BAKED GOODS - SWEETS
53105300	CAKE OR CUPCAKE, GERMAN CHOCOLATE	BAKED GOODS - SWEETS
53114100	CAKE OR CUPCAKE, LEMON	BAKED GOODS - SWEETS

FDCD	DRXFCSD	CATEGORY
53114000	CAKE OR CUPCAKE, LEMON, WITHOUT ICING OR FILLING	BAKED GOODS - SWEETS
53115200	CAKE OR CUPCAKE, MARBLE	BAKED GOODS - SWEETS
53115100	CAKE OR CUPCAKE, MARBLE, WITHOUT ICING OR FILLING	BAKED GOODS - SWEETS
53100100	CAKE OR CUPCAKE, NFS	BAKED GOODS - SWEETS
53115320	CAKE OR CUPCAKE, NUT, WITH ICING OR FILLING	BAKED GOODS - SWEETS
53115310	CAKE OR CUPCAKE, NUT, WITHOUT ICING OR FILLING	BAKED GOODS - SWEETS
53115410	CAKE OR CUPCAKE, OATMEAL	BAKED GOODS - SWEETS
53115450	CAKE OR CUPCAKE, PEANUT BUTTER	BAKED GOODS - SWEETS
53116510	CAKE OR CUPCAKE, PUMPKIN	BAKED GOODS - SWEETS
53116500	CAKE OR CUPCAKE, PUMPKIN, WITHOUT ICING OR FILLING	BAKED GOODS - SWEETS
53116520	CAKE OR CUPCAKE, RED VELVET	BAKED GOODS - SWEETS
53117200	CAKE OR CUPCAKE, SPICE	BAKED GOODS - SWEETS
53117100	CAKE OR CUPCAKE, SPICE, WITHOUT ICING OR FILLING	BAKED GOODS - SWEETS
53118110	CAKE OR CUPCAKE, STRAWBERRY	BAKED GOODS - SWEETS
53121270	CAKE OR CUPCAKE, WHITE WITH CHOCOLATE ICING, BAKERY	BAKED GOODS - SWEETS
53121272	CAKE OR CUPCAKE, WHITE WITH CHOCOLATE ICING, FROM MIX	BAKED GOODS - SWEETS
53120270	CAKE OR CUPCAKE, WHITE WITH WHITE ICING, BAKERY	BAKED GOODS - SWEETS
53120272	CAKE OR CUPCAKE, WHITE WITH WHITE ICING, FROM MIX	BAKED GOODS - SWEETS
53121275	CAKE OR CUPCAKE, WHITE, NO ICING	BAKED GOODS - SWEETS
53120275	CAKE OR CUPCAKE, WHITE, WITHOUT ICING OR FILLING	BAKED GOODS - SWEETS
53124110	CAKE OR CUPCAKE, ZUCCHINI	BAKED GOODS - SWEETS
53101100	CAKE, ANGEL FOOD	BAKED GOODS - SWEETS
53101200	CAKE, ANGEL FOOD, WITH ICING OR FILLING	BAKED GOODS - SWEETS
53103000	CAKE, BOSTON CREAM PIE	BAKED GOODS - SWEETS
53105396	CAKE, CHOCOLATE, FLOURLESS	BAKED GOODS - SWEETS
53106500	CAKE, CREAM	BAKED GOODS - SWEETS
53110000	CAKE, FRUIT CAKE	BAKED GOODS - SWEETS
53119000	CAKE, PINEAPPLE, UPSIDE DOWN	BAKED GOODS - SWEETS
53116000	CAKE, POUND	BAKED GOODS - SWEETS
53116270	CAKE, POUND, CHOCOLATE	BAKED GOODS - SWEETS
53116020	CAKE, POUND, WITH ICING OR FILLING	BAKED GOODS - SWEETS
53116650	CAKE, QUEZADILLA, EL SALVADORIAN STYLE	BAKED GOODS - SWEETS

FDCD	DRXFCSD	CATEGORY
53116600	CAKE, RICE FLOUR, WITHOUT ICING OR FILLING	BAKED GOODS - SWEETS
53122080	CAKE, SHORTCAKE, BISCUIT TYPE, WITH FRUIT	BAKED GOODS - SWEETS
53122070	CAKE, SHORTCAKE, BISCUIT TYPE, WITH WHIPPED CREAM AND FRUIT	BAKED GOODS - SWEETS
53123080	CAKE, SHORTCAKE, SPONGE TYPE, WITH FRUIT	BAKED GOODS - SWEETS
53118100	CAKE, SPONGE	BAKED GOODS - SWEETS
53118200	CAKE, SPONGE, WITH ICING OR FILLING	BAKED GOODS - SWEETS
53123070	CAKE, STRAWBERRY SHORTCAKE	BAKED GOODS - SWEETS
53118500	CAKE, TORTE	BAKED GOODS - SWEETS
53118550	CAKE, TRES LECHE	BAKED GOODS - SWEETS
55501000	CHINESE PANCAKE	BAKED GOODS - SWEETS
51165000	COFFEE CAKE, YEAST TYPE	BAKED GOODS - SWEETS
53202000	COOKIE, ALMOND	BAKED GOODS - SWEETS
53240000	COOKIE, ANIMAL	BAKED GOODS - SWEETS
53240010	COOKIE, ANIMAL, WITH FROSTING OR ICING	BAKED GOODS - SWEETS
53203000	COOKIE, APPLESAUCE	BAKED GOODS - SWEETS
53205260	COOKIE, BAR, WITH CHOCOLATE	BAKED GOODS - SWEETS
53200100	COOKIE, BATTER OR DOUGH, RAW	BAKED GOODS - SWEETS
53203500	COOKIE, BISCOTTI	BAKED GOODS - SWEETS
53204860	COOKIE, BROWNIE, FAT FREE, NS AS TO ICING	BAKED GOODS - SWEETS
53204000	COOKIE, BROWNIE, NS AS TO ICING	BAKED GOODS - SWEETS
53204840	COOKIE, BROWNIE, REDUCED FAT, NS AS TO ICING	BAKED GOODS - SWEETS
53204100	COOKIE, BROWNIE, WITH ICING OR FILLING	BAKED GOODS - SWEETS
53204010	COOKIE, BROWNIE, WITHOUT ICING	BAKED GOODS - SWEETS
53241500	COOKIE, BUTTER OR SUGAR	BAKED GOODS - SWEETS
53244010	COOKIE, BUTTER OR SUGAR, WITH CHOCOLATE ICING OR FILLING	BAKED GOODS - SWEETS
53241600	COOKIE, BUTTER OR SUGAR, WITH FRUIT AND/OR NUTS	BAKED GOODS - SWEETS
53244020	COOKIE, BUTTER OR SUGAR, WITH ICING OR FILLING OTHER THAN CH	BAKED GOODS - SWEETS
53205250	COOKIE, BUTTERSCOTCH, BROWNIE	BAKED GOODS - SWEETS
53206000	COOKIE, CHOCOLATE CHIP	BAKED GOODS - SWEETS
53206020	COOKIE, CHOCOLATE CHIP, MADE FROM HOME RECIPE OR PURCHASED A	BAKED GOODS - SWEETS
53206030	COOKIE, CHOCOLATE CHIP, REDUCED FAT	BAKED GOODS - SWEETS
53260030	COOKIE, CHOCOLATE CHIP, SUGAR FREE	BAKED GOODS - SWEETS
53207000	COOKIE, CHOCOLATE OR FUDGE	BAKED GOODS - SWEETS
53207020	COOKIE, CHOCOLATE OR FUDGE, REDUCED FAT	BAKED GOODS - SWEETS
53210000	COOKIE, CHOCOLATE WAFER	BAKED GOODS - SWEETS

FDCD	DRXFCSD	CATEGORY
53207050	COOKIE, CHOCOLATE, WITH CHOCOLATE FILLING OR COATING, FAT FR	BAKED GOODS - SWEETS
53209005	COOKIE, CHOCOLATE, WITH ICING OR COATING	BAKED GOODS - SWEETS
53215500	COOKIE, COCONUT	BAKED GOODS - SWEETS
53220030	COOKIE, FIG BAR	BAKED GOODS - SWEETS
53220040	COOKIE, FIG BAR, FAT FREE	BAKED GOODS - SWEETS
53222010	COOKIE, FORTUNE	BAKED GOODS - SWEETS
53803050	COOKIE, FRUIT, BABY FOOD	BAKED GOODS - SWEETS
53220000	COOKIE, FRUIT-FILLED BAR	BAKED GOODS - SWEETS
53223000	COOKIE, GINGERSNAPS	BAKED GOODS - SWEETS
53261000	COOKIE, GLUTEN FREE	BAKED GOODS - SWEETS
53224000	COOKIE, LADYFINGER	BAKED GOODS - SWEETS
53224250	COOKIE, LEMON BAR	BAKED GOODS - SWEETS
53225000	COOKIE, MACAROON	BAKED GOODS - SWEETS
53208200	COOKIE, MARSHMALLOW PIE, CHOCOLATE COVERED	BAKED GOODS - SWEETS
53208000	COOKIE, MARSHMALLOW, CHOCOLATE-COVERED	BAKED GOODS - SWEETS
53228000	COOKIE, MERINGUE	BAKED GOODS - SWEETS
53230000	COOKIE, MOLASSES	BAKED GOODS - SWEETS
53201000	COOKIE, NFS	BAKED GOODS - SWEETS
53233000	COOKIE, OATMEAL	BAKED GOODS - SWEETS
53233040	COOKIE, OATMEAL, REDUCED FAT, NS AS TO RAISINS	BAKED GOODS - SWEETS
53260200	COOKIE, OATMEAL, SUGAR FREE	BAKED GOODS - SWEETS
53233100	COOKIE, OATMEAL, WITH CHOCOLATE AND PEANUT BUTTER, NO BAKE	BAKED GOODS - SWEETS
53233010	COOKIE, OATMEAL, WITH RAISINS	BAKED GOODS - SWEETS
53234000	COOKIE, PEANUT BUTTER	BAKED GOODS - SWEETS
53260600	COOKIE, PEANUT BUTTER, SUGAR FREE	BAKED GOODS - SWEETS
53234100	COOKIE, PEANUT BUTTER, WITH CHOCOLATE	BAKED GOODS - SWEETS
53236000	COOKIE, PIZZELLE	BAKED GOODS - SWEETS
53236100	COOKIE, PUMPKIN	BAKED GOODS - SWEETS
53237000	COOKIE, RAISIN	BAKED GOODS - SWEETS
53251100	COOKIE, RUGELACH	BAKED GOODS - SWEETS
53260400	COOKIE, SUGAR OR PLAIN, SUGAR FREE	BAKED GOODS - SWEETS
53242000	COOKIE, SUGAR WAFER	BAKED GOODS - SWEETS
53209010	COOKIE, SUGAR WAFER, CHOCOLATE-COVERED	BAKED GOODS - SWEETS
53260500	COOKIE, SUGAR WAFER, SUGAR FREE	BAKED GOODS - SWEETS
53246000	COOKIE, TEA, JAPANESE	BAKED GOODS - SWEETS
53803250	COOKIE, TEETHING, BABY	BAKED GOODS - SWEETS
53247000	COOKIE, VANILLA WAFER	BAKED GOODS - SWEETS

FDCD	DRXFCSD	CATEGORY
53247050	COOKIE, VANILLA WAFER, REDUCED FAT	BAKED GOODS - SWEETS
53247500	COOKIE, VANILLA WITH CARAMEL, COCONUT, AND CHOCOLATE COATING	BAKED GOODS - SWEETS
53235500	COOKIE, WITH PEANUT BUTTER FILLING, CHOCOLATE-COATED	BAKED GOODS - SWEETS
53270100	COOKIES, PUERTO RICAN STYLE	BAKED GOODS - SWEETS
54103000	CRACKERS, BREAKFAST BISCUIT	BAKED GOODS - SWEETS
53520170	DOUGHNUT HOLES	BAKED GOODS - SWEETS
53520140	DOUGHNUT, CAKE TYPE, CHOCOLATE ICING	BAKED GOODS - SWEETS
53520100	DOUGHNUT, CAKE TYPE, PLAIN	BAKED GOODS - SWEETS
53520130	DOUGHNUT, CAKE TYPE, POWDERED SUGAR	BAKED GOODS - SWEETS
53520135	DOUGHNUT, CAKE TYPE, WITH ICING	BAKED GOODS - SWEETS
53520120	DOUGHNUT, CHOCOLATE	BAKED GOODS - SWEETS
53520160	DOUGHNUT, CHOCOLATE, WITH CHOCOLATE ICING	BAKED GOODS - SWEETS
53521210	DOUGHNUT, CUSTARD-FILLED	BAKED GOODS - SWEETS
53521230	DOUGHNUT, CUSTARD-FILLED, WITH ICING	BAKED GOODS - SWEETS
53521140	DOUGHNUT, JELLY	BAKED GOODS - SWEETS
53520000	DOUGHNUT, NFS	BAKED GOODS - SWEETS
53521110	DOUGHNUT, YEAST TYPE	BAKED GOODS - SWEETS
53521130	DOUGHNUT, YEAST TYPE, WITH CHOCOLATE ICING	BAKED GOODS - SWEETS
55301031	FRENCH TOAST STICKS	BAKED GOODS - SWEETS
55301040	FRENCH TOAST STICKS, FAST FOOD / RESTAURANT	BAKED GOODS - SWEETS
55301030	FRENCH TOAST STICKS, NFS	BAKED GOODS - SWEETS
55301050	FRENCH TOAST STICKS, PLAIN	BAKED GOODS - SWEETS
55301048	FRENCH TOAST STICKS, SCHOOL	BAKED GOODS - SWEETS
55300050	FRENCH TOAST, FAST FOOD / RESTAURANT	BAKED GOODS - SWEETS
55300020	FRENCH TOAST, FROZEN	BAKED GOODS - SWEETS
55301025	FRENCH TOAST, GLUTEN FREE	BAKED GOODS - SWEETS
55300010	FRENCH TOAST, NFS	BAKED GOODS - SWEETS
55301000	FRENCH TOAST, PLAIN	BAKED GOODS - SWEETS
55300060	FRENCH TOAST, SCHOOL	BAKED GOODS - SWEETS
55301015	FRENCH TOAST, WHOLE GRAIN	BAKED GOODS - SWEETS
55300055	FRENCH TOAST, WHOLE GRAIN, FROM FAST FOOD / RESTAURANT	BAKED GOODS - SWEETS
55801000	FUNNEL CAKE WITH SUGAR	BAKED GOODS - SWEETS
55801010	FUNNEL CAKE WITH SUGAR AND FRUIT	BAKED GOODS - SWEETS
52211010	JOHNNYCAKE	BAKED GOODS - SWEETS
53241510	MARIE BISCUIT	BAKED GOODS - SWEETS
52304040	MUFFIN, BRAN WITH FRUIT, LOWFAT	BAKED GOODS - SWEETS

FDCD	DRXFCSD	CATEGORY
52302600	MUFFIN, CHOCOLATE	BAKED GOODS - SWEETS
52302500	MUFFIN, CHOCOLATE CHIP	BAKED GOODS - SWEETS
51186010	MUFFIN, ENGLISH	BAKED GOODS - SWEETS
51186130	MUFFIN, ENGLISH, CHEESE	BAKED GOODS - SWEETS
51630200	MUFFIN, ENGLISH, MULTIGRAIN	BAKED GOODS - SWEETS
51404550	MUFFIN, ENGLISH, PUMPERNICKEL	BAKED GOODS - SWEETS
51302500	MUFFIN, ENGLISH, WHEAT BRAN	BAKED GOODS - SWEETS
51303010	MUFFIN, ENGLISH, WHEAT OR CRACKED WHEAT	BAKED GOODS - SWEETS
51303050	MUFFIN, ENGLISH, WHEAT OR CRACKED WHEAT, WITH RAISINS	BAKED GOODS - SWEETS
51303100	MUFFIN, ENGLISH, WHOLE GRAIN WHITE	BAKED GOODS - SWEETS
51303030	MUFFIN, ENGLISH, WHOLE WHEAT	BAKED GOODS - SWEETS
51303070	MUFFIN, ENGLISH, WHOLE WHEAT, WITH RAISINS	BAKED GOODS - SWEETS
51186160	MUFFIN, ENGLISH, WITH FRUIT OTHER THAN RAISINS	BAKED GOODS - SWEETS
51186100	MUFFIN, ENGLISH, WITH RAISINS	BAKED GOODS - SWEETS
52302010	MUFFIN, FRUIT	BAKED GOODS - SWEETS
52302020	MUFFIN, FRUIT, LOW FAT	BAKED GOODS - SWEETS
52301000	MUFFIN, NFS	BAKED GOODS - SWEETS
52304150	MUFFIN, OAT BRAN	BAKED GOODS - SWEETS
52304100	MUFFIN, OATMEAL	BAKED GOODS - SWEETS
52306010	MUFFIN, PLAIN	BAKED GOODS - SWEETS
52306500	MUFFIN, PUMPKIN	BAKED GOODS - SWEETS
52303500	MUFFIN, WHEAT	BAKED GOODS - SWEETS
52304010	MUFFIN, WHEAT BRAN	BAKED GOODS - SWEETS
52303010	MUFFIN, WHOLE WHEAT	BAKED GOODS - SWEETS
52306550	MUFFIN, ZUCCHINI	BAKED GOODS - SWEETS
55105000	PANCAKES, BUCKWHEAT	BAKED GOODS - SWEETS
55103100	PANCAKES, CHOCOLATE	BAKED GOODS - SWEETS
55100060	PANCAKES, CHOCOLATE, FAST FOOD / RESTAURANT	BAKED GOODS - SWEETS
55100025	PANCAKES, CHOCOLATE, FROZEN	BAKED GOODS - SWEETS
55105100	PANCAKES, CORNMEAL	BAKED GOODS - SWEETS
55103000	PANCAKES, FRUIT	BAKED GOODS - SWEETS
55100055	PANCAKES, FRUIT, FAST FOOD / RESTAURANT	BAKED GOODS - SWEETS
55100020	PANCAKES, FRUIT, FROZEN	BAKED GOODS - SWEETS
55106000	PANCAKES, GLUTEN FREE	BAKED GOODS - SWEETS
55100040	PANCAKES, GLUTEN FREE, FROM FROZEN	BAKED GOODS - SWEETS
55100005	PANCAKES, NFS	BAKED GOODS - SWEETS
55101000	PANCAKES, PLAIN	BAKED GOODS - SWEETS
55100050	PANCAKES, PLAIN, FAST FOOD / RESTAURANT	BAKED GOODS - SWEETS

FDCD	DRXFCSD	CATEGORY
55100010	PANCAKES, PLAIN, FROZEN	BAKED GOODS - SWEETS
55101015	PANCAKES, PLAIN, REDUCED FAT	BAKED GOODS - SWEETS
55103020	PANCAKES, PUMPKIN	BAKED GOODS - SWEETS
55100080	PANCAKES, SCHOOL	BAKED GOODS - SWEETS
55105200	PANCAKES, WHOLE GRAIN	BAKED GOODS - SWEETS
55100065	PANCAKES, WHOLE GRAIN, FAST FOOD / RESTAURANT	BAKED GOODS - SWEETS
55100030	PANCAKES, WHOLE GRAIN, FROZEN	BAKED GOODS - SWEETS
55105205	PANCAKES, WHOLE GRAIN, REDUCED FAT	BAKED GOODS - SWEETS
53391000	PIE SHELL	BAKED GOODS - SWEETS
53301000	PIE, APPLE	BAKED GOODS - SWEETS
53301070	PIE, APPLE, FAST FOOD	BAKED GOODS - SWEETS
53301080	PIE, APPLE, FRIED PIE	BAKED GOODS - SWEETS
53301500	PIE, APPLE, ONE CRUST	BAKED GOODS - SWEETS
53341000	PIE, BANANA CREAM	BAKED GOODS - SWEETS
53303500	PIE, BERRY	BAKED GOODS - SWEETS
53303510	PIE, BERRY, NOT BLACKBERRY, BLUEBERRY, BOYSENBERRY, HUCKLEBE	BAKED GOODS - SWEETS
53303570	PIE, BERRY, NOT BLACKBERRY, BLUEBERRY, BOYSENBERRY, HUCKLEBE	BAKED GOODS - SWEETS
53303000	PIE, BLACKBERRY, TWO CRUST	BAKED GOODS - SWEETS
53304000	PIE, BLUEBERRY	BAKED GOODS - SWEETS
53341500	PIE, BUTTERMILK	BAKED GOODS - SWEETS
53305000	PIE, CHERRY	BAKED GOODS - SWEETS
53305070	PIE, CHERRY, INDIVIDUAL SIZE OR TART	BAKED GOODS - SWEETS
53305010	PIE, CHERRY, ONE CRUST	BAKED GOODS - SWEETS
53341750	PIE, CHESS	BAKED GOODS - SWEETS
53342000	PIE, CHOCOLATE CREAM	BAKED GOODS - SWEETS
53343000	PIE, COCONUT CREAM	BAKED GOODS - SWEETS
53344000	PIE, CUSTARD	BAKED GOODS - SWEETS
53344070	PIE, CUSTARD, INDIVIDUAL SIZE OR TART	BAKED GOODS - SWEETS
53300170	PIE, INDIVIDUAL SIZE OR TART, NFS	BAKED GOODS - SWEETS
53305800	PIE, KEY LIME	BAKED GOODS - SWEETS
53305700	PIE, LEMON	BAKED GOODS - SWEETS
53381000	PIE, LEMON MERINGUE	BAKED GOODS - SWEETS
53381070	PIE, LEMON MERINGUE, INDIVIDUAL SIZE OR TART	BAKED GOODS - SWEETS
53300100	PIE, NFS	BAKED GOODS - SWEETS
53385500	PIE, OATMEAL	BAKED GOODS - SWEETS
53307000	PIE, PEACH	BAKED GOODS - SWEETS
53307070	PIE, PEACH, INDIVIDUAL SIZE OR TART	BAKED GOODS - SWEETS

FDCD	DRXFCSD	CATEGORY
53307050	PIE, PEACH, ONE CRUST	BAKED GOODS - SWEETS
53346000	PIE, PEANUT BUTTER CREAM	BAKED GOODS - SWEETS
53385000	PIE, PECAN	BAKED GOODS - SWEETS
53346500	PIE, PINEAPPLE CREAM	BAKED GOODS - SWEETS
53308000	PIE, PINEAPPLE, TWO CRUST	BAKED GOODS - SWEETS
53386000	PIE, PUDDING, FLAVORS OTHER THAN CHOCOLATE	BAKED GOODS - SWEETS
53347000	PIE, PUMPKIN	BAKED GOODS - SWEETS
53310000	PIE, RASPBERRY, ONE CRUST	BAKED GOODS - SWEETS
53310050	PIE, RASPBERRY, TWO CRUST	BAKED GOODS - SWEETS
53311000	PIE, RHUBARB, TWO CRUST	BAKED GOODS - SWEETS
53312000	PIE, STRAWBERRY	BAKED GOODS - SWEETS
53348070	PIE, STRAWBERRY CREAM, INDIVIDUAL SIZE OR TART	BAKED GOODS - SWEETS
53313000	PIE, STRAWBERRY-RHUBARB, TWO CRUST	BAKED GOODS - SWEETS
53360000	PIE, SWEET POTATO	BAKED GOODS - SWEETS
53387000	PIE, TOLL HOUSE CHOCOLATE CHIP	BAKED GOODS - SWEETS
53365000	PIE, VANILLA CREAM	BAKED GOODS - SWEETS
53118410	RUM CAKE, WITHOUT ICING	BAKED GOODS - SWEETS
52105100	SCONE	BAKED GOODS - SWEETS
52105200	SCONE, WITH FRUIT	BAKED GOODS - SWEETS
53108200	SNACK CAKE, CHOCOLATE	BAKED GOODS - SWEETS
53109220	SNACK CAKE, NOT CHOCOLATE, WITH ICING OR FILLING, REDUCED FA	BAKED GOODS - SWEETS
53109200	SNACK CAKE, WHITE	BAKED GOODS - SWEETS
55203600	WAFFLE, CHOCOLATE	BAKED GOODS - SWEETS
55200110	WAFFLE, CHOCOLATE, FAST FOOD / RESTAURANT	BAKED GOODS - SWEETS
55200050	WAFFLE, CHOCOLATE, FROZEN	BAKED GOODS - SWEETS
55203700	WAFFLE, CINNAMON	BAKED GOODS - SWEETS
55204000	WAFFLE, CORNMEAL	BAKED GOODS - SWEETS
55203000	WAFFLE, FRUIT	BAKED GOODS - SWEETS
55200120	WAFFLE, FRUIT, FAST FOOD / RESTAURANT	BAKED GOODS - SWEETS
55200040	WAFFLE, FRUIT, FROZEN	BAKED GOODS - SWEETS
55208000	WAFFLE, GLUTEN FREE	BAKED GOODS - SWEETS
55200090	WAFFLE, GLUTEN FREE, FROM FROZEN	BAKED GOODS - SWEETS
55200010	WAFFLE, NFS	BAKED GOODS - SWEETS
55201000	WAFFLE, PLAIN	BAKED GOODS - SWEETS
55200100	WAFFLE, PLAIN, FAST FOOD / RESTAURANT	BAKED GOODS - SWEETS
55200020	WAFFLE, PLAIN, FROZEN	BAKED GOODS - SWEETS
55200030	WAFFLE, PLAIN, REDUCED FAT	BAKED GOODS - SWEETS
55211050	WAFFLE, PLAIN, REDUCED FAT	BAKED GOODS - SWEETS

FDCD	DRXFCSD	CATEGORY
55200200	WAFFLE, SCHOOL	BAKED GOODS - SWEETS
55205000	WAFFLE, WHOLE GRAIN	BAKED GOODS - SWEETS
55200130	WAFFLE, WHOLE GRAIN, FAST FOOD / RESTAURANT	BAKED GOODS - SWEETS
55200060	WAFFLE, WHOLE GRAIN, FROZEN	BAKED GOODS - SWEETS
55200080	WAFFLE, WHOLE GRAIN, FRUIT, FROZEN	BAKED GOODS - SWEETS
55200070	WAFFLE, WHOLE GRAIN, REDUCED FAT	BAKED GOODS - SWEETS
55212000	WAFFLE, WHOLE GRAIN, REDUCED FAT	BAKED GOODS - SWEETS
26155110	FISH, TUNA, CANNED	CANNED TUNA
26155180	TUNA, CANNED, OIL PACK	CANNED TUNA
53222020	COOKIE, CONE SHELL, ICE CREAM TYPE, WAFER OR CAKE	ICE CREAM
13120140	ICE CREAM BAR, CHOCOLATE	ICE CREAM
13120050	ICE CREAM BAR, VANILLA	ICE CREAM
13120100	ICE CREAM BAR, VANILLA, CHOCOLATE COATED	ICE CREAM
53112100	ICE CREAM CAKE	ICE CREAM
13120110	ICE CREAM CANDY BAR	ICE CREAM
13120792	ICE CREAM CONE, CHOCOLATE, PREPACKAGED	ICE CREAM
13120740	ICE CREAM CONE, NFS	ICE CREAM
13120770	ICE CREAM CONE, SCOOPED, CHOCOLATE	ICE CREAM
13120775	ICE CREAM CONE, SCOOPED, CHOCOLATE, WAFFLE CONE	ICE CREAM
13120730	ICE CREAM CONE, SCOOPED, VANILLA	ICE CREAM
13120735	ICE CREAM CONE, SCOOPED, VANILLA, WAFFLE CONE	ICE CREAM
13120784	ICE CREAM CONE, SOFT SERVE, CHOCOLATE	ICE CREAM
13120788	ICE CREAM CONE, SOFT SERVE, CHOCOLATE, WAFFLE CONE	ICE CREAM
13120782	ICE CREAM CONE, SOFT SERVE, VANILLA	ICE CREAM
13120786	ICE CREAM CONE, SOFT SERVE, VANILLA, WAFFLE CONE	ICE CREAM
13120790	ICE CREAM CONE, VANILLA, PREPACKAGED	ICE CREAM
13120550	ICE CREAM COOKIE SANDWICH	ICE CREAM
13120510	ICE CREAM SANDWICH, CHOCOLATE	ICE CREAM
13120500	ICE CREAM SANDWICH, VANILLA	ICE CREAM
13120810	ICE CREAM SODA, CHOCOLATE	ICE CREAM
13120800	ICE CREAM SODA, FLAVORS OTHER THAN CHOCOLATE	ICE CREAM
13121400	ICE CREAM SUNDAE, CARAMEL TOPPING	ICE CREAM
13121100	ICE CREAM SUNDAE, FRUIT TOPPING	ICE CREAM
13121300	ICE CREAM SUNDAE, HOT FUDGE TOPPING	ICE CREAM
13121000	ICE CREAM SUNDAE, NFS	ICE CREAM

FDCD	DRXFCSD	CATEGORY
13110110	ICE CREAM, CHOCOLATE	ICE CREAM
13110112	ICE CREAM, CHOCOLATE, WITH ADDITIONAL INGREDIENTS	ICE CREAM
13110000	ICE CREAM, NFS	ICE CREAM
13110210	ICE CREAM, SOFT SERVE, CHOCOLATE	ICE CREAM
13110200	ICE CREAM, SOFT SERVE, VANILLA	ICE CREAM
13110100	ICE CREAM, VANILLA	ICE CREAM
13110102	ICE CREAM, VANILLA, WITH ADDITIONAL INGREDIENTS	ICE CREAM
13140115	LIGHT ICE CREAM BAR, CHOCOLATE	ICE CREAM
13140000	LIGHT ICE CREAM BAR, VANILLA	ICE CREAM
13140100	LIGHT ICE CREAM BAR, VANILLA, CHOCOLATE COATED	ICE CREAM
13142110	LIGHT ICE CREAM CONE, CHOCOLATE, PREPACKAGED	ICE CREAM
13142100	LIGHT ICE CREAM CONE, VANILLA, PREPACKAGED	ICE CREAM
13135010	LIGHT ICE CREAM SANDWICH, CHOCOLATE	ICE CREAM
13135000	LIGHT ICE CREAM SANDWICH, VANILLA	ICE CREAM
13130310	LIGHT ICE CREAM, CHOCOLATE	ICE CREAM
13130100	LIGHT ICE CREAM, NFS	ICE CREAM
13130300	LIGHT ICE CREAM, VANILLA	ICE CREAM
41810200	BACON STRIP, MEATLESS	MEAT SUBSTITUTES
41810400	BREAKFAST LINK, PATTIE, OR SLICE, MEATLESS	MEAT SUBSTITUTES
41810610	CHICKEN, MEATLESS, BREADED, FRIED	MEAT SUBSTITUTES
41810600	CHICKEN, MEATLESS, NFS	MEAT SUBSTITUTES
42204100	GRAVY, VEGETARIAN	MEAT SUBSTITUTES
27564420	HOT DOG SANDWICH, VEGETARIAN, ON BUN	MEAT SUBSTITUTES
41811400	HOT DOG, VEGETARIAN	MEAT SUBSTITUTES
41811600	LUNCHEON SLICE, MEATLESS-BEEF, CHICKEN, SALAMI OR TURKEY	MEAT SUBSTITUTES
59003000	MEAT SUBSTITUTE, CEREAL- AND VEGETABLE PROTEIN-BASED, FRIED	MEAT SUBSTITUTES
41811800	MEATBALL, MEATLESS	MEAT SUBSTITUTES
41811950	SWISS STEAK, WITH GRAVY, MEATLESS	MEAT SUBSTITUTES
41812450	VEGETARIAN CHILI, MADE WITH MEAT SUBSTITUTE	MEAT SUBSTITUTES
92101960	COFFEE, CAFE MOCHA, WITH NON-DAIRY MILK	NON-DAIRY MILK - IN PREP
92161002	COFFEE, CAPPUCCINO, WITH NON-DAIRY MILK	NON-DAIRY MILK - IN PREP
92102502	COFFEE, ICED LATTE, WITH NON-DAIRY MILK	NON-DAIRY MILK - IN PREP
92102505	COFFEE, ICED LATTE, WITH NON-DAIRY MILK, FLAVORED	NON-DAIRY MILK - IN PREP
92101903	COFFEE, LATTE, WITH NON-DAIRY MILK	NON-DAIRY MILK - IN PREP

FDCD	DRXFCSD	CATEGORY
92101906	COFFEE, LATTE, WITH NON-DAIRY MILK, FLAVORED	NON-DAIRY MILK - IN PREP
56207102	CREAM OF WHEAT, INSTANT, MADE WITH NON-DAIRY MILK, NO ADDED	NON-DAIRY MILK - IN PREP
92101923	FROZEN COFFEE DRINK, WITH NON-DAIRY MILK	NON-DAIRY MILK - IN PREP
56201360	GRITS, INSTANT, MADE WITH NON-DAIRY MILK, FAT ADDED	NON-DAIRY MILK - IN PREP
56203107	OATMEAL, INSTANT, PLAIN, MADE WITH NON-DAIRY MILK, FAT ADDED	NON-DAIRY MILK - IN PREP
56203106	OATMEAL, INSTANT, PLAIN, MADE WITH NON-DAIRY MILK, NO ADDED	NON-DAIRY MILK - IN PREP
56203077	OATMEAL, REGULAR OR QUICK, MADE WITH NON-DAIRY MILK, FAT ADD	NON-DAIRY MILK - IN PREP
56203076	OATMEAL, REGULAR OR QUICK, MADE WITH NON-DAIRY MILK, NO ADDE	NON-DAIRY MILK - IN PREP
56203075	OATMEAL, REGULAR OR QUICK, MADE WITH NON-DAIRY MILK, NS AS T	NON-DAIRY MILK - IN PREP
58161200	RICE, COOKED WITH COCONUT MILK	NON-DAIRY MILK - IN PREP
11350000	ALMOND MILK, SWEETENED	NON-DAIRY MILK - LIQUID
11350010	ALMOND MILK, SWEETENED, CHOCOLATE	NON-DAIRY MILK - LIQUID
11350020	ALMOND MILK, UNSWEETENED	NON-DAIRY MILK - LIQUID
11350030	ALMOND MILK, UNSWEETENED, CHOCOLATE	NON-DAIRY MILK - LIQUID
11513310	CHOCOLATE MILK, MADE FROM DRY MIX WITH NON-DAIRY MILK	NON-DAIRY MILK - LIQUID
11513750	CHOCOLATE MILK, MADE FROM SYRUP WITH NON-DAIRY MILK	NON-DAIRY MILK - LIQUID
11370000	COCONUT MILK	NON-DAIRY MILK - LIQUID
42401010	COCONUT MILK, USED IN COOKING	NON-DAIRY MILK - LIQUID
11514150	HOT CHOCOLATE / COCOA, MADE WITH DRY MIX AND NON-DAIRY MILK	NON-DAIRY MILK - LIQUID
11300100	NON-DAIRY MILK, NFS	NON-DAIRY MILK - LIQUID
11360000	RICE MILK	NON-DAIRY MILK - LIQUID
11320000	SOY MILK	NON-DAIRY MILK - LIQUID
11321000	SOY MILK, CHOCOLATE	NON-DAIRY MILK - LIQUID
11320100	SOY MILK, LIGHT	NON-DAIRY MILK - LIQUID
11321100	SOY MILK, LIGHT, CHOCOLATE	NON-DAIRY MILK - LIQUID
11320200	SOY MILK, NONFAT	NON-DAIRY MILK - LIQUID
42401100	YOGURT, COCONUT MILK	NON-DAIRY YOGURT
41420380	YOGURT, SOY	NON-DAIRY YOGURT
12210520	COFFEE CREAMER, SOY, LIQUID	OTHER - COFFEE CREAMER, SOY
14502000	IMITATION CHEESE	OTHER - IMITATION CHEESE
27450130	CRAB SALAD MADE WITH IMITATION CRAB	OTHER - IMITATION CRAB

FDCD	DRXFCSD	CATEGORY
27250520	IMITATION CRAB MEAT	OTHER - IMITATION CRAB
57316385	CEREAL (GENERAL MILLS CHEERIOS PROTEIN)	OTHER - MISC PROTEIN ADDED ITEMS
78101110	FRUIT AND VEGETABLE SMOOTHIE, ADDED PROTEIN	OTHER - MISC PROTEIN ADDED ITEMS
78101118	FRUIT AND VEGETABLE SMOOTHIE, NON-DAIRY, ADDED PROTEIN	OTHER - MISC PROTEIN ADDED ITEMS
11553120	FRUIT SMOOTHIE, WITH WHOLE FRUIT AND DAIRY, ADDED PROTEIN	OTHER - MISC PROTEIN ADDED ITEMS
64134020	FRUIT SMOOTHIE, WITH WHOLE FRUIT, NO DAIRY, ADDED PROTEIN	OTHER - MISC PROTEIN ADDED ITEMS
53720500	NUTRITION BAR (SNICKERS MARATHON PROTEIN BAR)	OTHER - MISC PROTEIN ADDED ITEMS
53720610	NUTRITION BAR (SOUTH BEACH LIVING HIGH PROTEIN BAR)	OTHER - MISC PROTEIN ADDED ITEMS
95120020	NUTRITIONAL DRINK OR SHAKE, HIGH PROTEIN, LIGHT, READY-TO-DR	OTHER - MISC PROTEIN ADDED ITEMS
95110020	NUTRITIONAL DRINK OR SHAKE, HIGH PROTEIN, READY-TO-DRINK (SL	OTHER - MISC PROTEIN ADDED ITEMS
95120010	NUTRITIONAL DRINK OR SHAKE, HIGH PROTEIN, READY-TO-DRINK, NF	OTHER - MISC PROTEIN ADDED ITEMS
95201200	NUTRITIONAL POWDER MIX (EAS WHEY PROTEIN POWDER)	OTHER - MISC PROTEIN ADDED ITEMS
95201700	NUTRITIONAL POWDER MIX (KELLOGG'S SPECIAL K20 PROTEIN WATER)	OTHER - MISC PROTEIN ADDED ITEMS
95201500	NUTRITIONAL POWDER MIX, HIGH PROTEIN (HERBALIFE)	OTHER - MISC PROTEIN ADDED ITEMS
95210020	NUTRITIONAL POWDER MIX, HIGH PROTEIN (SLIM FAST)	OTHER - MISC PROTEIN ADDED ITEMS
95220010	NUTRITIONAL POWDER MIX, HIGH PROTEIN, NFS	OTHER - MISC PROTEIN ADDED ITEMS
95230020	NUTRITIONAL POWDER MIX, PROTEIN, LIGHT, NFS	OTHER - MISC PROTEIN ADDED ITEMS
95230030	NUTRITIONAL POWDER MIX, PROTEIN, NFS	OTHER - MISC PROTEIN ADDED ITEMS
95230010	NUTRITIONAL POWDER MIX, PROTEIN, SOY BASED, NFS	OTHER - MISC PROTEIN ADDED ITEMS
41440000	TEXTURED VEGETABLE PROTEIN, DRY	OTHER - MISC PROTEIN ADDED ITEMS

7.5 APPENDIX D: Summary of Published Studies Relevant to *P. pulmonarius* Food Safety

Summary of Published Studies with Data or Information Relevant to the Food Safety of *P. pulmonarious*				
Study Title (reference)	**Study Design**	**Type of Test Article**	**Outcomes/ Endpoints Measured**	**Reported Results**
In Vitro Studies				
Antiatherogenic Potential of Extracts from the Gray Oyster Medicinal Mushroom, *Pleurotus pulmonarius* (Agaricomycetes), *In Vitro* (Abidin, M.H.Z. *et al.* 2018. *Int. J. Medic. Mushrooms.* 20(3):283–290)	*In vitro* study of potential for affecting atherogenic mechanisms (same lab as the 2016 study below) Different mushroom extracts were tested versus a control (water) Test concentrations=10 mg extract/ml	Extracts of *P. pulmonarius* fruiting bodies • crude aqueous (CA) • crude hot water (CHW) • hexane (H) • methanol/ dichloromethane (MDC)	Responses (oxidative stress; nitric oxide bioavailability and endocan expression) in human aortic endothelial cells (HAECs) or in enzyme assays (ACE inhibition and HMG-CoA reductase inhibition)	Authors cite to wide medicinal use of edible mushrooms including *P. pulmonarius* species *In vitro* data supports medicinal use of this species
Protective effect of antioxidant extracts from grey oyster mushroom, *Pleurotus pulmonarius* (Agaricomycetes), against human low-density lipoprotein oxidation and aortic endothelial cell damage (Abidin, M.H.Z. *et al.* 2016. *Int. J. Medic. Mushrooms* 18(2):109-121)	*In vitro* study of antioxidant activity and potential ability to affect coronary disease Different mushroom extracts were tested versus a control (water)	Extracts of *P. pulmonarius* fruiting bodies • crude aqueous (CA) and protein fractions (protein fractions PF30, PF60, PF90) • crude hot water (CHW) • partially purified polysaccharide fraction (PP) • hexane (HF) • methanol/ dichloromethane (MD)	*In vitro* antioxidant assays included: Folin-Ciocalteu, 1,1-diphenyl-2-picrylhydrazyl free radical scavenging, metal chelating, cupric ion reducing antioxidant capacity, and lipid peroxidation inhibition Effects on human aortic endothelial cell (HAEC) viability and hydrogen peroxide-induced endothelial cell damage	Toxicity endpoint of cell viability: "*Based on the results (Fig. 5), cell viability was not less than 80% when treated with up to 200 µg/mL concentration of the extracts. The extracts possessed IC50 values higher than 200 µg/mL. It is proven that HF, WF, CA, and CHW extracts were not toxic to the HAEC.*" Studies support antioxidant

Summary of Published Studies with Data or Information Relevant to the Food Safety of *P. pulmonarious*				
Study Title (reference)	**Study Design**	**Type of Test Article**	**Outcomes/ Endpoints Measured**	**Reported Results**
		• water fraction (WF)		activity of various types of *P. pulmonarius* extracts
Antioxidant, antifungal, and anticancer activities of Se enriched *Pleurotus spp.* mycelium extracts (Milovanovic, I. *et al.* 2014. *Arch. Biol. Sci.* 66(4):1379-1388)	*In vitro* studies of anti-fungal and anti-cancer potential, with and without selenium enrichment	Extracts of *P. pulmonarius* strain HAI 573 mycelia (aqueous extraction and filtration)	Cytotoxicity to cells using extracts of the mushroom mycelia (12.5, 25, 50, 100 and 200 µg/ml) Minimal inhibitory concentrations (MICs) against fungal organisms Radical-scavenging activity in relation to phenol and flavonoid content of the extract	The authors report: *"Cytotoxic activity against both HeLa and LS174 cell lines was very low…"* Selenium enrichment influenced the anti-fungal activity of *P. pulmonarius* extracts (reductions) Radical scavenging capacity of extracts directly correlated with phenol/ flavonoid content.
Antioxidant and immunomodulating activities of exo- and endopolysaccharide fractions from submerged mycelia cultures of culinary-medicinal mushrooms (Jeong, S.C. *et al.* 2013. *Int. J. Med.*	*In vitro* assays to evaluate and compare antioxidant and immunomodulatory effects of exopolysaccharides (EXPs) and endopolysaccharides (ENPs) of submerged cultures of	Extracts of *P. pulmonarius* mycelia (aqueous extract)	Antioxidant activity assays: biological assay using *Saccharomyces cerevisiae;* DPPH radical scavenging activity; chelating ability for ferrous ions; and ferric reducing antioxidant power. Immunomodulation assessed by	Authors cite to wide medicinal use of edible mushrooms including *P. pulmonarius* species. Mycelia extract of *P. pulmonarius* exhibited both antioxidant and

Summary of Published Studies with Data or Information Relevant to the Food Safety of *P. pulmonarious*				
Study Title (reference)	**Study Design**	**Type of Test Article**	**Outcomes/ Endpoints Measured**	**Reported Results**
Mushrooms 15(3):251–266)	mushroom mycelia		production of IFN-γ, IL-2, IL-4 and IL-5, and macrophage enzyme activity	immunomodulatory activity.
In Vivo Animal Studies				
Evaluation of antidiabetic activity of *P. pulmonarius* against streptozotocin-nicotinamide induced diabetic Wistar albino rats (Balaji, P. *et al.* 2020. *Saudi J. Biol. Sci.* 27:913–924)	*In vivo* rat study of anti-diabetic activity of mushroom extracts in a rat model of diabetes LD50 values determined and then extracts administered orally for 4 weeks (200 and 400 mg extract/kg bw; both extracts tested separately)	*P. pulmonarius* mycelia extracts (aqueous and organic)	Normalization of blood glucose and insulin levels Levels of HDL and cholesterol in blood Pathology in pancreas. Liver and kidney	Only the aqueous extract exhibited activity to normalize glucose and insulin levels and reduced blood lipids Reversal of pathological changes in pancreas, liver, and kidney in untreated animals No reported adverse outcomes in the rats treated with single doses of either *P. pulmonarius* extract: *"The results of the acute toxicity study of HWE [hot water extract] and AE [acetone extract] showed no mortality even after 72 hrs at 2000 mg/kg. The AE showed*

Summary of Published Studies with Data or Information Relevant to the Food Safety of *P. pulmonarious*				
Study Title (reference)	**Study Design**	**Type of Test Article**	**Outcomes/ Endpoints Measured**	**Reported Results**
				mild sedation at 2000 mg/kg but however, HWE didn't alter any of the general behavior. No fatality or noxious reactions were found during and after the treatment period with both extracts."
Water extract from *Pleurotus pulmonarius* with antioxidant activity exerts in vivo chemoprophylaxis and chemosensitization for liver cancer (Xu, W.W. *et al.* 2014. *Nutrit. Cancer* 66(6):989-998)	*In vivo* study in mice of the effects of mushroom extracts on experimental liver cancer prevention and progression Prevention group received oral *P. pulmonarius* extract (200 mg/kg bw) daily for 2 weeks before cancer induction (control received no extract) Sensitization group had cancer induction before oral daily administration of the extract (50 mg/kg bw) for 28 days (control	Extracts of *P. pulmonarius* fruiting bodies (aqueous)	Tumor induction and tumor progression were monitored as was body weight and histopathology of "critical organs"	Toxicity evaluation: *"…critical organs of mice were collected and subjected to H&E staining for checking their histological features. No obvious difference in these organs was observed in the Pp-treated [P. pulmonarius only animals] and control groups… There was also no significant change in the body weight of the animals between treated groups and control group from the beginning until*

Summary of Published Studies with Data or Information Relevant to the Food Safety of *P. pulmonarious*				
Study Title (reference)	**Study Design**	**Type of Test Article**	**Outcomes/ Endpoints Measured**	**Reported Results**
	received no extract)			*the end of the experiments (Fig. 3B and Fig. 4B).”*
Extract of *Pleurotus pulmonarius* suppresses liver cancer development and progression through inhibition of VEGF-Induced PI3K/AKT signaling pathway (Xu, W. *et al.* 2012. *PLoS ONE* 7(3): e34406)	*In vivo* study in mouse xenograft model of anti-tumor potential of extract of *P. pulmonarius* Mushroom extract administered orally (200 mg/kg bw) or i.p. (50 mg/kg bw) three times a week for 27 days (total of nine doses) [*In vitro* mechanistic data also provided but not discussed here]	*P. pulmonarius* fruiting body extracts (aqueous)	The tumor volumes and body weight were measured every three days in all mice Histopathology of organs at sacrifice	Authors report the extract was non-toxic: *“Furthermore, no obvious side effects or changes were observed by comparing the histological features of internal organs including the liver, lung and kidney (Fig. 7A) and body weight of the animals (Fig. 7B) in the treatment and control group suggesting the safety and clinical implication of PP [P. pulmonarius] as an anticancer agent.”*
Glucans from the edible mushroom *Pleurotus pulmonarius* inhibit colitis-associated colon carcinogenesis in mice (Lavi, I. *et al*. 2012. *J.*	*In vivo* study of anti-cancer activity of mushroom extracts in a mouse model of colorectal cancer Mice fed a daily diet (80 days) containing 2 or 20 mg of	Extracts of *P. pulmonarius* fruiting bodies and mycelia (aqueous extracts of both)	Monitored: • formation of aberrant crypt foci • expression of proliferating cell nuclear antigen • number of apoptotic cells in colon	Both extracts reported to inhibit events linked to colon carcinogenesis: “*We conclude that P. pulmonarius FBE and ME inhibit colitis-associated colon carcinogenesis*

Summary of Published Studies with Data or Information Relevant to the Food Safety of *P. pulmonarious*				
Study Title (reference)	**Study Design**	**Type of Test Article**	**Outcomes/ Endpoints Measured**	**Reported Results**
Gastroenterol. 47:504–518)	FBE (fruiting body extract) or ME (mycelia extract) per mouse		• expression of TNF-a in colon tissue	*induced in mice through the modulation of cell proliferation, induction of apoptosis, and inhibition of inflammation."* No toxicity of the extracts is mentioned in the narrative of the paper.
Anti-leukemic and Immunomodulatory Effects of Fungal Metabolites of *Pleurotus pulmonarius* and *Pleurotus ostreatus* on Benzene-induced Leukemia in Wister Rats (Olefumi, A.E. et al. 2012. *Korean J. Hematol.* 47:67-73)	*In vivo* study in a rat model of leukemia (intravenous injection of benzene) An aqueous solution of mycelial fungal "metabolites" (20 mg/mL) orally administered (0.2 mL) before, during, and after leukemia induction One treatment group (n=16 rats) served as the control group for adverse reactions and toxicity that may be linked to intake of mycelia (Group D; only treated with the mycelia and	Extracts of *P. pulmonarius* mycelia (aqueous)	Hematological parameters at baseline and after leukemia induction Immunomodulatory potential assessed by a phagocytic assay	Results in the mycelia only group (control group) reported included: *"Results of various hematological parameters in group D revealed that there was no adverse reaction or toxicity experienced by the rats as a result of administration of the 2 metabolites."; "The healthy animals treated with P. ostreatus and P. pulmonarius metabolite presented in group D showed*

Summary of Published Studies with Data or Information Relevant to the Food Safety of *P. pulmonarious*				
Study Title (reference)	**Study Design**	**Type of Test Article**	**Outcomes/ Endpoints Measured**	**Reported Results**
	given a control diet)			*no significant alterations in erythrocyte counts, leukocyte counts, and hematological indices, when compared with the healthy animals in group F who received commercial feed and water only."*
Antimicrobial and anti-inflammatory potential of polysaccharide from *Pleurotus pulmonarius* LAU 09 (Adebayo E.A. *et al.* 2012. *Afr. J. Microbiol. Res.* 6(13):3315-3323)	*In vivo* study rat studies to investigate the potential anti-inflammatory actions of polysaccharides from *P. pulmonarius* Polysaccharides prepared by submerged fermentation culture of mycelia mat; harvested after 10 days, filtered, centrifuged and precipitated with acetone. Polysaccharide activity (6 mg/kg oral administration for 15 days) tested in a rat model of inflammation	Extract of *Pleurotus pulmonarius* mycelia (strain known as LAU 09; GenBank accession number JF736658) (acetone/organic extract)	Paw edema and sperm motility were monitored after 15 days	No polysaccharide toxicity reported for any dose level.

Summary of Published Studies with Data or Information Relevant to the Food Safety of *P. pulmonarius*				
Study Title (reference)	**Study Design**	**Type of Test Article**	**Outcomes/ Endpoints Measured**	**Reported Results**
	(paw edema induced by formalin or carrageenan).			
Antinociception of β-d-glucan from *Pleurotus pulmonarius* is possibly related to protein kinase C inhibition (Baggio, C.H. *et al.* 2012. *Int. J. Biol. Macromol.* 50:872– 877)	Follow-on study to Baggio et al. 2010. See next listing. Mice injected i.p. with mushroom extract (doses from 0.1 to 100 mg/kg bw) and acute effects monitored	Extracted B-glucan from milled *P. pulmonarius* (aqueous)	Anti-nociceptive activity assessed 30 minutes after intraplantar (paw) injections of capsaicin, cinnamaldehyde, menthol, acidified saline and phorbol myristate acetate (PMA) Also monitored PKC activation in skin and connective tissue	Anti-nociceptive activity *in vivo* reported at doses as low as 1 mg/kg B-glucan extract No polysaccharide toxicity reported for any dose level.
Antinociceptive effects of (1/3),(1/6)-linked b-glucan isolated from *Pleurotus pulmonarius* in models of acute and neuropathic pain in mice: evidence for a role for glutamatergic receptors and cytokine pathways (Baggio, C.H. *et al.* 2010. *J. Pain* 11(10): 965-971)	*In vivo* studies of mushroom polysaccharides in mouse pain models Mice injected i.p. (single or repeated doses) with B-glucan (30 mg/kg)	Extracted B-glucan from milled *P. pulmonarius* (aqueous)	ACUTE PAIN: intra-plantar injection of glutamate; intrathecal injection of glutamatergic receptor agonists, substance P, IL-1b, TNF-a CHRONIC NEUROAPTHIC PAIN: partial sciatic nerve ligation	Extract inhibited acute and neuropathic pain in mice via inhibition of glutamate receptors and IL-1b pathways The IC50 in the acute pain model = 0.34 mg/kg Chronic treatment with extract (2X/day for 7 days) reversed mechanical allodynia caused by partial sciatic nerve ligation

Summary of Published Studies with Data or Information Relevant to the Food Safety of *P. pulmonarious*				
Study Title (reference)	**Study Design**	**Type of Test Article**	**Outcomes/ Endpoints Measured**	**Reported Results**
				(chronic pain treatment). Doses effective to treat pain did not affect locomotor activity of mice (30 mg/kg; measure of toxicity) No extract toxicity reported
Orally administered glucans from the edible mushroom *Pleurotus pulmonarius* reduce acute inflammation in dextran sulfate sodium-induced experimental Colitis (Lavi, I. *et al.* 2010. *Br. J. Nutrit.* 103:393–402)	*In vivo* study in mouse colitis model (induced by 3.5% dextran sulfate sodium); doses of 2 and 20 mg/mouse extracts tested in mice with induced colitis Extract doses approximated dietary fiber intakes in humans Control groups: 20 days administration of extract in diet, no colitis induction, and (a) hot water extract (HWS) at 20 mg/mouse/day, mycelial extract (ME) at 20 mg/mouse/day	Extracts of *P. pulmonarius* fruiting bodies (aqueous; HWS group) and mycelia (aqueous; ME group)	Endpoints monitored: • Colon damage (macroscopic and histological) • Changes in colon length • Levels in colon of MPO activity and mRNA levels of IL-1b IL-10	Dietary HWS and ME reversed histological and biochemical alterations in experimentally induced colitis No adverse histopathological changes in colons of mice receiving only extracts for 29 days in diet; no adverse effects on inflammatory modulators in these same groups

Summary of Published Studies with Data or Information Relevant to the Food Safety of *P. pulmonarious*				
Study Title (reference)	**Study Design**	**Type of Test Article**	**Outcomes/ Endpoints Measured**	**Reported Results**
Clastogenicity potential screening of *Pleurotus pulmonarius* and *Pleurotus ostreatus* metabolites as potential anticancer and antileukaemic agents using micronucleus assay (Akanni, E.O. *et al.* 2010. *Br. J. Pharmacol. Toxicol.* 1(2):56-61)	*In vivo* genotoxicity study (micronucleus assay) Wister rats injected i.p. (16 and 64 mg/kg; 3.2 and 12.8% of the LD50) and then sacrificed (24, 48 and 72 h post treatment); both positive and negative controls were included	Unspecified extracts of *P. pulmonarius*	Rate of micronucleus formation over three time points compared to positive and negative controls	Extracts of *P. pulmonarius* were negative Only the positive control was associated with clastogenicity Provides evidence for a lack of mutagenic potential *in vivo*
Mushroom polysaccharide extracts delay progression of carcinogenesis in mice (Wasonga, C.G.O. *et al.* 2008. *J. Exp. Therapeut. Oncol.* 7:147-152)	*In vivo* study in a mouse model of liver cancer (DRNA injection followed by DENA in drinking water from week 3 to week 12) Mushroom extract mixed in the diet (mixed in a ratio of 1:4 with pellets; 25% of the diet); controls included no treatment and positive controls	Extracts of dried *P. pulmonarius* (aqueous)	Development/progression of liver cancer assessed by histological analysis and tumor markers (LDH and sialic acid)	Extracts of *P. pulmonarius* delayed progression of liver cancer Mice treated with mushroom extract alone (no carcinogen pretreatment) did not exhibit liver histopathology (same outcome as untreated controls) No other toxicity endpoints reported

Summary of Published Studies with Data or Information Relevant to the Food Safety of *P. pulmonarious*				
Study Title (reference)	**Study Design**	**Type of Test Article**	**Outcomes/ Endpoints Measured**	**Reported Results**
Effect of usuhiratake (*P. pulmonarius*) on sneezing and nasal rubbing in BALB/c mice (Yatsuzuka, R. *et al.* 2007. *Biol. Pharm. Bull.* 30(8):1557-1560)	*In vivo* study in BALB/c mice (repeated oral doses of *P. pulmonarius* powder in solution; 200 or 500 mg/kg/day) for 4 weeks	Powder of *P. pulmonarius* (whole mushroom)	Monitored sneezing and nasal rubbing behavior after antigen stimulation of histamine release as well as total IgE levels Monitored release of histamine from rat mast cells (*in vitro* method)	No effects on IgE, indicating direct effects on histamine release, consistent with *in vitro* study data (rat mast cell)
Hypoglycemic activity of aqueous extract of *Pleurotus pulmonarius* in alloxan-induced diabetic mice (Badole, S.L. *et al.* 2006. *Pharmaceut. Biol.* 44(6):421–425)	*In vivo* study of mushroom extract in diabetic and non-diabetic (untreated) mice Oral doses of 250, 500, and 1000 mg/kg bw (untreated and alloxan-treated mice) up to 28 days Acute oral toxicity also determined in untreated mice (doses up to 5000 mg/kg bw)	Dried *P. pulmonarius* powder extract (aqueous)	Serum glucose and body weight measured One group subjected to an oral glucose tolerance test (OGTT) Mortality after acute doses up to 5000 mg/kg	No mortality and no toxic effects reported at doses up to 5000 mg/kg (single dose) Repeated doses for 28 days reversed body weight loss, improved serum glucose levels, and reduced mortality compared to diabetic control mice No reports of other toxic effects attributed to the extract
Antioxidant, anti-inflammatory and antitumor activities of culinary-medicinal mushroom	*In vivo* studies in mice tumor models (ascites and solid tumor models) Mice injected i.p. with mushroom	Extract of *P. pulmonarius* fruiting bodies (methanol; organic)	Tumor regression was monitored	Significant tumor regression at all doses of the *P. pulmonarius* extract in both tumor models

Summary of Published Studies with Data or Information Relevant to the Food Safety of *P. pulmonarious*				
Study Title (reference)	**Study Design**	**Type of Test Article**	**Outcomes/ Endpoints Measured**	**Reported Results**
Pleurotus pulmonarius (Fr.) Quel. (Agaricomycetideae) (Jose, N. *et al.* 2002. *Int. J. Med. Mushrooms* 4:329-335)	extract (250, 500 and 1000 mg/kg bw) 24 hours after tumor inoculation (5 doses in the ascites model; for 10 days in the solid tumor model)			No extract toxicity reported for any dose level.

Form Approved: OMB No. 0910-0342; Expiration Date: 08/31/2025
(See last page for OMB Statement)

DEPARTMENT OF HEALTH AND HUMAN SERVICES
Food and Drug Administration

GENERALLY RECOGNIZED AS SAFE *(GRAS)* NOTICE (Subpart E of Part 170)

FDA USE ONLY

GRN NUMBER	DATE OF RECEIPT
001220	Oct 12, 2024
ESTIMATED DAILY INTAKE	INTENDED USE FOR INTERNET
NAME FOR INTERNET	
KEYWORDS	

Transmit completed form and attachments electronically via the Electronic Submission Gateway *(see Instructions)*; OR Transmit completed form and attachments in paper format or on physical media to: Office of Food Additive Safety *(HFS-200)*, Center for Food Safety and Applied Nutrition, Food and Drug Administration,5001 Campus Drive, College Park, MD 20740-3835.

SECTION A – INTRODUCTORY INFORMATION ABOUT THE SUBMISSION

1. Type of Submission *(Check one)*

☒ New ☐ Amendment to GRN No.________ ☐ Supplement to GRN No. ________

2. ☒ All electronic files included in this submission have been checked and found to be virus free. *(Check box to verify)*

3 Most recent presubmission meeting *(if any)* with FDA on the subject substance *(yyyy/mm/dd)*: 2024-05-29

4 For Amendments or Supplements: Is your amendment or supplement submitted in response to a communication from FDA? *(Check one)* ☐ Yes ☐ No If yes, enter the date of communication *(yyyy/mm/dd)*: ________

SECTION B – INFORMATION ABOUT THE NOTIFIER

1a. Notifier

Name of Contact Person	Position or Title
Thibault Godard	Chief Science Officer

Organization *(if applicable)*: Mushlabs GmbH

Mailing Address *(number and street)*: Humboldtstr. 59, 22083

City	State or Province	Zip Code/Postal Code	Country
Hamburg	none	22083	Germany

Telephone Number	Fax Number	E-Mail Address
2814935702		thibault@infiniteroots.com

1b. Agent or Attorney *(if applicable)*

Name of Contact Person	Position or Title
Laura Plunkett	Co-founder

Organization *(if applicable)*: BioPolicy Solutions LLC

Mailing Address *(number and street)*: 1127 Eldridge Parkway Suite 300-335

City	State or Province	Zip Code/Postal Code	Country
Houston	Texas	77077	United States of America

Telephone Number	Fax Number	E-Mail Address
281-493-5702		lmp@bpsllc.us

SECTION C – GENERAL ADMINISTRATIVE INFORMATION

1. Name of notified substance, using an appropriately descriptive term

Phoenix Oyster Mushroom Mycelium

2. Submission Format: *(Check appropriate box(es))*

- [x] Electronic Submission Gateway
- [ ] Electronic files on physical media
- [ ] Paper

If applicable give number and type of physical media

3. For paper submissions only:

Number of volumes

Total number of pages

4. Does this submission incorporate any information in CFSAN's files? *(Check one)*

- [ ] Yes *(Proceed to Item 5)*
- [x] No *(Proceed to Item 6)*

5. The submission incorporates information from a previous submission to FDA as indicated below *(Check all that apply)*

- [ ] a) GRAS Notice No. GRN
- [ ] b) GRAS Affirmation Petition No. GRP
- [ ] c) Food Additive Petition No. FAP
- [ ] d) Food Master File No. FMF
- [ ] e) Other or Additional *(describe or enter information as above)*

6. Statutory basis for conclusions of GRAS status *(Check one)*

- [x] Scientific procedures *(21 CFR 170.30(a) and (b))*
- [ ] Experience based on common use in food *(21 CFR 170.30(a) and (c))*

7. Does the submission (including information that you are incorporating) contain information that you view as trade secret or as confidential commercial or financial information? (see 21 CFR 170.225(c)(8))

- [ ] Yes *(Proceed to Item 8*
- [x] No *(Proceed to Section D)*

8. Have you designated information in your submission that you view as trade secret or as confidential commercial or financial information *(Check all that apply)*

- [ ] Yes, information is designated at the place where it occurs in the submission
- [ ] No

9. Have you attached a redacted copy of some or all of the submission? *(Check one)*

- [ ] Yes, a redacted copy of the complete submission
- [ ] Yes, a redacted copy of part(s) of the submission
- [ ] No

SECTION D – INTENDED USE

1. Describe the intended conditions of use of the notified substance, including the foods in which the substance will be used, the levels of use in such foods, and the purposes for which the substance will be used, including, when appropriate, a description of a subpopulation expected to consume the notified substance.

Alternative source of protein in food

2. Does the intended use of the notified substance include any use in product(s) subject to regulation by the Food Safety and Inspection Service (FSIS) of the U.S. Department of Agriculture?

(Check one)

- [ ] Yes
- [x] No

3. If your submission contains trade secrets, do you authorize FDA to provide this information to the Food Safety and Inspection Service of the U.S. Department of Agriculture?

(Check one)

- [ ] Yes
- [ ] No , you ask us to exclude trade secrets from the information FDA will send to FSIS.

SECTION E – PARTS 2 -7 OF YOUR GRAS NOTICE

(check list to help ensure your submission is complete – PART 1 is addressed in other sections of this form)

☒ PART 2 of a GRAS notice: Identity, method of manufacture, specifications, and physical or technical effect (170.230).

☒ PART 3 of a GRAS notice: Dietary exposure (170.235).

☒ PART 4 of a GRAS notice: Self-limiting levels of use (170.240).

☒ PART 5 of a GRAS notice: Experience based on common use in foods before 1958 (170.245).

☒ PART 6 of a GRAS notice: Narrative (170.250).

☒ PART 7 of a GRAS notice: List of supporting data and information in your GRAS notice (170.255)

Other Information

Did you include any other information that you want FDA to consider in evaluating your GRAS notice?

☐ Yes ☒ No

Did you include this other information in the list of attachments?

☐ Yes ☐ No

SECTION F – SIGNATURE AND CERTIFICATION STATEMENTS

1. The undersigned is informing FDA that Mushlabs GmbH

(name of notifier)

has concluded that the intended use*(s)* of Phoenix Oyster Mushroom Mycelium

(name of notified substance)

described on this form, as discussed in the attached notice, is (are) not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on your conclusion that the substance is generally recognized as safe recognized as safe under the conditions of its intended use in accordance with § 170.30.

2. Mushlabs GmbH *(name of notifier)* agrees to make the data and information that are the basis for the conclusion of GRAS status available to FDA if FDA asks to see them; agrees to allow FDA to review and copy these data and information during customary business hours at the following location if FDA asks to do so; agrees to send these data and information to FDA if FDA asks to do so.

Mushlabs GmbH, Humboldtstr.59, 22083, Hamburg, Germany

(address of notifier or other location)

The notifying party certifies that this GRAS notice is a complete, representative, and balanced submission that includes unfavorable, as well as favorable information, pertinent to the evaluation of the safety and GRAS status of the use of the substance.The notifying party certifies that the information provided herein is accurate and complete to the best or his/her knowledge. Any knowing and willful misinterpretation is subject to criminal penalty pursuant to 18 U.S.C. 1001.

3. Signature of Responsible Official, Agent, or Attorney	Printed Name and Title	Date (mm/dd/yyyy)
Laura M Plunkett Digitally signed by Laura M Plunkett Date: 2024.09.12 14:46:07 -05'00'	Laura M Plunkett, Agent for Mushlabs GmbH	09/12/2024

SECTION G – LIST OF ATTACHMENTS

List your attached files or documents containing your submission, forms, amendments or supplements, and other pertinent information. Clearly identify the attachment with appropriate descriptive file names (or titles for paper documents), preferably as suggested in the guidance associated with this form. Number your attachments consecutively. When submitting paper documents, enter the inclusive page numbers of each portion of the document below.

Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Form3667.pdf	Administrative
	PhoenixOysterMushroomMyceliumGRN12Sep2024.pdf	GRAS Notice
	Coverletter.pdf	Incoming Correspondence
	COSM_3667_21611_MushlabsGmbH.pdf	Administrative
	PhoenixOysterMushroomMyceliumGRN12Sep2024.pdf	Administrative